Chapter 7

Airborne Molecular Contamination: Contamination on Substrates and the Environment in Semiconductors and Other Industries

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Chapter Outline

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7.1 INTRODUCTION

A Japanese book to systematically describe airborne molecular contaminants (AMCs) from various viewpoints, such as primary sources, analytical methods to determine their concentrations on surfaces, in the air, and at the source, monitoring methods, chemical properties, detrimental mechanisms, and removal methods, was published 10 years ago.[1] This is one of the few books that systematically describe AMCs and their effects. In the present chapter, the authors intend to focus on the activities that have occurred since the book was published 10 years ago.

At that time, there were only a very small number of people in Japan who understood what AMCs were. In the past 10 years, the number of reports on AMCs has increased considerably and outstanding progress has been made in various fields. Studies to establish principles to systematically describe relevant phenomena, as well as to formulate equations by means of modeling, are also increasing. The authors believe that the study of AMCs is entering a new paradigm.

In 2001, a cleanroom was planned to be installed at the National Institute for Environmental Studies for the study of endocrine disruptors (Environmental Hormones). It had been previously doubted that many AMCs, including ester phthalates, exhibited the effects of an endocrine disruptor. In order to evaluate an endocrine disruptor, it is necessary to conduct the evaluation of the AMCs in an ultra cleanroom. All the knowledge and technology described in this chapter were used and considered to construct a cleanroom to evaluate potential endocrine actions. After construction of the cleanroom was completed, the quantification of AMCs that have a dramatic effect on endocrine actions, including ester phthalates, resulted in concentrations below ng/m^3 levels. This result is evidence to support the usefulness and validity of our research into clean technology as a countermeasure to mitigate AMCs in cleanroom environments. The technology to reduce AMCs in cleanrooms was demonstrated to be possible. From this point onwards, the subject of clean technology has shifted to cost-effective and efficient countermeasures to mitigate AMCs.

7.1.1 Background

Scientific personnel and research and development activities to study airborne molecular contaminants (AMCs) are increasing these days. Ten years ago reports on particle removal filters such as HEPA (high efficiency particulate air) and ULPA (ultra low penetration air) constituted the majority of the activities in the air cleanliness field. This trend has drastically changed over the past 10 years. AMCs have become a major topic in the study of cleanliness in cleanroom air and on Si wafer surface.

The trends in the study of AMCs in Japan over the past 5–10 years are summarized below by quoting the proportion of papers reported at the Annual Technical Meeting on Air Cleaning and Contamination Control held by the Japan Air Cleaning Association (JACA).

1. Total presentation papers: about 100.
2. The papers on AMCs account for 65%.
3. Adding the papers on dwellings (sick house syndrome, indoor facilities, including hospitals, and bio-cleanrooms) of 15%, the papers on AMCs account for over 80%.
4. The papers on analytical methods for AMCs in the air, AMCs on Si wafer surface and outgassing account for 30%.
5. The papers on removal methods for AMCs account for 20%. Recently, reports on other removal methods than chemical filtration, such as use of UV (ultraviolet)/photoelectron radiation, UV radiation/catalyst, and TiO_2 catalysts, have been increasing.
6. The papers that report new contaminant sources are not exhausted, but the number of papers reporting on new contaminants (even including those in indoor air environment) is decreasing. For example, dioctylphthalate (DOP; synonym: bis(2-ethylhexyl) phthalate) is often detected, and it is very frequently reported as a cause of deleterious effects. The DOP concentration calculated based on the currently available data from all DOP sources is still only half of the DOP concentration actually measured in the air. This means the material balance has not yet been correctly calculated.
7. Air cleanliness in a wider range of places, such as art and other museums and large computer rooms in banks, as well as bio-cleanrooms and indoor dwellings, is now being studied.
8. Studies to establish generic models and equations by interpreting various phenomena and data are increasing to 20%. Some of these quantitative equations feature outstanding compatibility with actual data, universality and usefulness. They help analyze and interpret large amounts of data appropriately. It has been reported recently that a cleanroom with high cleanliness has been developed in

which phthalate and phosphate concentrations are as low as 10 ng/m^3 or lower. These reports indicate that actual cleanliness is getting closer to a target cleanliness level. It is not impossible to improve cleanliness by one or two orders of magnitude to achieve the ultimate target in the future.

These trends indicate that the study of AMCs has been regarded as a target theme in Japan, and that the study of AMCs is moving through the following evolutionary stages, namely from frontier technology and know-how to science (a new paradigm).

1. Know-how for countermeasures against deleterious effects
2. Developments in analytical chemistry methods for qualification and quantification.
3. Establishment of database and reports on phenomenological mechanisms and findings.
4. Models and equations that have a wide-range versatility and applicability.
5. Science.

The present authors feel that the study of AMCs is in the transition period from Stage 3 to 4. Therefore, this chapter needs to not only generally introduce frontier technology reports, but also to describe the laws and chemistry to summarize the behavior of AMCs which will then be applied to a wide range of industrial and environmental chemistry.

To comprehensively describe here new technologies and to systematically explain them are intrinsically contradictory to each other. It was difficult to avoid some issues. There are a number of literature references and there is some overlap when systematically introducing theories, interpretations and data. It is very much expected that a more systematic book on science and technology of AMCs will be published in the near future. In the meanwhile, the authors expect the readers to keep in mind that this chapter describes a situation which is premature as science and that therefore it describes a precursor of the science of AMCs.

Guides and primers on AMCs include SEMATECH Technology Transfer Report,[2] as well as published papers[3–5] and proceedings of recent SEMICON Europe conferences.[6] Guides and primers published in Japan regarding AMCs include the 1997 AMC book,[1] plenary lectures at the Japan Aerosol Society,[7–9] the special issue of JACA,[10] Realize Inc. Seminar,[11] and review papers.[12,14] For basic information on the analytical methods for AMCs, the papers of Fabry,[13–15] Kaiser,[16] and Miki[17] are informative. Trends in standardization of the analytical methods will be discussed in Section 7.5.6.2.

7.1.2 Changes in Semiconductor Integration

The semiconductor industry has been growing by increasing the density of semiconductor devices.[2] Currently, the industry is exploring technologies to fabricate 256 M bit DRAM (dynamic random access memory) with 0.15 μm geometry on 300 mm wafers. As the density of semiconductor devices increases, higher cleanliness levels must be achieved on the Si wafer surface.[18]

Accordingly, the air in the manufacturing environment is also required to achieve higher cleanliness levels. Also, in cleanrooms used by other advanced industries, such as magnetic disk drives[19–22] and liquid crystal display,[23] as well as the brewery and biotechnology industries, contamination caused by AMCs and by particles becomes an issue that must be overcome.[24]

7.1.3 Changes in Target Contaminant

One of the biggest contributors to advancing semiconductor manufacturing technology was cleanliness improvement in the manufacturing environment. In the 1980s, particles were regarded as a major contaminant. Particle contamination was successfully controlled by development of cleanrooms equipped with HEPA and ULPA filters and even SULPA (super ultra low penetration air) filters. To manufacture 1 M bit DRAM, the cleanliness of the manufacturing environment must be controlled at ISO (International Standards Organization) Class 3 for particles of 0.5 μm or larger. Currently, the highest cleanliness of a cleanroom in ISO Class 3 is 0.1 μm, i.e. no more than 1000 particles of 0.1 μm size in one cubic meter.

Particle contamination has the following features.

- Particle contamination can be detected by means of electron microscopic examination.
- Particle contamination physically affects devices, e.g. defective interconnect and particle adhesion.
- Particle contamination can be overcome (cleanliness in terms of particle can be increased) by improving the filter performance.

In the 1990s, the detrimental effects caused by contaminants other than particles started to be reported, characterized by the following features.

- These contaminants were too small to be detected by means of electron microscopic examination.
- Some chemical reactions contribute to these effects.
- The behavior and detrimental effects of each contaminant are different and, therefore, a unique mitigation solution must be developed for each contaminant.

It was gradually realized that these new contamination effects were attributable to AMCs.

7.2 DEFINITIONS, TYPES, AND SOURCES OF AMCs

7.2.1 Definition of "Airborne Molecular Contamination"

For the purpose of this chapter, the following definition will be used in order to make it easier to describe the complex issues related to AMCs. "Contamination mainly comes from airborne molecular contaminants which are unable to be removed with particle filters such as HEPA and ULPA."

In order to explore the mechanisms of AMC contamination and their potential solutions, it is necessary to investigate not only AMCs in a narrow classification, but also metal vapor and mist which may not exist in the air in a gaseous phase. At present, measures are being taken against not only AMCs, but also other contamination caused by chemical reaction of metals and contaminants.

The International Technology Roadmap for Semiconductors (ITRS) Roadmap (1999, 2001, 2007, 2011, and 2013)[18] sets a requirement on the cleanliness of a Si wafer surface and the air in the manufacturing environment by specifying not only classical contaminants which are in a gaseous phase at room temperature, but also metals which are referred to as AMCs. In other words, the ITRS Roadmap uses the term AMCs to refer to all contaminants that degrade air cleanliness and that cannot be removed with particle filters such as HEPA and ULPA filters.

Airborne Molecular Contamination, the title of this chapter, is comparable to particle contamination. The most common film contamination occurs when organic impurities which are in a gaseous phase in the air adhere to the Si wafer surface to form a contaminant film on the surface. In order to come up with comprehensive understanding and practical solutions, however, it is necessary to interpret the airborne molecular contamination more extensively including the following cases.

1. Airborne Molecular Contamination includes inorganic compounds and metal organic compounds as well as organic compounds, such as hydrogen fluoride, boron trifluoride, ammonia, cyclosiloxane, and tributyl phosphate. Recently it has also included metal contaminants.
2. Airborne Molecular Contamination is much more limited in concentration than molecular layer contamination. What typically causes problems is not contamination that covers the entire surface of a Si wafer, glass or a hard disk, but a local contaminant island with thickness of less than 1/100 or 1/1000 of a molecular layer. In other words, it is usually not a "film" of contaminants.
3. Airborne Molecular Contamination is detected as particle on Si substrate (Si wafer), while it does not exist as a particle in the air. Rather, it is generated from gaseous contaminant sources. For example:
   - Hydrochloric acid in the air corrodes Al interconnect, eventually generating corrosion products such as aluminum oxide.
   - Ammonia in the air gets adsorbed onto a Si wafer surface. It then reacts with sulfur dioxide gas in the air to generate ammonia sulfate which is eventually detected as an impurity.
   - Cyclosiloxanes adhering to a glass surface react with UV radiation used in subsequent lithography process, and consequently through photochemical reaction (decomposition), SiO_2 particles are generated on the glass surface.

- Phosphorus oxide solidified on SiO_2 layer absorbs NH_3 and moisture in the air. By repeating the fusion-precipitation process, compounds are generated whose moisture and NH_3 concentrations are in equilibrium with these concentrations in the air. During this process, isolated spots and/or continuous lines are formed, which may be incorrectly identified as particle contamination.

4. Airborne Molecular Contamination may degrade the underlying surface. For example:
- When the perfluoro oil lubricant layer of ten plus angstroms thick on the disk surface in a HDD (hard disk drive) gets contaminated with siloxane or alcohol, the head crash phenomenon occurs in too many localized areas to determine which areas are precisely contaminated.[19,20,25] It is hypothesized that the impurities change the conformation of the perfluoro oil lubricant which is present not as a film but as multiple molecular layers on the surface. Three dimensionally, fluorine and oxygen are no longer randomly distributed, exhibiting self-insoluble tendency which is often observed in paint technology. In this case, AMCs affect film conformation on HDD substrate surface.
- Since ammonia and amines react with protons on the surface of photoresist used in the lithography process, protons on the surface decrease in number. As a result, the lithography process does not proceed satisfactorily, and photoresist residues remain like T-shaped bridges on the surface. This is referred to as the T-top phenomenon.
- Hexamethyldisilazane (HMDS) which is used to change intrinsically hydrophilic Si wafer surface to lipophilic surface so as to enhance photoresist adhesion, as well as trimethylsilanol which is a derivative generated when HMDS reacts with moisture in the air, adhere to the glass surface to generate lipophilic spots. Discoloration takes place when pigments selectively adhere to these lipophilic spots.
- Boron compounds and ester phosphate adhere to the Si wafer surface and diffuse into the Si wafer during subsequent manufacturing processes to change the dopant doses.

What is common to the above degradation mechanisms is that these effects are caused not by film-type contamination, but by contaminants in cleanroom air which cannot be removed with particle removal filters. Therefore, this chapter uses the term airborne molecular contamination when comparing it to contamination by a particle. As a generic term that refers to all contaminants that cannot be removed with particle removal filters, however, this chapter uses AMCs similarly to its usage in SEMATECH Technology Transfer No. 95052812A-TR.[2] Contamination generated on Si wafer due to these AMCs will be referred to as film contamination. This chapter uses the term AMC contamination when the contamination is attributed to AMCs even if the contaminants are of particulate shape.

7.2.2 Examples of AMC-Induced Problems in the Manufacturing Process

Below is a list of AMCs and the problems caused by them in the device manufacturing process.[1,26–42]

- Boron compounds: shift of threshold voltage and drop of driving voltage.
- SO_X: interconnect corrosion.
- NH_3 and amines: lithography pattern defect and T-top phenomenon.
- HMDS (hexamethyldisilizane): fogged glass and lens.
- LMCS (low molecular weight polycyclodimethylsiloxane): fogged glass and lens.
- DOP (dioctylphthalate): dielectric breakdown voltage failure of gate oxide.
- TCEP (trichloroethyl phosphate): shift of threshold voltage.

Sources of these contaminants are mainly non-natural compounds, including construction materials of the cleanroom and chemicals used in the manufacturing process. Usually AMC concentrations are lower in outdoor air than in cleanroom air. In a sense, AMC contamination can be regarded as a contamination side effect which is induced by using the cleanroom to suppress particle contamination.

7.2.3 Nature of AMC-Induced Effects

The deleterious effects induced by AMCs are quite different from those induced by particles.

- AMC concentration in a cleanroom does not show a visible day-to-day fluctuation. It is a gradual drift from a month to another.
- By setting manufacturing conditions at current AMC contamination levels, therefore, it is possible to maintain the manufacturing yield to some extent. In these cases, AMC contamination may not be recognized.
- In observing that the appropriate manufacturing conditions are different between cleanrooms, engineers often recognize for the first time that the differences are attributed to the difference in chemical contamination level in cleanrooms.
- In order to work out solutions to eliminate the deleterious effects of different contaminants, it is necessary to study each individual chemical contaminant in terms of source, generation mechanism, state in the air, mechanism to adhere to a clean surface, mechanism to cause these effects, and physical and chemical characteristics.
- One of the major sources of AMCs contamination is construction material of the cleanroom. Therefore, it is important to select appropriate materials prior to start of cleanroom construction. Even if cleanliness of the cleanroom is found to be inadequate, it is not easy to replace construction materials

after cleanroom construction is completed as cleanroom operation needs to be suspended for a long time. AMC contamination can be significantly decreased by replacing common construction materials with low outgassing cleanroom construction materials.

- The required AMC concentration is extremely low of the order of ng/m^3 order. This concentration is far lower than the chemical concentration that affects the human body. It is only very hazardous toxic gases (sarin, soman, nickel carbonyl, etc.) that are found to be dangerous at ppt (parts per trillion) or ng/m^3 level. There are few formal (and practical) analytical methods reported for these concentration levels. The environmental regulations in Japan[43] use the unit of μg/m^3 for all contaminants except dioxin. AMC contamination in semiconductor wafer fabs having no odor is hardly sensed by people working there. When a water droplet of 10 μl is deposited on a substrate such as clean glass or a Si wafer that was exposed to cleanroom air, the surface of the substrate gradually changes from hydrophilic to hydrophobic due to AMCs in the air deposited on the surface. The change in the contact angle of the water droplet as a function of exposure time is one of the few phenomena, other than analytical results, that provide evidence of AMC contamination.
- A practical chemical filter capable of removing AMCs to maintain cleanliness of less than the order of μg/m^3 for a long time is still under development. A chemical filter to selectively remove certain target low-concentration molecules is harder to develop than a particle removal filter, and it also requires chemical consideration. The removal efficiency of a particle filter is 99.9999%, while that of an AMC chemical filter is around 99%. Chemical filters for AMC contamination feature low efficiency and short lifetime and they are costly. When the manufacturing process is affected by particle contamination, such a solution as filter replacement (upgrade to more advanced particle removal filter) is effective. In the case of AMC contamination, however, installing a chemical filter is not necessarily the best solution as AMCs vary a lot in type. Although the performance of chemical filters has increased remarkably, it is necessary to improve it further technically and economically. Improvement of chemical filters requires research and development of trace-amount analysis. Some chemical filters feature removal efficiency of 99.99% in removing high-concentration AMCs. This level of chemical filter performance, however, is not sufficient to solve the problems of low-concentration AMCs.

What is critical in maintaining cleanliness of the cleanroom air in the semiconductor industry is the ultimate AMC concentration at the downstream location of the chemical filter. Section 7.5.3 describes a case where the removal efficiency of a chemical filter is different between high-concentration AMC environment and an extremely low-concentration AMC environment due to the difference in adsorption mechanism of the chemical filter for NH_3 removal in which oxides of phosphorus have been impregnated in the filter as an adsorbing center. When multiple types of AMCs coexist, it is necessary to consider the fruit-basket phenomenon described in Section 7.4.3.4.4. AMC contamination must be fully understood in terms of source, behavior in the air, contamination mechanism on substrates such as Si wafer, and the mechanisms that can cause problems. Based on this understanding, it is then necessary to develop appropriate mitigation solutions. The objectives of this chapter are (1) to describe the development of qualitative and quantitative analyses for various AMCs for different sources such as air, substrates and cleanroom construction materials, (2) to study the behavior of AMCs by using established analytical methods, and (3) to describe solutions that have been worked out based on personal investigation.

7.2.4 Classification of Airborne Molecular Contaminants

The International Technology Roadmap for Semiconductors (ITRS) 1999 Edition classifies AMCs into five groups of acids, bases, condensables, dopants and metals, and specifies the required AMC concentration in the air for each group as shown in Table 7.1.[18] The required AMC concentration on a Si wafer is defined in Tables 7.2 and 7.3.

7.2.4.1 SEMATECH Technology Transfer Report No. 95052812A-TR

SEMATECH Technology Transfer Report No. 9505212A-TR[2] published in 1995 focused on chemical characteristics of four groups of AMCs: molecular acids (MA), molecular bases (MB), molecular condensables (MC), and molecular dopants (MD). Table 7.4 shows a summary of this classification. Condensables are those compounds whose boiling point is higher than room temperature and which are detected on substrate such as Si wafer. Water is excluded from the group of condensables according to the definition in the report. Many of organic impurities which often cause problems fall in this group. Shiramizu and Kitajima[44] reported 0.2 ng/cm^2 of DOP on silicon wafer surface degrades the surface properties, leading to harmful effects of lowering TDDB (time-dependent dielectric breakdown). Dopants are defined as compounds of boron, phosphorus, or others, which affect the electrical characteristics of Si wafer. The required cleanliness level for the AMCs is set at 1/100 of the minimum concentration of the compound reported to have caused problems.

7.2.4.2 International Technology Roadmap for Semiconductor (ITRS) 1999

Table 7.1 from ITRS 1999 includes metals as part of AMCs to be controlled in the manufacturing environment for the 130 nm technology node in 2002.

The 130 nm device production is to be implemented in the manufacturing environment that contains various AMCs with trace concentration at μg/m^3,

TABLE 7.1 ITRS Roadmap Defect Prevention and Elimination Technology Requirements[18]

Year	1999	2000	2001	2002	2003	2004	2005	2008	2011	2014
Technology Node	*180 nm*			*130 nm*			*100 nm*	*70 nm*	*50 nm*	*35 nm*
Wafer environmental control										
Particles										
Critical particle size (nm)	90	90	75	65	60	55	50	35	25	18
Particles of critical size ($/m^{-3}$)	12	10	8	5	4	3	2	1	1.00E + 00	1
AMCs (ppt M)										
Lithography—Bases (as amine, amide, or NH_3)	1000	1000	750	750	750	750	750	<750	<750	<750
Gate—metals (as Cu, $E = 2 \times 10^{-5}$)	0.3	0.3	0.25	0.2	0.2	0.15	0.1	<0.07	<0.07	<0.07
Gate—organics (as MW > 250, $E = 1 \times 10^{-3}$)	200	170	130	100	90	80	70	70	50	<50
Organics (as CH_2) (*: Cn = 18)	3600	3000	2400	1800	1620	1440	1260	1260	900	<900
Salicidation contact—acids	10	10	10	10	10	10	10	10	10	10
Salicidation contact—bases	40	32	24	20	16	12	10	4	<4	<4
Dopants P or B	<10	<10	<10	<10	<10	<10	<10	<10	<10	<10

TABLE 7.2 ITRS Roadmap Starting Materials Technology Requirements[18]

Year	1999	2000	2001	2002	2003	2004	2005	2008	2011	2014
Technology Node	*180 nm*			*130 nm*			*100 nm*	*70 nm*	*50 nm*	*35 nm*
DRAM 1/2 PITCH (nm)	180	165	150	130	120	110	100	70	50	35
Wafer diameter (mm)	200	300	300	300	300	300	300	300	300	450
Particle: Front surface particle size (nm) latex sphere equivalent	≥90	≥82.5	≥75	≥65	≥60	≥55	≥50	≥35	≥25	≥17.5
Particles (cm^{-2})	≤0.13	≤0.12	≤0.12	≤0.14	≤0.13	≤0.12	≤0.10	≤0.10	≤0.10	≤0.10
Particles (number/wafer)	≤38	≤84	≤80	≤95	≤89	≤84	≤72	≤73	≤72	≤165
Metal: Critical surface metals (atoms/cm^{-2})	$\leq 1.8 \times 10^{10}$	$\leq 1.4 \times 10^{10}$	$\leq 1.2 \times 10^{10}$	$\leq 8.8 \times 10^{9}$	$\leq 6.8 \times 10^{9}$	$\leq 5.8 \times 10^{9}$	$\leq 4.9 \times 10^{9}$	$\leq 4.2 \times 10^{9}$	$\leq 3.6 \times 10^{9}$	$\leq 3.4 \times 10^{9}$
Silicon-on-insulator wafer										
Si final device layer thickness (nm) (tolerance ± 5%)	30–200	30–200	30–200	30–200	30–200	20–100	20–100	20–100	20–100	20–100

TABLE 7.3 ITRS Roadmap Surface Preparation Technology Requirements[18]

Year	1999	2000	2001	2002	2003	2004	2005	2008	2011	2014
Technology Node	*180 nm*			*130 nm*			*100 nm*	*70 nm*	*50 nm*	*35 nm*
DRAM 1/2 PITCH (nm)	180	165	150	130	120	110	100	70	50	35
Wafer diameter (mm)	200	300	300	300	300	300	300	300	300	450
Front end of line										
Particle size (nm)	90	82.5	75	65	60	55	50	35	25	18
Critical surface metals (atoms/cm^{-2})	$\leq 9 \times 10^9$	$\leq 7 \times 10^9$	$\leq 6 \times 10^9$	$\leq 4.4 \times 10^9$	$\leq 3.4 \times 10^9$	$\leq 2.9 \times 10^9$	$\leq 2.5 \times 10^9$	$\leq 2.1 \times 10^9$	$\leq 1.8 \times 10^9$	$\leq 1.7 \times 10^9$
Organics/polymers (C atoms/cm^2)	7.3×10^{13}	6.6×10^{13}	6.0×10^{13}	5.3×10^{13}	4.9×10^{13}	4.5×10^{13}	4.1×10^{13}	2.8×10^{13}	2.0×10^{13}	1.4×10^{13}

TABLE 7.4 Projected AMC Limits for the 0.25 μm Process SEMATECH Technology Transfer No. 95052812A-TR[2]

Process Step	Max. Residence Time (hour)	MA (ppt M)	MB (ppt M)	MC (ppt M)	MD (ppt M)
Pre-gate oxidation	4	13,000	13,000	1000 (0.06*)	0.1
Salicidation	1	180	13,000	35,000	1000
Contact formation	24	5	13,000	2000	100,000
DUV Lithography	2	10,000	1000	100,000	10,000

MA: molecular acids; MB: molecular bases; MC: molecular condensables;
MD: molecular dopants; DUV: deep ultraviolet.
**Calculated by Shiramizu and Kitajima based on Japanese data.*[44]

ng/m^3, and/or pg/m^3 levels. These concentration levels are much lower than those stipulated in work environment codes and environmental regulations. Compounds with this level of concentration have no odor. The concentration of pg/m^3 is used when discussing dioxin in the air.

Similar classification is included in STRJ 1999 published by the Semiconductor Technology Roadmap Committee in Japan (STRJ).[45]

This classification is appropriate for the semiconductor industry to control cleanliness, identify contaminant sources, explore the mechanisms of contamination problems, and work out mitigation solutions. Some researchers suggest that a classification using organic groups and inorganic groups is appropriate for other industries and research areas. The present chapter refers to the most common ITRS 1999 classification when describing leading-edge cleanliness research.

7.3 ANALYSIS METHODS

Unlike particle contamination, AMC contamination requires a number of enabling technologies such as trace analysis, material characterization, and study of chemical reaction mechanisms. For AMCs in the air, however, very few technologies have been reported; a method to simultaneously analyze AMCs of several tens of micrograms per cubic meter is the only one that has been reported.[46] A major obstacle to developing an innovative AMC analysis technology is that the blank test value usually becomes high when a large volume of sample is taken in which a number of chemicals coexist, resulting in large noise in the analytical signal. Ohmi and Miki[47] analyzed specific metal ions at a level μg/m^3 order in air. They took a large volume of ultra pure water exposed to cleanroom air, concentrated it to a few ml, and analyzed the concentrated solution with ICP-AES (inductively coupled plasma—atomic emission spectrometry)/ICP-MS (inductively coupled plasma mass spectroscopy). They also took precautions to prevent contamination during each handling step. They suggested the possibility that contaminant analysis of lower concentration in air can be performed when the blank value or its variation is low and is controlled.[47–49]

It has now become possible to study AMCs by examining accurate analysis results obtained with trace analysis. As cutting edge trace analysis and local surface analysis need to be fully utilized for this purpose, it is recommended the reader refer to the introductory guides of these analysis methods.[13,50–55] For metal analysis on silicon wafer surface, chemical analysis and instrumental analysis methods have been described.[56–59]

The remainder of this section discusses analysis methods required for the study of AMC in the following order.

1. Methods to analyze chemical contaminants of the order of ng/m^2 adhering to Si wafer and glass substrate surfaces, including qualification/quantification by instrument and/or surface analysis quantification.
2. Methods to analyze chemical contaminants at μg/m^3 or ng/m^3 levels in the air.
3. Outgassing evaluation methods for cleanroom construction materials and plastics which constitute the main sources for generation of AMCs.

7.3.1 Substrate Surface Analysis

7.3.1.1 Quantitative Analysis

Quantitative analysis of contaminants on a substrate surface often uses solvent cleaning method and wafer thermal desorption gas chromatography/mass spectrometry (WTD-GC/MS). To analyze specific local surface areas with contaminants or defects rather than the entire surface, local surface analysis methods such as time-of-flight secondary ion mass spectrometry (TOF-SIMS), X-ray photoelectron spectroscopy (XPS), and Fourier transform infrared spectroscopy (FT-IR) are applied.

7.3.1.1.1 Solvent washing method

The substrate surface is cleaned with an appropriate solvent to dissolve the contaminants on the surface in the solvent. The composition of the solvent is adjusted to prepare a final sample solution. The sample solution is analyzed with ICP-MS, ion chromatography (IC), CE, CE/MS (capillary electrophoresis/mass spectrometry), or GC/MS.[54,57–68] Recently, Iikawa and Momoji[64,68] reported a CE/MS method for cation and anion analysis in the wafer cleaned solution,

TABLE 7.5 Amine and Ammonia Concentration (μg/m^3) in Air in Various Cleanrooms

Compound	C/R-1	C/R-2	R-1
NH_4^+	1.4	3.5	1.4
Methylamine	0.03	<0.01	<0.01
Amine-a*	0.06	0.07	1.5
Amine-b*	0.05	0.08	0.06
Amine-c*	0.02	0.04	<0.01
Total Amines	0.13	0.17	1.5

**Calculated using a tetramethylammonium standard working curve.*

which is rapid and accurate, and provides a lower determination limit in routine analysis. The results from CE/MS analysis for ammonia and amines are shown in Table 7.5 and Figure 7.1.

It is critical to select the most appropriate solvent for a target contaminant. Several solvents are proposed which feature high purity, dissolve contaminants

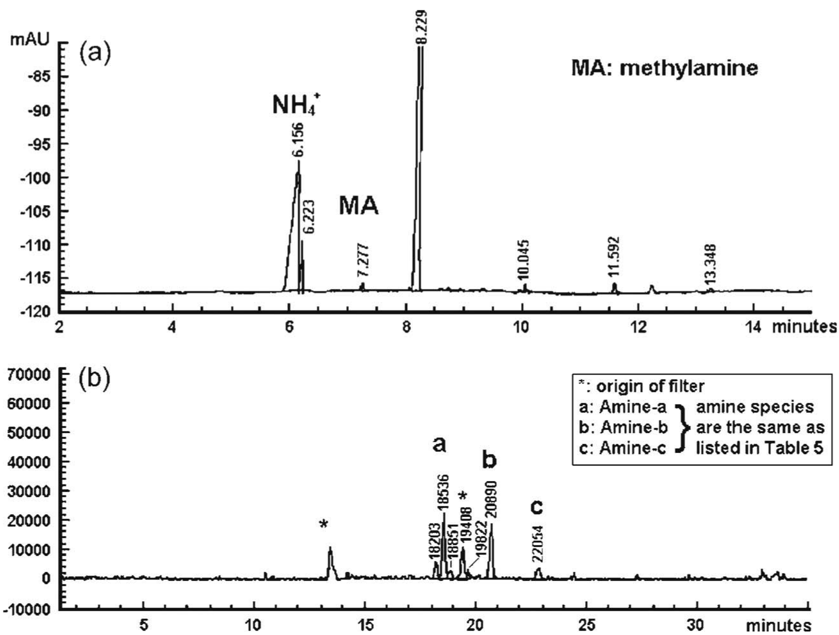

FIGURE 7.1 Electropherogram of the air in cleanroom C/R-1 (in Table 7.5) by (a) CE (upper figure) and (b) CE/MS (lower figure).

effectively, can be applied to a wide range of contaminants, are easy to be concentrated, and are resistant to spectral interferences. It is necessary to select the most appropriate solvent for each target contaminant as these solvents have their merits and demerits. This method is further segmented by the solvation method, including solvent vapor cleaning method, solvent droplet extraction method, and the like.[67–77]

7.3.1.1.2 Wafer thermal desorption—gas chromatography/mass spectrometry (WTD-GC/MS)

In this method, a sample is heated in a cell and the gas evaporated from the sample is introduced into GC/MS to measure the concentration. There are two types of WTD-GC/MS.

- Static headspace—gas chromatography—mass spectrometry (SHS-GC/MS) in which the gas emitted from the cell is introduced directly into the GC/MS.
- Dynamic headspace—gas chromatography—mass spectrometry (DHS-GC/MS) has an adsorption tube to collect the gas. The gas collected in the tube is then concentrated. The adsorption tube is heated again to regenerate the gas and an appropriate carrier gas is added prior to introduction into the GC/MS.

The solvent cleaning method is highly accurate, but it may not meet a target determination limit in some cases. DHS-GC/MS is suitable for trace analysis as it features high sensitivity. Figure 7.2 shows the GC/MS chromatograms for DHS and SHS outgassing tests. The identification limit appears to be governed by whether it is possible to prevent inclusion of the gas evaporated from the back surface of the substrate and to collect the gas evaporated only from front surface of the substrate. This method is often used to analyze a Si wafer surface that has been exposed to the air for an extended period.

The witness plate method is one of the methods used to measure particulate contamination in the air. A Si wafer or a glass substrate is exposed to cleanroom air for a given period of time and the particle count on its surface is measured. WTD-GC/MS is analogous to the witness plate method for analysis of AMCs for the following reasons.

- For some organic compounds, the lowest determination limit measured with WTD-GC/MS is lower than that measured with the air analysis described below. Very useful information can be obtained with the database of the sticking probability which is the relationship between AMC concentration in air and on silicon wafer surface exposed for 24 hours. Sticking probability is discussed later in Section 7.4.3.4.
- What is generally required is information on a target AMC to contaminate a Si wafer surface in the manufacturing process.
- When the cleanroom is located far away from the analytical laboratory, it is easier to transport a Si wafer exposed to cleanroom air than to transport sampled cleanroom air.

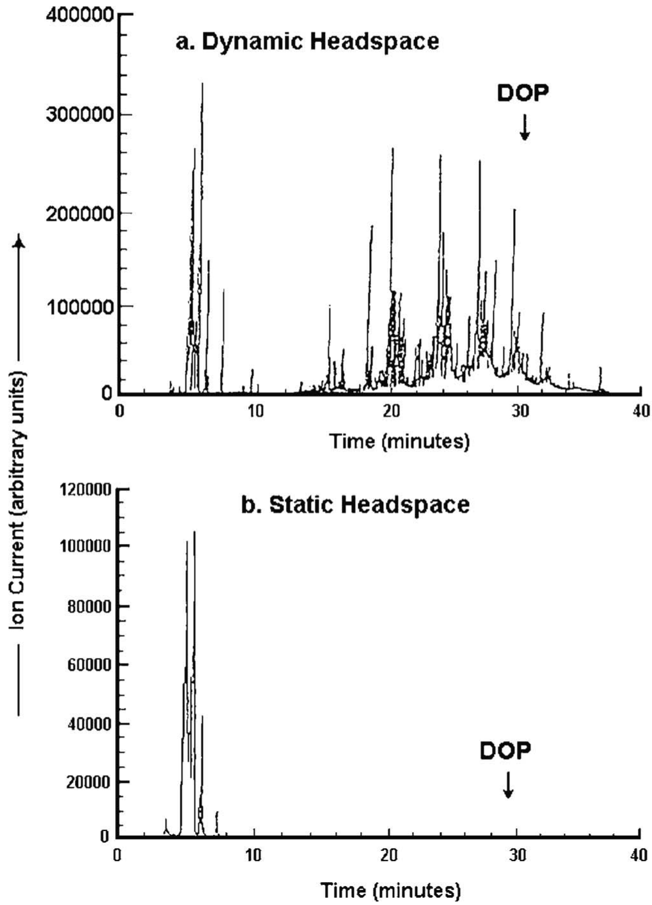

FIGURE 7.2 GC/MS chromatograms of a polypropylene sheet sample from (a) dynamic headspace outgassing tests and (b) static headspace outgassing tests.

Figure 7.3 shows a schematic diagram of WTD-GC/MS system. Recently, Ishiwari et al. have reported on the use of a small Si plate exposure method that is rapid and convenient.[69,71]

However, the common plastic wafer carrier is not appropriate to transport Si wafers exposed to cleanroom air because the plastic carrier also seriously contaminates the wafer with AMCs. For instance, in order to measure cleanliness levels at 0.001 ng/cm^2, it is essential to select a container which maintains the blank value on a 300 mm Si wafer below 1 ng. An all-aluminum carrier has been developed as a single or multiple wafer carrier that is free from AMC generation. Figures 7.4 and 7.5 show the carriers for one wafer and for 26 wafers, respectively. A large database is available on the data taken through the analytical procedure combining the all-aluminum carrier and WTD-GC/MS.[78]

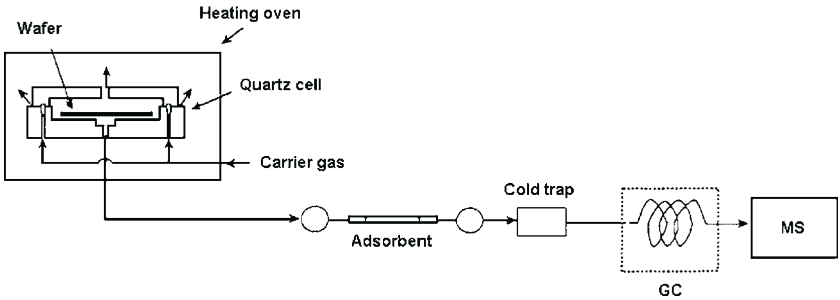

FIGURE 7.3 Schematic of the setup for wafer thermal desorption GC/MS (WTD-GC/MS).

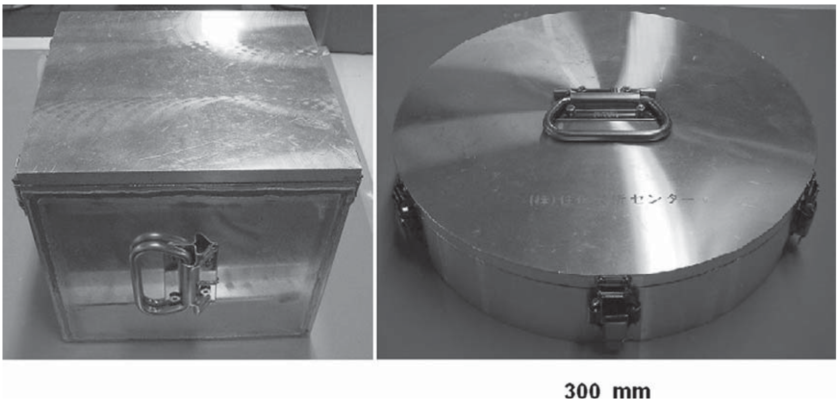

FIGURE 7.4 Example of a single sample wafer carrier made of aluminum.

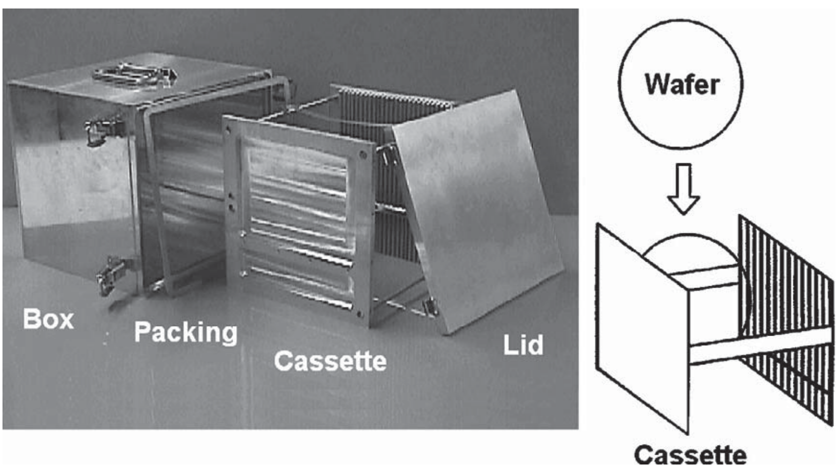

FIGURE 7.5 Example of a multiple sample wafer carrier made of aluminum.

7.3.1.2 Surface (Instrumental) Analysis

X-ray micro-analysis (XMA), total refection X-ray fluorescence (TRXRF), and XPS are available methods to analyze inorganic compounds such as NH_3, NO_x, and SO_x. For the analysis of organic compounds, TOF-SIMS, FT-IR, ion mobility spectrometry (IMS), and thermal desorption spectrometry/atmospheric pressure ionization mass spectrometry (TDS-API-MS) are available.[25–29,54,56–58,79–88]

Recently, Beckhoff et al.[56] reported metal analysis by TXRF and NEXAFS with Synchrotron Orbital Ray (SOR) in the BESSY II accelerator which indicates improvement of the determination limit. It suggests the possibility of achieving lower a determination limit for the low-Z elements, such as C, O, N, Mg, Na, and Al. The SOR of BESSY II has an absolute detection limit of 0.3–1.3 pg, corresponding to approximately 2×10^7 to 1×10^8 atoms/cm^2.

Wang et al.[57] reported the determination limit for metal analysis 3.6×10^4 atoms/cm^2 can be attained "theoretically" with synchrotron radiation total reflection X-ray fluorescence (SRTXRF) using a pre-concentration process such as vapor phase decomposition (VPD). They discussed the actual determination limit of each contaminant by controlling the contaminant concentration of DI water and chemicals.

Figures 7.6–7.8 show the concentration of AMCs measured with TOF-SIMS and XPS on the same Si wafer. In-depth AMC profiles are different among these methods although the same sample was measured. When AMCs adhering to glass that cause hazing were analyzed, siloxanes, CN^-, CNO^-, NO_2^-, and NO_3^-

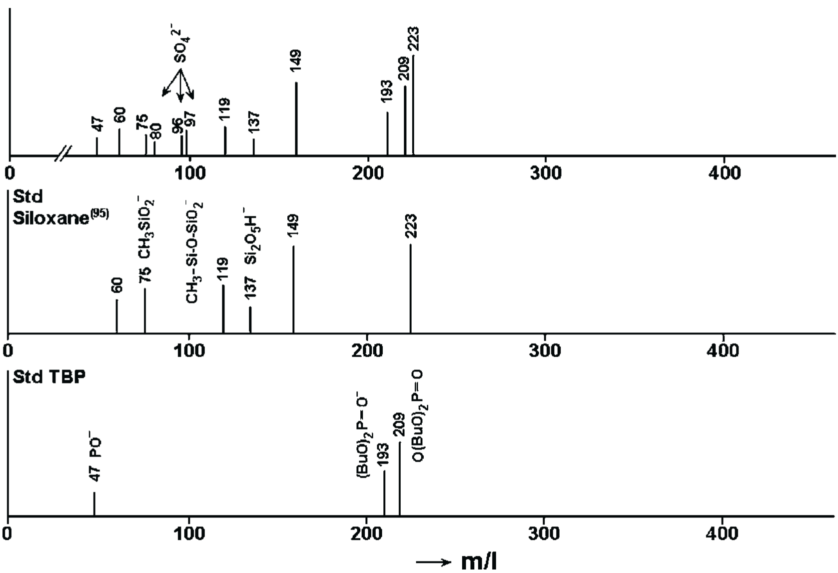

FIGURE 7.6 Typical spectra from TOF-SIMS analysis of a wafer surface.

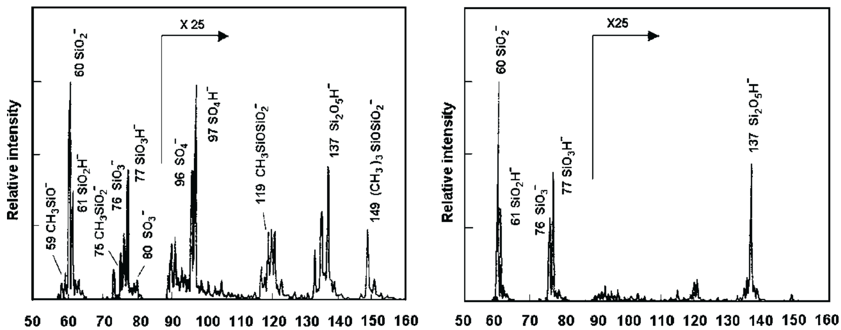

FIGURE 7.7 TOF-SIMS spectra of a wafer surface exposed to cleanroom environment.

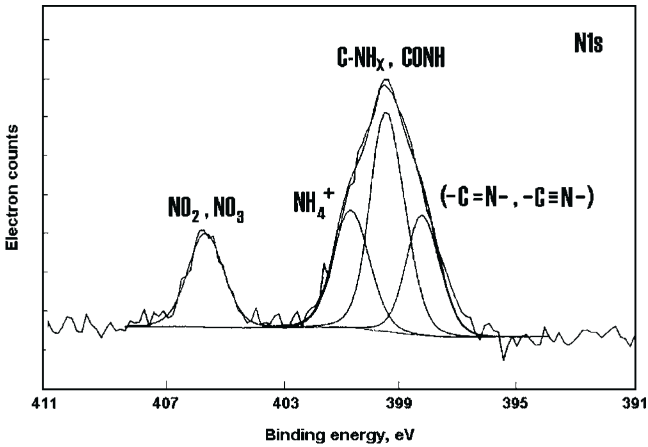

FIGURE 7.8 XPS spectra analysis of a wafer surface exposed to cleanroom environment.

were detected with TOF-SIMS, while C–NH_x, CONH, NO_2, NO_3, and NH_4^+ were detected with XPS. Based on consideration of the spectra from the two methods and the behavior of AMCs in a vacuum tool, it is suggested that NH_4NO_3, NH_4NO_2, and similar compounds were possibly generated.

7.3.2 Air Analysis

For the analysis of AMCs, several reports have been presented.[21,89–96] JACA standard No. 35A-2003 (revised version of No. 35-2000) has been published by the Japan Air Cleaning Association (JACA).[90]

Figure 7.9 outlines the sampling and analysis methods for AMCs. A sample is taken in a solid form with collection-with-filter method, or it is taken as gas sample with an impinger or an adsorption tube. The sample is concentrated and its composition is adjusted before it is evaluated with trace analysis methods such as ICP-MS, ion chromatography, and GC/MS. The analysis method for each analyte category defined by SEMATECH will be discussed below.

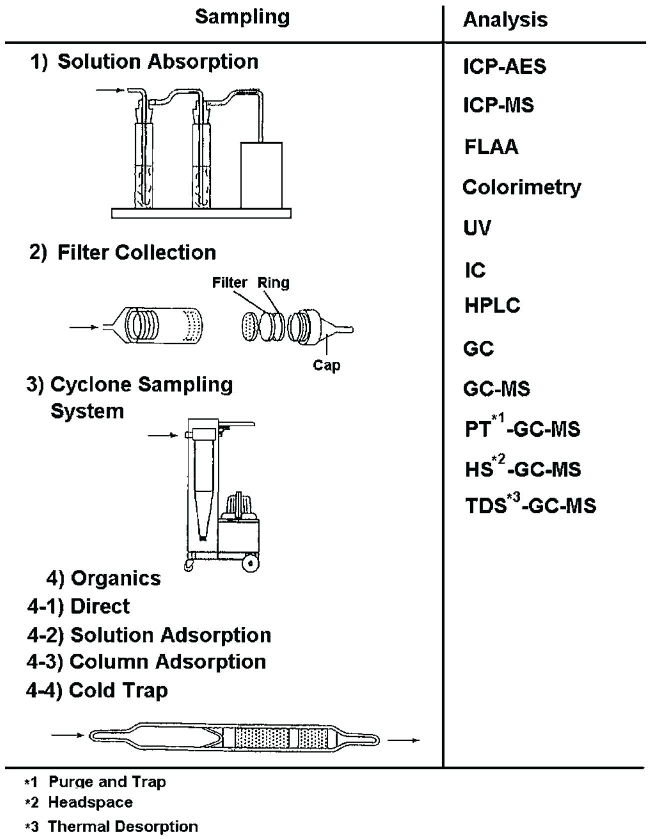

FIGURE 7.9 Outline of the different sampling and analysis methods for AMCs.

7.3.2.1 Acids[80–84]

SO_2, NO, NO_2, HCl, Cl_2, and HF are acids gases that are often detected. They are mainly collected with the absorption-to-solution method using an impinger and are subsequently analyzed with ion chromatography. Figure 7.10 shows the absorption-to-solution sampling system and Figure 7.11 shows typical chromatograms from IC analysis.

Two different absorption solutions were tested: ultra pure water and/or ultra pure water spiked with H_2O_2. Table 7.6 compares the results of the analysis of each solution. When the H_2O_2 concentration is high, it appears that NO decreases and NO_2 increases. However, NO_x, the sum of NO and NO_2, remains unchanged. Cl_2 features low saturated solubility in ultra pure water. When 0.1% H_2O_2 is spiked to ultra pure water, however, Cl_2 recovery goes up sufficiently due to enhancement of the dissolution rate and the saturated solubility (or because Cl_2 is converted not to ClO^- but to Cl^-). Tables 7.7 and 7.8 show results from various cleanroom air analyses.

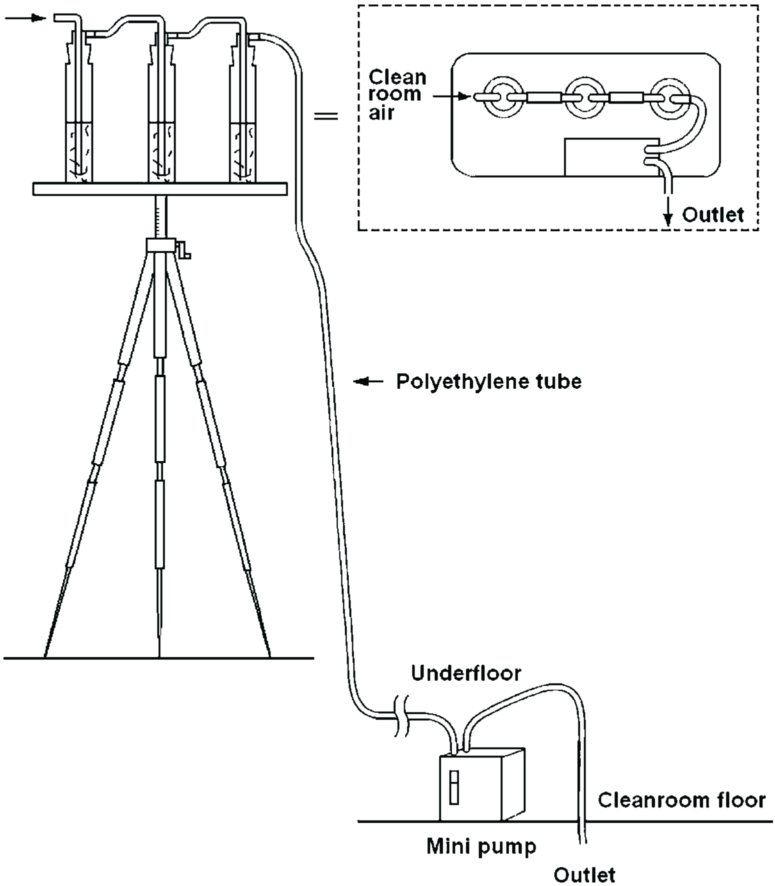

FIGURE 7.10 Schematic of the absorption-to-solution sampling system.

7.3.2.2 Bases: Ammonia (NH_3) and Amines

Ammonia is derived from outdoor air as well as air inside the cleanroom. Just like acids, ammonia together with several amines are collected with the absorption-to-solution method and analyzed with ion chromatography. Figure 7.12 shows the typical ion chromatogram. Amines are analyzed, like organic compounds described below, with GC/MS.

7.3.2.3 Condensables: Organic Compounds

Organic compounds are mainly analyzed with GC/MS. Organic compounds in cleanroom air are collected with activated carbon, TENAX and other adsorbents.[90–92] After going through pretreatment such as concentration, they are analyzed with GC/MS. Organic compounds are identified and quantified from the obtained peaks and their intensities. To evaluate overall contaminant

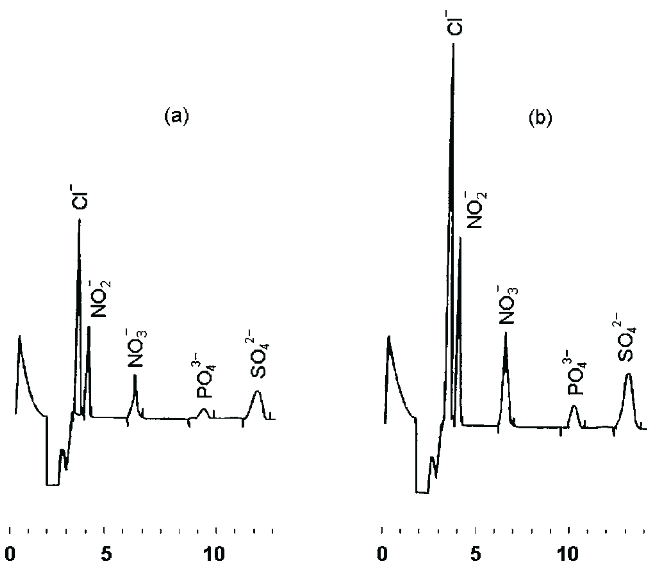

FIGURE 7.11 Ion chromatograms of a standard solution with known concentration (a) 5 ng/ml and (b) 10 ng/ml.

TABLE 7.6 IC Analysis Results of Room Air in Various Absorbing Solutions

AIR Sample	Absorption Solution	Anions (ng/m^3)				
		Cr	NO_2^-	NO_3^-	*Total* NO_x	SO_4^{3-}
Room A	H_2O	1000	13,000	900	18,000	1900
	H_2O + 0.01% H_2O_2	1100	12,000	1900	18,000	3300
	H_2O + 0.1% H_2O_2	1200	8000	6200	17,000	3500
Room B	H_2O	17	220	20	240	34
	H_2O + 0.01% H_2O_2	18	210.00	30	240	59
	H_2O + 0.1% H_2O_2	12	140	110	250	63

TABLE 7.7 Results of IC Analysis of Various Air Samples

Sample Air Location	Anions (ng/m^3)				
	Cr^-	NO_2^-	NO_3^-	SO_4^{2-}	PO_4^{3-}
Outdoor	76	9	900	8070	<10
Cleanroom A	670	<5	580	3560	<10
Cleanroom B	90	<5	660	1890	<10
Cleanroom C	20	6	150	2420	600
Cleanroom D	<10	<5	<5	5	<10
Cleanroom E	<10	<5	<35	1300	1300
Analysis laboratory	105	10	1100	11,200	20

TABLE 7.8 Results of IC Analysis of Cleanroom Air for Various Processes

Cleanroom	Anions (ng/m^3)				
	F^-	Cl^-	NO_3^-	SO_4^{2-}	PO_4^{3-}
Process A	<10	160	18,000	12,000	<10
Process B	<10	570	10,000	20,000	<10
Process C	<10	150	3000	56,000	<10
Process D	4000	1100	3000	24,000	3000

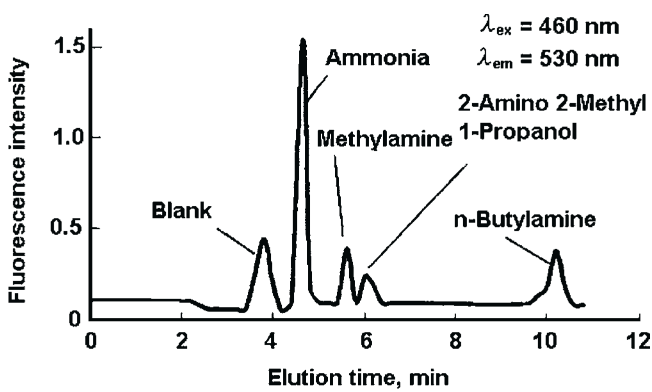

FIGURE 7.12 Ion chromatograms of ammonia and amines from cleanroom air in an absorption solution.

TABLE 7.9 Mass Numbers for Analytical Ions for Different Compounds for the SIM Method of GC/MS

Compound	Analytical Ions (m/z)	
	Quantification	*Identification*
Hexamethylcyclotrisiloxane (D_3)	207	96
Octamethylcyclotetrasiloxane (D_4)	281	265
Decamethylcyclopentasiloxane (D_5)	73	267
Dodecamethylcyclohexasiloxane (D_6)	73	429
Dibutylphthalate (DBP)	149	—
Di-(2-ethylhexyl) phthalate (DOP)	149	167
Tributylphosphate (TBP)	99	155
Tri-(2-chloroethyl) phosphate (TCEP)	249	83

distribution, total ion monitoring (TIM) should be used to detect abnormal compounds which do not exist under normal conditions. Quantification of regular contaminants which always exist is carried out in Selective Ion Monitoring (SIM) mode to obtain precise information. Table 7.9 shows the ion mass numbers used for determination of various AMCs in SIM mode.

By reducing the variation and carefully controlling the blank value, it is possible to perform routine analysis at a level of 0.01 $\mu g/m^3$. To control the accuracy and reliability, it is important to arrange multiple adsorption tubes in series and compare the ratio of recovery between the first tube and the second tube and the recovery ratio between the second tube and the third tube.

Figure 7.13 shows a typical adsorption tube. Figure 7.14 shows a TIM chromatogram. Table 7.10 shows the results of identification of 50 plus peaks of compounds that are often detected with GC/MS. The identification was based on fragmentation with synthetic standard samples, as well as published data.[93] Taking DOP as an example, Figures 7.15 and 7.16 show mass spectra of a

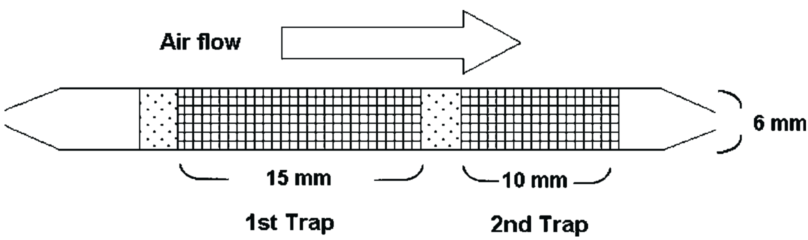

FIGURE 7.13 Typical adsorbent tube.

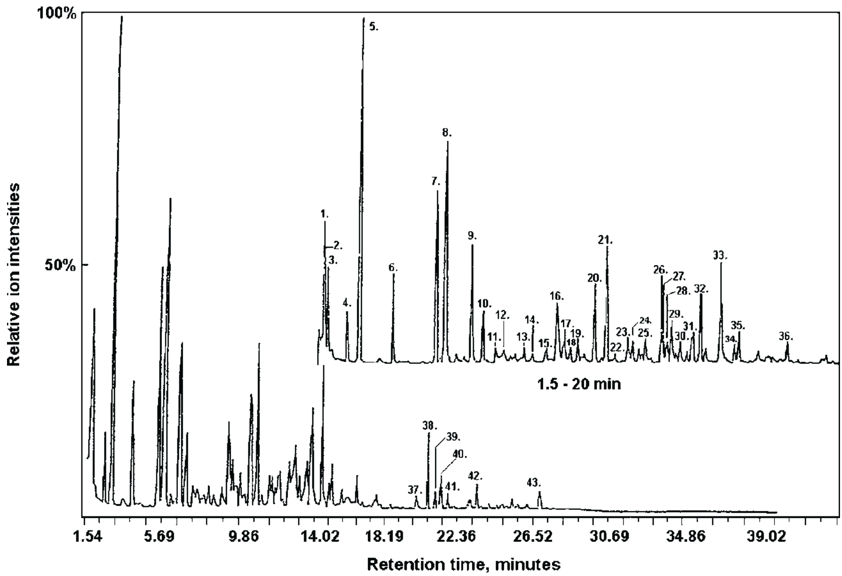

FIGURE 7.14 Total ion chromatogram of cleanroom air.

standard sample and the calibration curve in SIM mode, respectively. For gases generated from urethane sheet, it is possible to identify not only each peak but also groups of peaks.

Organic compounds, which are detected as regular contaminants and are likely to adhere to a Si wafer surface and likely cause problems in a manufacturing environment, are discussed below.

7.3.2.3.1 Siloxanes

The monomers of siloxanes are HMDS mentioned previously. Their dimers are hexamethyldisiloxanes, which are rarely observed compared with other siloxanes. The primary siloxanes in cleanroom air that can cause problems are the cyclosiloxanes (D_n) at the trimer (D_3) and above, where n represents the number of polymerization of $(CH_3)_2(SiO)_n$. Most commonly reported polymerization numbers include $n = 3, 4, 5, 6$, and 12. HMDS is a reagent in the silicon fabrication process. Cyclosiloxanes mainly come from silicone sealants.

7.3.2.3.2 Ester phthalates

Di-(2-ethyl hexyl) phthalates or DOP are typical phthalates. Other compounds that are often detected are dibutyl phthalates (DBP) and diethyl phthalates (DEP).

When AMCs on Si wafer surface are analyzed with WTD-GC/MS, 2-ethyl hexanol (2-EH) and phthalic anhydride are detected in addition to DOP. It appears that these compounds exist in the air besides DOP. Therefore, chemical

TABLE 7.10 Results of Analysis of Condensable Compounds in Cleanroom Air

Compound	Reliability*	Calculation Method	Concentration (µg/m³)			Origin**
			Cleanroom air		*Ambient air*	
			A	*B*		
Tetrachloromethane	93.5	Toluene	<1	<1	<3	A
2,3-Dimethylpentane	84.7	Toluene	<1	1	<3	
1-Methoxy-2-propanol	87	Toluene	18	5	<3	C
2-Hexanone	86.8	Toluene	<1	1	2	A
Toluene	–	STD	221	89	48	A, C
Acetic acid, 2-methylpropyl	89.7	Toluene	2	<1	3	A
Ethylbenzene	94.2	Toluene	72	25	5	A, C
Xylene	93.2	Toluene	121	36	5	A, C
Xylene	92.3	Toluene	39	11	<3	A, C
2,4,6-Trimethyldecane	91.2	Toluene	12	5	4	A, C
2-Butoxyethanol	88.5	Toluene	7	2	<3	C
2-Ethoxyethylacetate	81.6	Toluene	2	<1	<3	C
2,6-Dimethyloctane	91.3	Toluene	1	<1	<3	C

Continued

TABLE 7.10 Results of Analysis of Condensable Compounds in Cleanroom Air—Cont'd

Compound	Reliability	Calculation Method	Concentration (μg/m^3)			Origin
			Cleanroom air		*Ambient air*	
			A	*B*		
4-Propylheptane	90.1	Toluene	2	<1	<3	C
Propylbenzene	91.8	Toluene	2	1	<3	C
1-Ethyl-3-methylbenzene	96.3	Toluene	19	7	11	A, C
1,3,5-Trimethylbenzene	93.7	Toluene	9	4	<3	C
2,6-Dimethyloctane	89.9	Toluene	3	1	<3	C
1-Ethyl-3-methylbenzene	95.9	Toluene	4	2	<3	C
1-Ethyl-3-methylbenzene	94.7	Toluene	23	9	5	A, C
Decane	90.9	Toluene	26	10	12	A, C
Octamethylcyclotetrasiloxane (D_4)	–	STD	0.5	0.2	<3	C
1,3,5-Trimethylbenzene	92.2	Toluene	7	2	<3	C
2,3-Dimethylnonane	86.5	Toluene	3	2	<3	C
2-Ethyl-1-hexanol	88.8	Toluene	8	2	<3	C
1,2-Diethoxybenzene	86.5	Toluene	10	4	<3	C

1-Methylpropylbenzene	91.5	Toluene	11	3	<3	C
1,2-Diethylbenzene	91.1	Toluene	6	4	<3	C
1-Ethyl-3,5-dimethylbenzene	93.3	Toluene	16	7	<3	C
1-Methylpropylbenzene	71.8	Toluene	3	2	<3	C
1-Ethyl-2,4-dimethylbenzene	93.8	Toluene	17	7	<3	C
2-Ethyl-1,3-dimethylbenzene	93.1	Toluene	24	10	<3	C
6-Ethyl-2-methyldecane	93	Toluene	26	12	6	C
1,2,3,4-Tetramethylbenzene	92	Toluene	6	1	<3	C
1,2,3,4-Tetramethylbenzene	91.2	Toluene	10	4	<3	C
Decamethylcyclopentasiloxane (D_5)	–	STD	2.3	0.4	<3	C
Pentadecane	91.3	Toluene	2	<1	<3	C
Dodecamethylcyclohexasiloxane (D_6)	–	STD	9.1	1.8	<3	C
Propanoic acid, 2-methyl-, 2,2-dimethyl-1 (2-hydroxy-1-)	91.8	Toluene	<1	<1	<3	C
Butanoic acid, 1-methylpropylester	79.5	Toluene	<1	<1	<3	C
2-Methyltetradecane	92.1	Toluene	2	<1	<3	C
3-Isopropoxy-1,1,1,7,7,	67.5	Toluene	5	4	<3	C
7-Hexamethyl-3,5,5-						
Cyclohexane, 1,1-methylene(oxy)bis-	77.5	Toluene	2	3	<3	C
1,1,3,4-Tetrachloro-2,2,3,4, 4-hexafluorobutane	67.6	Toluene	<1	<1	5	A

Continued

TABLE 7.10 Results of Analysis of Condensable Compounds in Cleanroom Air—Cont'd

Compound	Reliability	Calculation Method	Concentration (μg/m³)			Origin
			Cleanroom air		*Ambient air*	
			A	*B*		
Bis(2-ethylhexyl)phthalate	78.4	Toluene	<1	<1	3	C
Trimethylsilanol	–	STD	2	<1	<3	C
2-Propanol	83.1	Toluene	<1	<1	<3	C
2-Methyl-1-propanol	85.8	Toluene	4	<1	<3	C
1-Butanol	85.1	Toluene	4	1	<3	C
Ethanol, 2-methoxy, acetate	80.6	Toluene	3	1	<3	C
Cyclohexanone	85.1	Toluene	<1	<1	<3	C
2-Butanone oxime	–	STD	5	96	<3	C
Phosphoric acid, trimethyl ester	86.5	Toluene	<1	<1	<3	C
Benzenemethanol, alpha, alpha-dimethyl-	86.7	Toluene	<1	<1	<3	
1-(2-Butoxyethoxy)ethanol	78.3	Toluene	<1	<1	<3	C

**Reliability: acquired mass spectrum compared with standard mass spectrum from library data base.*
***Origin: A = ambient air; C = cleanroom air.*

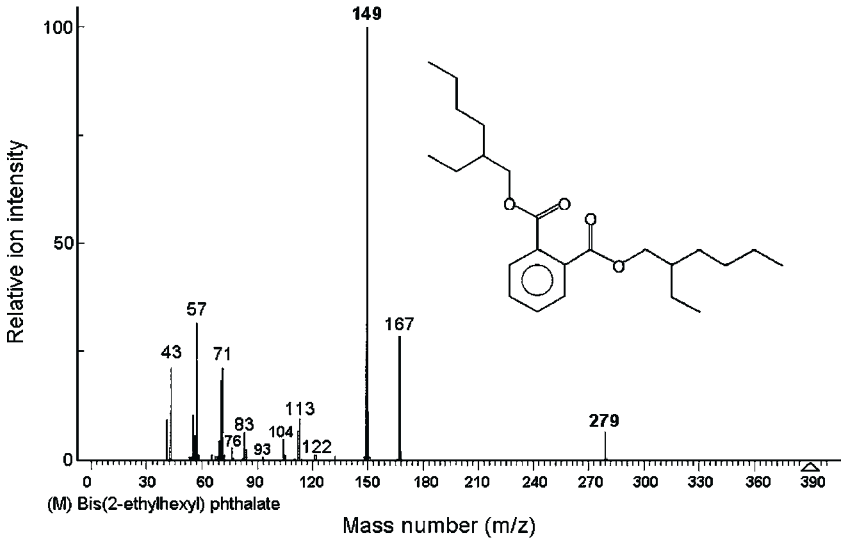

FIGURE 7.15 Mass spectrum of di (2-ethylhexyl) phthalate.

LIVE GRAPH
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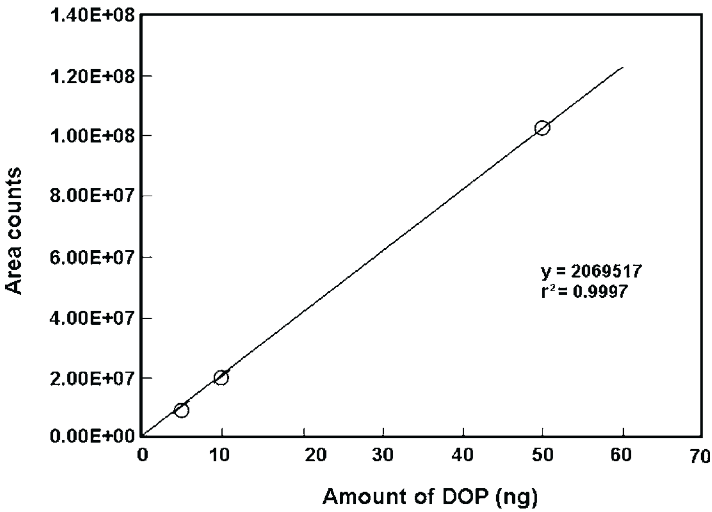


FIGURE 7.16 Calibration curve for di (2-ethylhexyl) phthalate.

filters have reportedly been developed to capture them. However, these AMCs are almost never detected when a Si wafer is cleaned with a solvent and the solvent is analyzed. Instead, DOP accounts for the majority in this analysis. Figure 7.17 shows the results of GC/MS analyzing the same sample to compare the two analysis methods. These two compounds (2-EH and phthalic anhydride) are the pyrolysates of DOP. Therefore, it is necessary to closely evaluate the

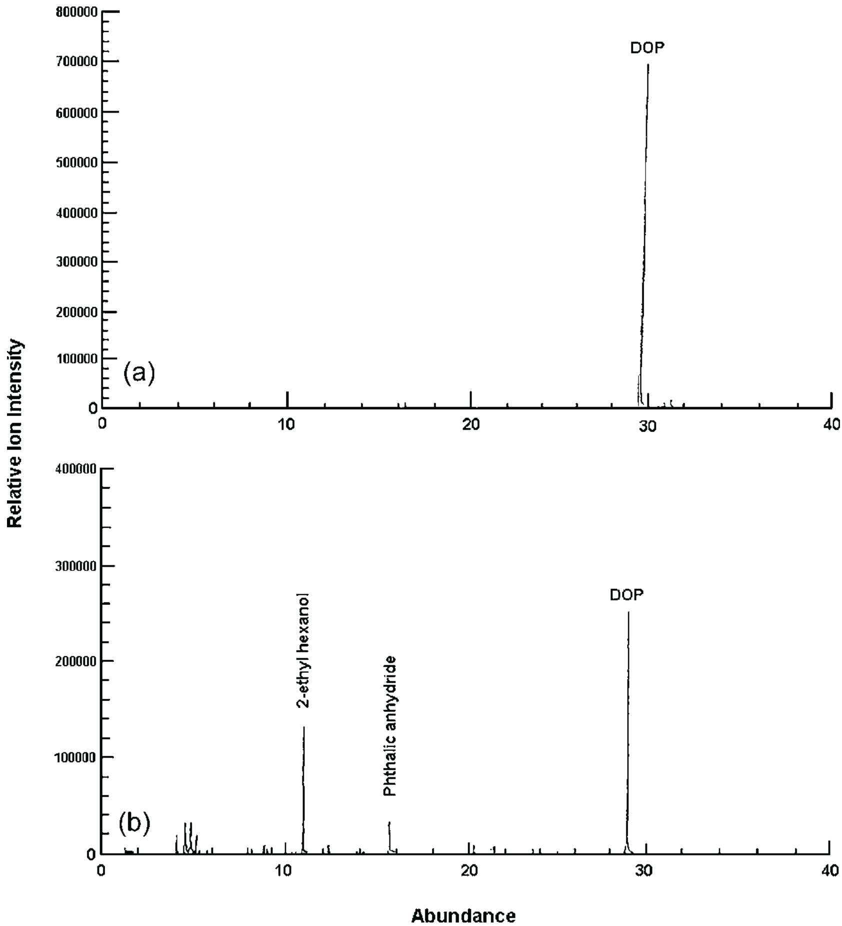


FIGURE 7.17 Total ion chromatogram for (a) solvent extraction GC/MS and (b) thermal desorption GC/MS.

measured data and the conditions of each cleanroom to decide whether or not a chemical filter must be used.[21,75]

7.3.2.3.3 Ester phosphates

Ester phosphates including tributyl phosphates (TBP) are used as fire retardant sand plasticizers. TBP, triethyl phosphate (TEP), tris(2-chloroethyl) phosphate (TCEP), tris(1-chloro-2-propyl) phosphate (TCPP), tricresyl phosphate (TCrP) and similar compounds are often detected.

The above-mentioned organic compounds, namely siloxanes, phthalates, and phosphates, are present in large volume. They are almost always detected in cleanroom air and are often detected on Si wafer surface.[21,75,91–95] These compounds represent the primary significant condensables (organic compounds) that contribute to contamination. They need to be quantified in order to be reported, together with bases such as NH_3, HMDS and BF_3, in the routine analysis of cleanroom air and Si wafer surface. In addition, gases generated from urethane resin and butylated hydroxytoluene (BHT; 2,6-di-tert-butyl-*p*-cresol) are often detected in cleanroom air and on Si wafer surface. They are significant secondary condensables that constantly require close monitoring and control.

For the primary and secondary significant condensables, it is critical to know whether or not their concentrations are within the control limits. For the other condensables, the fact that they are detected often constitutes critical information. By comparing the information with relevant historic data, this information will tell us whether they have been detected before and how they behave. Anomalous peaks detected in the TIM mode of GC/MS are subject to mass spectra analysis (fragmentation) in which comparison with reference materials and search of available databases are performed to identify their molecular structures.

7.3.2.4 Dopants: Boron, Phosphorus, and Metals

Dopants are primarily analyzed by combination of the absorption-to-solution method and ICP-MS (or FL-AAS, flame atomic absorption spectroscopy). The detection limits for boron and phosphorus are 0.1 and 20 ng/m^3, respectively.[90] Similar to boron, the metals are sampled into solution and analyzed with ICP-MS. Both boron and metals might be released into the air not in a form of AMC but in the form of particle.

In order to distinguish gaseous AMCs from particles, two sampling methods, namely the absorption-to-solution method and the collection-with-filter method, are adopted. The absorption-to-solution method collects both AMCs and particles. An outline of the method is shown in Figure 7.18, while the impinger is shown in Figure 7.19. On the other hand, the collection-with-filter method selectively collects particles only. The apparatus is shown schematically in Figure 7.20 and the filter is shown in Figure 7.21. Comparison of the two sets of data provides information on the volume of gaseous contaminants. The correlation map for such a comparison is shown in Figure 7.22. Figure 7.23 shows the algorithms used to analyze the data from the absorption-to-solution method with ICP-MS analysis and to calculate the concentration of contaminants in the air. Section 7.4.1.4 presents specific data and examples of the analyses. Imai et al.[96] report a series of analysis of organic ester phosphates by GC/MS and total phosphorus high resolution ICP-MS (sector/double focus type), which has the capability for organic and inorganic phosphorus analysis after with microwave acid digestion. High recovery of phosphorous compounds has been achieved (Table 7.11). The mass spectra are shown in

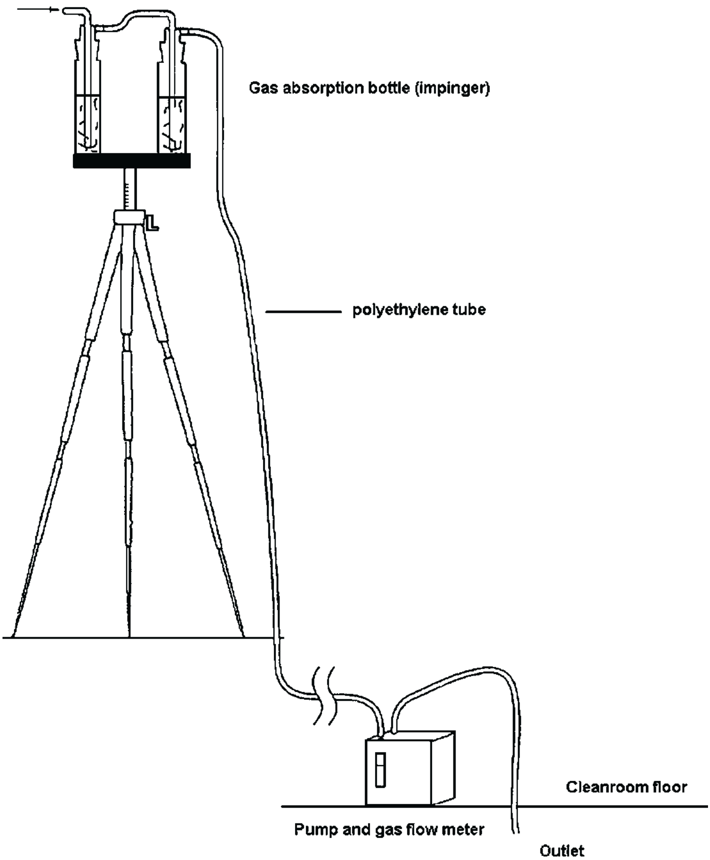


FIGURE 7.18 Absorption-to-solution sampling system used for dopants.

Figure 7.24 for low and high resolution analysis. Like condensables, organophosphates are also analyzed with GC/MS.

Tables 7.12 and 7.13 show the lowest determination limit of the above analysis methods for AMCs in air and on Si wafer surface. API-MS and IMS also provide useful information.

7.3.3 Outgassing Evaluation Method for Construction Materials

The most important consideration in order to achieve a clean manufacturing environment and a clean surface is the technology for selecting proper construction materials.

The outgassing method is used as a screening method for construction materials. It measures the volume of AMC gases generated from a sample.[25,97–108] This method is designed in an attempt to improve sensitivity and throughput. Gas generated from a heated sample is introduced to an adsorption tube in which

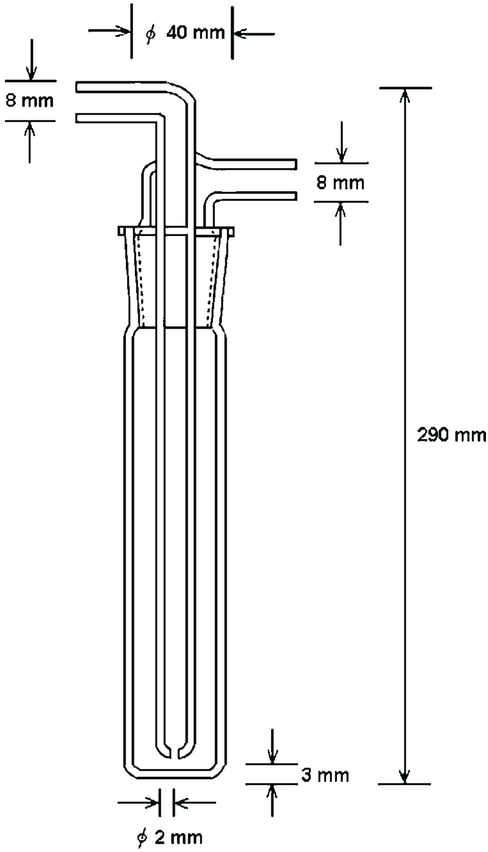


FIGURE 7.19 Schematic diagram of an impinger for the absorption-to-solution sampling system used for dopants.

the gas is cooled down to be collected in a collection tube. The collection tube is then heated to release the gas and the carrier gas transports the gas to GC/MS. Figure 7.25 is a schematic diagram of the method. This method is primarily applied to organic compounds (condensables, dopants including phosphorus compounds, and bases including amino compounds). For analysis of acids and bases, this method is combined with other methods such as ICP-MS, IC, and CE.[53–59]

7.3.3.1 IEST WG 031[25,99]

7.3.3.1.1 Screening Test Method

In the Institute of Environmental Sciences and Technology (IEST) WG031 document it is required to perform rough screening of multiple types of samples. A screening test is described in which small samples are rapidly heated to 150 °C

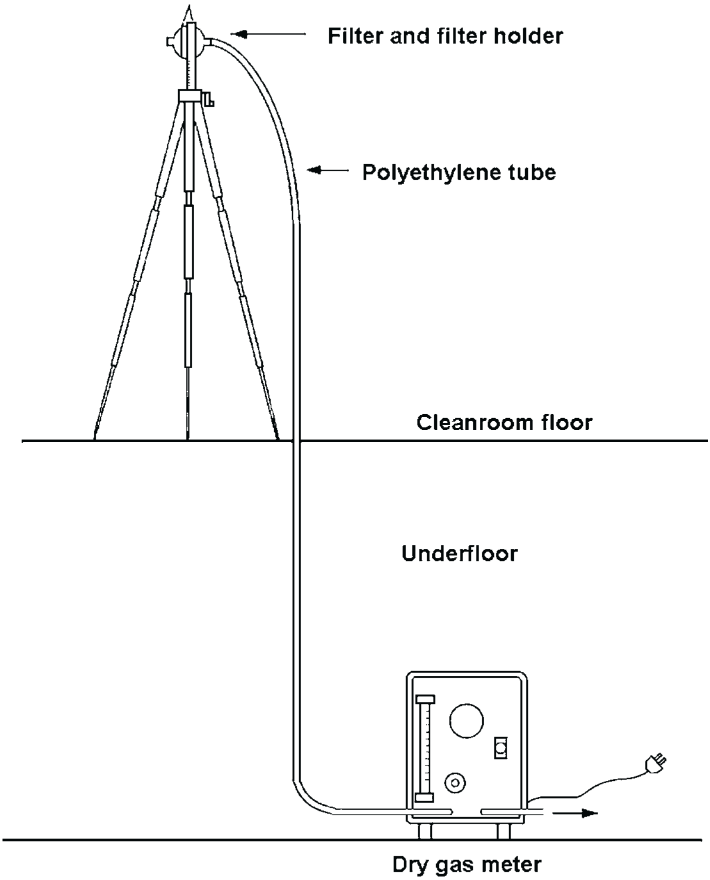


FIGURE 7.20 Typical collection-with-filter sampling system.

to release all the gases from the samples and the gases are measured. From the results of the screening test, the samples that generate large volume of gases are rejected as candidate cleanroom construction materials.

7.3.3.1.2 Engineering Test Method

The samples selected with the screening test method as favorable candidates for cleanroom construction materials are then closely analyzed with an engineering test method. This method looks at larger samples and measures gas generation at room temperature.

It has been proposed to combine the screening test method and the engineering test method, and the proposal is currently being deliberated.[25] The proposed method should be capable of rapidly analyzing a large number of samples.

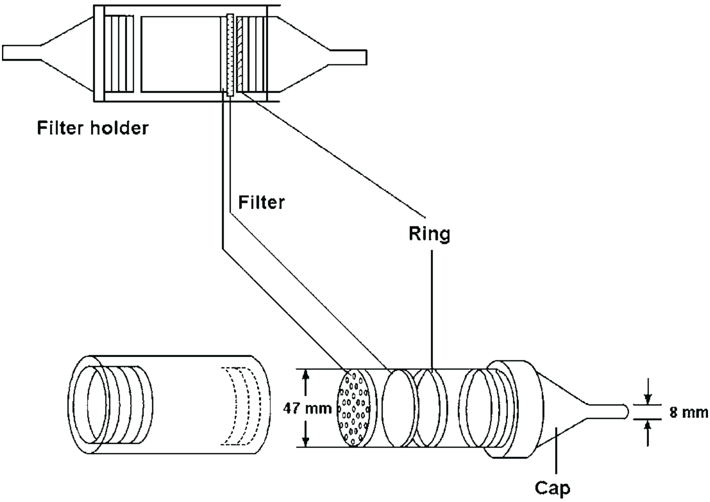


FIGURE 7.21 Filter and filter holder for the collection-with-filter sampling system.

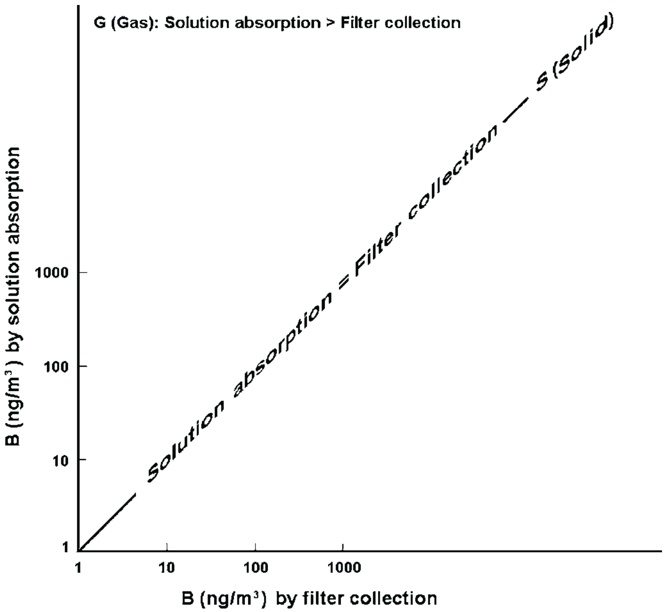


FIGURE 7.22 Schematic model of the analysis results for gas and for solid by the absorption-to-solution and the collection-with-filter sampling systems.

However, since some polymers start decomposing below 150 °C, it is important to note that the state of the interface between the sample surface and the atmosphere may be changed. In addition, at temperatures above 100 °C the coexisting moisture in the air may also vary between moisture, liquid water and steam.

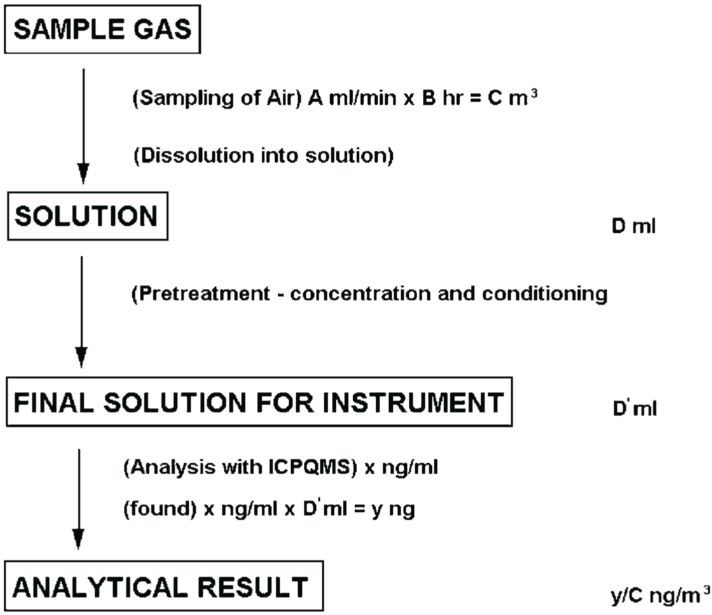


FIGURE 7.23 Outline of the analysis procedure for absorption-to-solution ICP-MS method.

TABLE 7.11 Recovery of Organophosphate Compounds by Microwave Acid Digestion for High Resolution ICP-MS

Compound	Recovery (%)
Triethylphosphate (TEP)	107
Tri-(2-chloroethyl) phosphate (TCEP)	83
Tributylphosphate (TBP)	95
Triphenylphosphate (TPP)	93
Tritolylphosphate (TCP)	71

These conditions are significantly different from those at room temperature. It has also been discussed that engineering test data cannot be directly derived from screening test data, and that the engineering test needs to be standardized.

7.3.3.2 JACA 34-1999[100–103]

JACA 34-1999 specifies five outgassing measurement methods for different applications as described below, including a screening step and

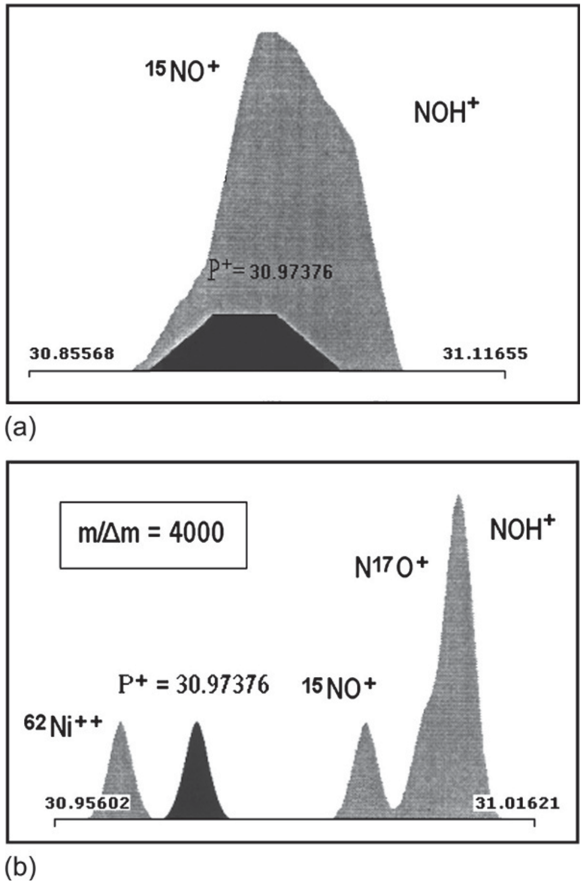


FIGURE 7.24 Mass spectra of P+ at (a) low resolution (upper figure) and (b) high resolution (lower figure).

quantitative estimation of gas volume. The methods are shown schematically in Figures 7.26–7.30.

7.3.3.2.1 Static Headspace Method

A sample is confined for a while in a sealed container. The gas concentration in the air inside the container is to be measured after concentration in the air is in equilibrium with that on the sample surface.

7.3.3.2.2 Dynamic Headspace: Screening Test Method

Three samples, (a) a small disk-shaped sample, (b) a large plate-shaped sample, and (c) a combined sample, are heated under conditions which trigger evaporation at an equilibrium state between gases and the sample surface. Gases

TABLE 7.12 Determination Limit for AMCs on a Si Wafer Surface

Contaminants	Determination Limit for 300 mm Wafer	Method
Non-metallic Compounds	*pg/cm^2*	*WTD-GC/MS*
Siloxanes		
D_3, D_4, D_5	1	
D_6	1	
D_7, D_8	1	
D_{12}	1	
Phthalates		
Diethylphthalate (DEP)	1	
Dibutylphthalate (DBP)	1	
Di(2-ethylhexyl)phthalate (DOP)	1	
Phosphates		
Triethylphosphate (TEP)	1	
Tributylphosphate (TBP)	1	
Triphenylphosphate (TPP)	1	
Tricholoroethylphosphatae (TCEP)	1	
Tricresylphosphate (TCrP)	1	
Butylhydroxytoluene (BHT)	1	
Dibutyl adipate (DBA)	1	
Di(2-ethylhexyl)adipate (DOA)	1	
Metals	atoms/cm^2	HF washing ICP-MS
B	3.00E + 09	
Na	7.00E + 08	
Al	7.00E + 08	
K	6.00E + 08	
Ca	4.00E + 08	
Cr	8.00E + 08	
Mn	3.00E + 08	

Continued

TABLE 7.12 Determination Limit for AMCs on a Si Wafer Surface—Cont'd

Contaminants	Determination Limit for 300 mm Wafer	Method
Non-metallic Compounds	*pg/cm^2*	*WTD-GC/MS*
Fe	5.00E + 08	
Co	3.00E + 08	
Ni	3.00E + 08	
Cu	3.00E + 08	
Zn	1.00E + 08	
Mo	2.00E + 08	
Sn	3.00E + 08	
Pt	1.00E + 08	
Au	1.00E + 08	
Ta	1.00E + 08	
W	1.00E + 08	

generated at several arbitrary temperatures are collected in adsorption tubes to be concentrated. These adsorption tubes are then heated to generate AMCs, and volume of the generated AMCs is measured.

7.3.3.2.3 Dynamic Headspace: Engineering Test Method

The ambient conditions of target cleanroom are reflected in the test conditions when setting the humidity and the temperature of the test environment. Gases generated under the test environment are measured. A formula is applied to the measurement data to estimate the volume of gases generated under different temperature conditions.

7.3.3.2.4 Substrate Surface Absorption: Thermal Desorption Test Method

A sample and a Si wafer are confined in a sealed container for a known time period. AMC gases generated by heating the substrates are then analyzed and the AMC concentration on the Si wafer surface is measured. This method is practical and is often used to measure gases generated from a Si wafer carrier and assembly in a mini-environment. This method is also referred to as a witness plate method for outgassing. In principle, this method follows the

TABLE 7.13 Determination Limit for AMCs in Air

Type	Contaminants	Unit	Determination Limit	Sampling	Analysis Method
Metals	Li, Be, B, Na, Mg, Al, K, Ca, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Mo, Cd, Sb, Ba, Pb	ng/m^3	1	Solution absorption	ICP-MS
(High sensitivity)	P	ng/m^3	2	Cycone sampling	ICP-MS
	B	ng/m^3	0.1		ICP-MS
	Na, Mg, Al, K, Ca, Ti, V, Cr, Mn, Fe, Co, Ni, Cu, Zn, Mo, Cd, Pb	ng/m^3	0.01		ICP-MS
Inorganic ions	NH_4^+	$\mu g/m^3$	0.05	Solution absorption	IC
	F^-	$\mu g/m^3$	0.05		
	Cl^-	$\mu g/m^3$	0.05		
	I^-	$\mu g/m^3$	0.05		
	NO_3^-	$\mu g/m^3$	0.05		Nox
	PO_4^{3-}	$\mu g/m^3$	0.1		IC
	SO_4^{2-}	$\mu g/m^3$	0.05		
(High sensitivity)	NH_4^+	ng/m^3	20	Solution absorption	IC with conc.
	F^-	ng/m^3	5		
	Cl^-	ng/m^3	5		

	NO_3^-	ng/m^3	5		
	PO_4^{3-}	ng/m^3	20		
	SO_4^{2-}	ng/m^3	5		
Organics and condensables	Formic acid	$\mu g/m^3$	0.1	Solution absorption	IC
	Acetic acid	$\mu g/m^3$	0.1		
	Oxalic acid	$\mu g/m^3$	0.5		
	Methylamine	$\mu g/m^3$	0.1		
	Dimethylamine	$\mu g/m^3$	0.1		
	Trimethylamine	$\mu g/m^3$	0.1		
	TMAH	$\mu g/m^3$	0.1		
	Ethanoholamine	$\mu g/m^3$	0.05		
	Cyclohexylamine	$\mu g/m^3$	0.05		
	2-Amino-2-methyl propanol	$\mu g/m^3$	0.1		
	TMSiOH	$\mu g/m^3$	0.1	Column absorption	GC/MS
	M2	$\mu g/m^3$	0.1		
	D_3 (hexamethylcyclo siloxane)	$\mu g/m^3$	0.1		
	D_4	$\mu g/m^3$	0.1		
	D_5	$\mu g/m^3$	0.1		

Continued

TABLE 7.13 Determination Limit for AMCs in Air—Cont'd

Type	Contaminants	Unit	Determination Limit	Sampling	Analysis Method
	D_6	μg/m^3	0.1		
	D_7	μg/m^3	0.1		
	D_8	μg/m^3	0.1		
	DBP	μg/m^3	0.1		
	DOP*	μg/m^3	0.1		
	Total condensables	μg/m^3		TENAX	GC/MS
	Qualitative analysis			TENAX	MS library

The determination limit for high sensitivity analysis is <0.01 μg/m^3.

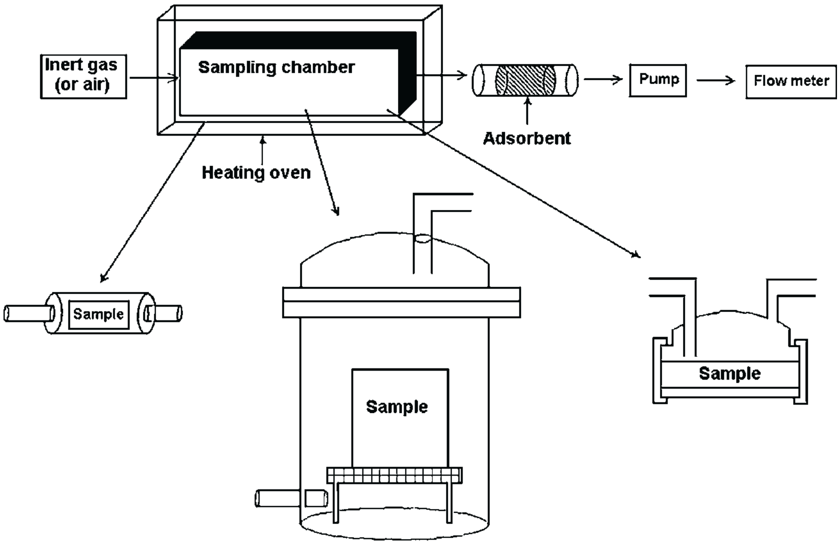


FIGURE 7.25 Schematic of the outgassing apparatus for the dynamic headspace screening test.

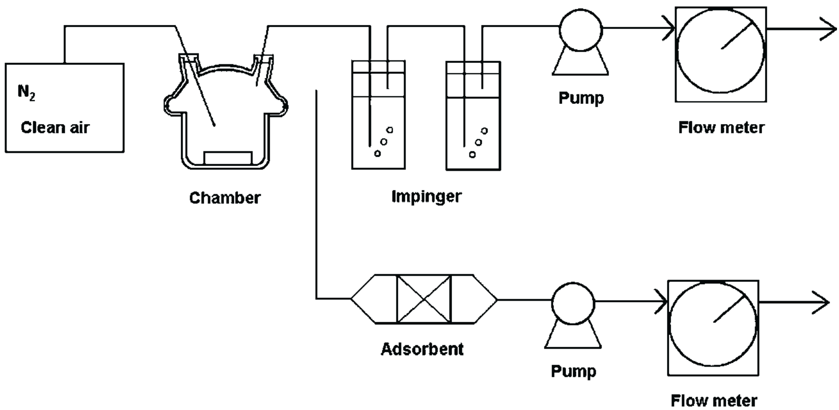


FIGURE 7.26 JACA sampling system for air in a test chamber.

same procedure as the analysis of contamination level on Si wafer. Figure 7.30 shows this method schematically.

Figure 7.31 shows a flowchart of the four outgassing methods described above for creating a low-contamination wafer carrier. To begin with, small samples are analyzed with the dynamic headspace screening test. The samples with low gas generation are selected with the dynamic headspace screening test. After that, large-sized plate-shaped samples and assemblies are selected as favorable candidates for construction materials with the screening test and

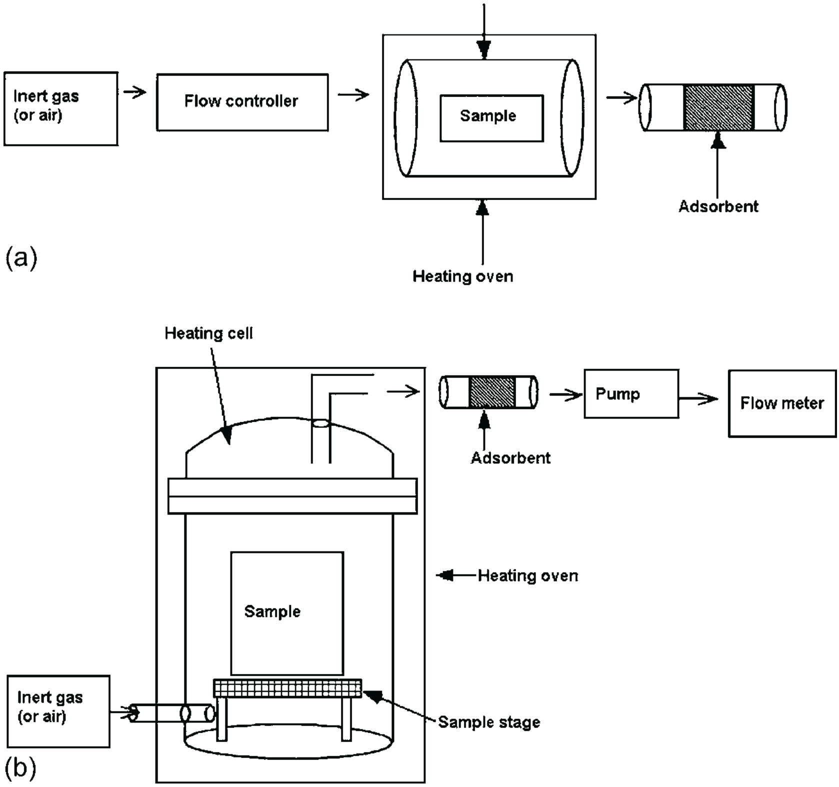


FIGURE 7.27 Examples of JACA outgassing systems using (a) a microchamber (upper figure) and (b) a small chamber (lower figure).

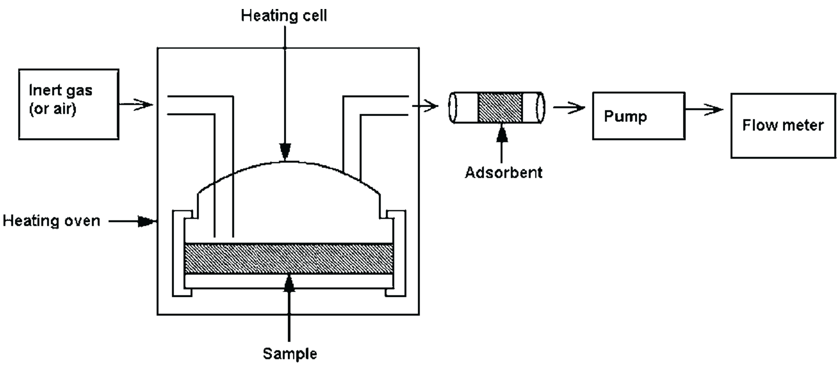


FIGURE 7.28 Example of an outgassing system for a one-sided surface.

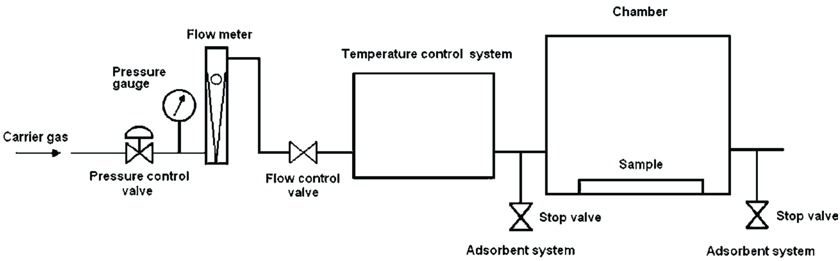


FIGURE 7.29 Schematic of the JACA outgassing engineering test system.

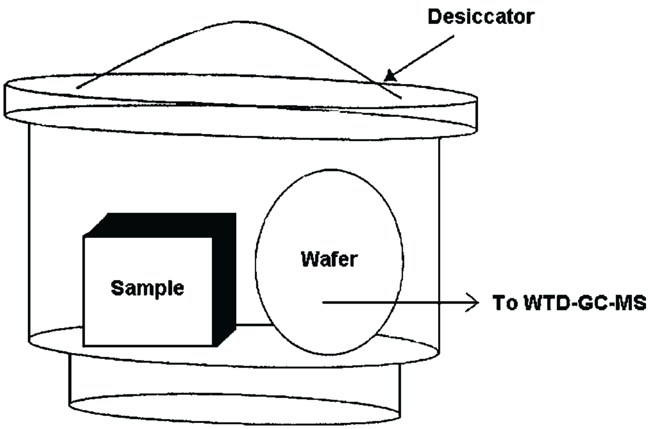


FIGURE 7.30 Outgassing apparatus for the substrate surface adsorption/thermal desertion method (JACA method 4).

the engineering test in order to achieve high quality manufacturing process environment. In order to demonstrate the integrity of a Si wafer storage container and the mini-environment, the Si wafer is confined in them and the inside air is measured with the substrate surface absorption thermal desorption test method.[102–108]

In order to evaluate a mini-environment in a cleanroom located far away from the analytical laboratory, a carrier not subject to contamination from transportation is extremely useful such as the all-aluminum carrier described in Section 7.3.1.1.2.[78]

7.3.3.2.5 Onsite Measuring Method

This method is useful to identify gas generation sources in a constructed cleanroom and to find out which portions of cleanroom should be replaced and to quantify the improvements. A sampling device is designed to make a sampling point a closed chamber. The sampling device is mounted on the wall, floor,

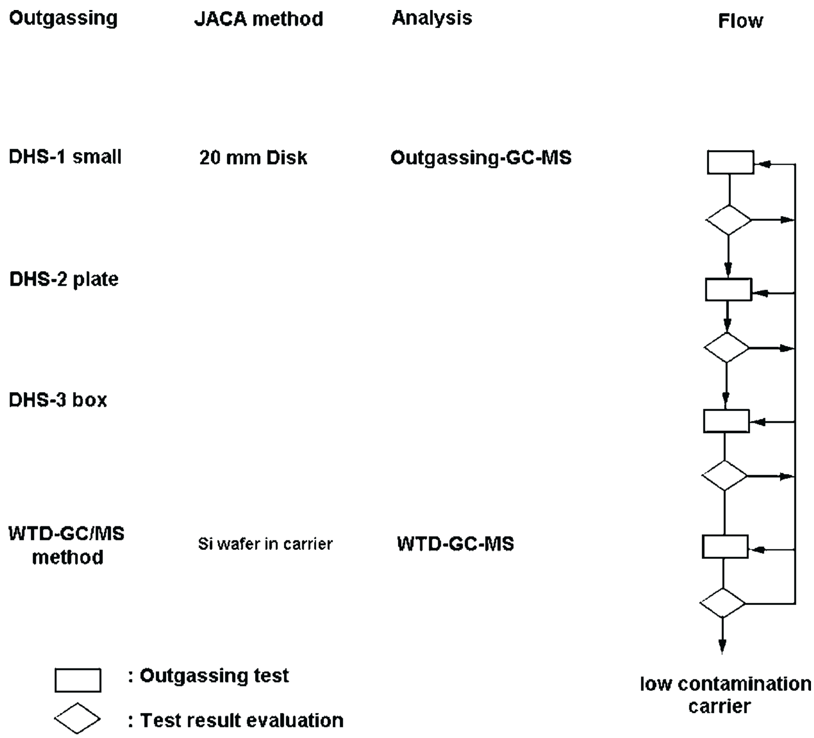


FIGURE 7.31 Flowchart for creation of a low-contamination wafer carrier.

ceiling and/or other areas to collect ambient air. The collected air is introduced into a collection tube, and AMCs in the tube are analyzed. There are two different methods: a double-tube chamber method, and a method to confine and collect ambient air.[109–111] There have been several papers published that evaluate the two methods in terms of cycle time, ease of use, calculation of gas generation per unit time and unit area, estimation of contamination contribution, sampling from ceilings, and the effect of substitution with low outgassing plates. Figures 7.32 and 7.33 show schematic diagrams for these outgassing evaluation methods.

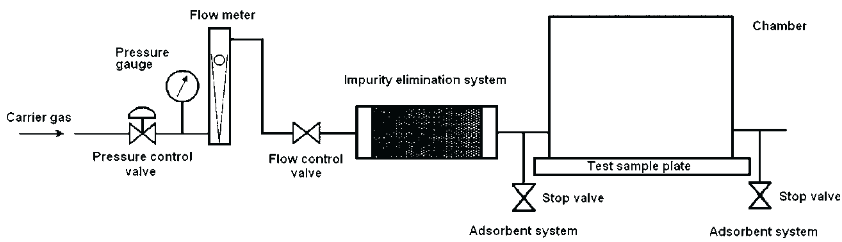


FIGURE 7.32 Example of an on-site system for outgassing evaluation.

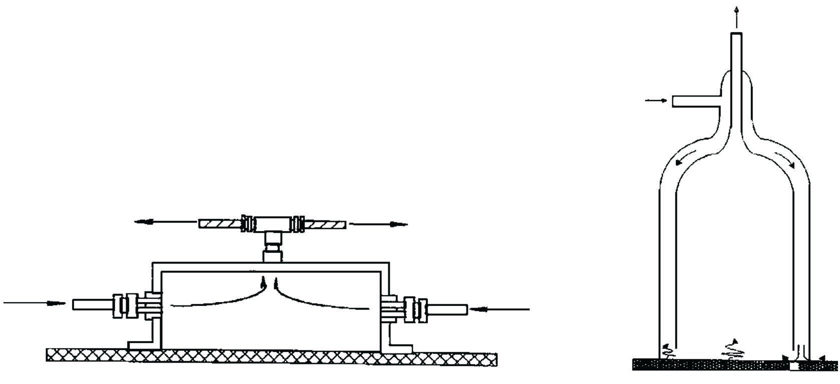


FIGURE 7.33 Example of the JACA chamber for on-site outgassing evaluation.

Section 7.4.3.3 compares the IEST and JACA methods and also discusses the outgassing phenomenon in more detail. For any outgassing evaluation method, it is critical to make measurements at appropriate temperatures between room temperature and 150 °C that do not induce thermal decomposition of the sample, or glass transformation, or hydrothermal reaction on the sample surface. For plastics, in particular, special consideration should be given to chlorinated polymers and silicone polymers.

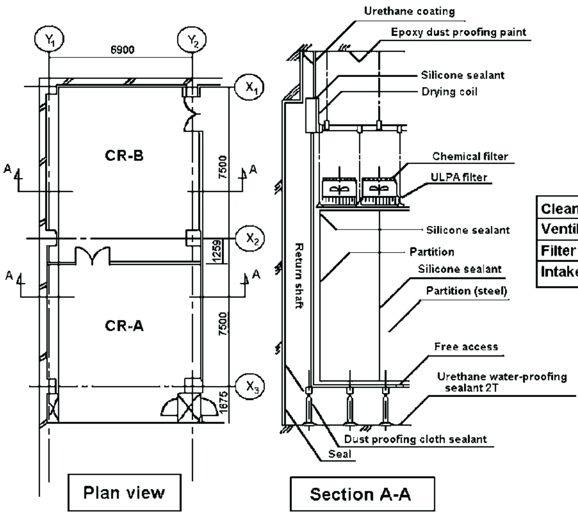
Ion mobility spectroscopy has a high sensitivity for AMCs. Various reports have been presented on outgassing evaluation and Si wafer contamination analysis. It has been proposed in SEMI E46-95 for application to mini-environments.[112–116]

7.4 NATURE OF AIRBORNE MOLECULAR CONTAMINATION AND ITS EFFECTS

In this section, the sources and properties of contaminants and contamination mechanisms on substrate surfaces will be discussed.

7.4.1 Investigation of the Properties of AMCs

Cleanroom air discussed in this section takes in about 5% of fresh outdoor air which is treated with preliminary filtration for particle removal. Relative humidity of the fresh outdoor air is adjusted at 40–50%. The outdoor air is introduced from the ceiling of the cleanroom through filters at an average flow rate of 40 cm/second in a form of laminar flow. The remaining 95% of cleanroom air is circulated air. The ratio is expressed as a function of the with air age. Therefore, the cleanroom air is susceptible to the effects of AMCs generated inside the cleanroom. Figures 7.34 and 7.35 show schematic diagrams of the design, construction, materials used, and the air flow in a typical cleanroom.



Specifications

Cleanliness	ISO Class 10
Ventilation rate	600 times/hr
Filter	Standard ULPA filter
Intake of ambient air	30 m^3/hr

FIGURE 7.34 Example of the plan view and a section A-A of a typical cleanroom.

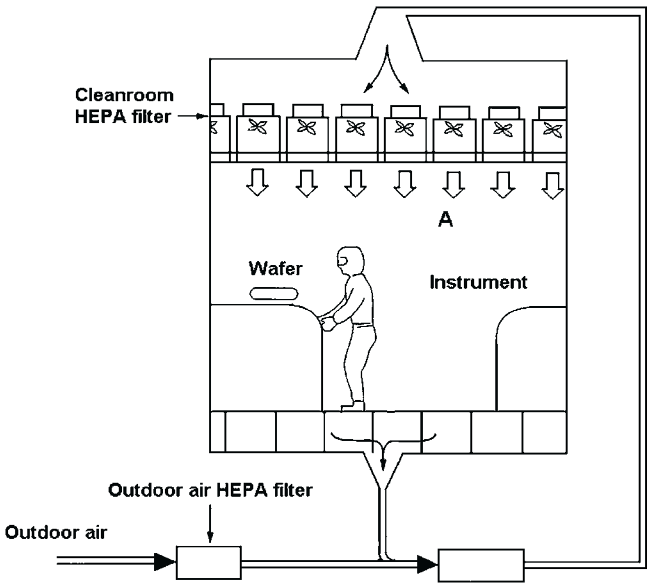


FIGURE 7.35 Flow of air in a typical cleanroom.

7.4.1.1 Effects of Rinsing the Outdoor Air

In those areas where inexpensive water is available in large volume (e.g. some areas in Asia), fresh outdoor air is often rinsed in a washer before being introduced into the cleanroom. With respect to air cleanliness, the rinsing process is often found to be effective in removing AMCs from outdoor air. In particular, this process is effective in removing acid gases and NH_3.[89,12,30] Outdoor air in the vicinity of farmland and rice paddies, which is considered clean, contains NH_3. Outdoor air in the vicinity of factories contains SO_x and NO_x, and the air near roads contains NO_x, oxidants, and hydrocarbons. The air rinsing process is considered effective in removing these contaminants. Although this process is effective in removing SO_x, it is less effective in removing carbonyl sulfide (COS) and other AMCs which are difficult to absorb in water. It is essential to standardize a method to sample and quantify NH_3 present as mist that is appropriate to the operating environment in order to measure how much NH_3 is removed by rinsing the air.

Some acids and bases are introduced into the cleanroom together with outdoor air, while some of them are generated in the cleanroom and mixed with circulated air. Condensables and dopants are mainly generated in the cleanroom and mixed in the circulated air.

The remainder of this section discusses these sources and describe the behavior of various AMCs. Several publications report on AMC contamination on Si wafer surface, while increasing numbers of papers report on AMC contamination on the surface of glass, optical lenses, LCD and HDD, and their prevention and control.

7.4.1.2 Acids

Acid contaminants include HF, HCl, Cl_2, NO_x species (NO is present in minor amounts in the cleanroom), and SO_x (SO_3 can be present in minor amounts in the cleanroom). They are mainly brought in together with outdoor air, while acid gases are sometimes generated in manufacturing process. Acids tend to induce problems such as corrosion, and they can lead to the formation of abnormal particles on substrates and wirings.

7.4.1.3 Bases

Bases include NH_3, amino alcohol, *N*-methyl pyrollidone and tetramethyl ammonium hydroxide (TMAH). Some amines are used as rust inhibitors for industrial water.

7.4.1.3.1 NH_3

As noted earlier, it is essential to standardize a method to sample and quantify NH_3 present as mist that is appropriate to the operating environment in order to measure how much NH_3 is removed by rinsing the air. When NH_3 is precipitated as salt, the surface may form a haze or develop other unacceptable features.[60,131]

7.4.1.3.2 HMDS

Among amines, hexamethyldisilazane (HMDS), in particular, is found to cause contamination problems. HMDS is one of the primary sources of ammonia, and it is by itself an amine and an organic compound.

HMDS is used in the lithography process to treat the surface of a Si wafer (silane coupling agent). HMDS, reacting (silane coupling reaction) with the surface of metals and ceramics, forms hydrophilic trimethylsilanol (TMSiOH) on the surface. This makes it easier to coat a Si wafer surface with photoresist.

$$\begin{aligned}&2\{-\mathrm{Si}-\mathrm{OH}\}+(\mathrm{CH_3})_3-\mathrm{Si}-\mathrm{NH}-\mathrm{Si}-(\mathrm{CH_3})_3\\&\quad=2\{\mathrm{Si}-\mathrm{OH}-\mathrm{Si}(\mathrm{CH_3})_3\}+\mathrm{NH_3}\end{aligned}$$

HMDS also reacts with moisture in the air and gradually converts to NH_3 and methyl silanol.

$$(\mathrm{CH_3})_3-\mathrm{Si}-\mathrm{NH}-\mathrm{Si}-(\mathrm{CH_3})_3+\mathrm{H_2O}=2(\mathrm{CH_3})_3-\mathrm{Si}-\mathrm{OH}+\mathrm{NH_3}$$

Therefore, HMDS must be considered as one of the primary sources of NH_3. When excessive HMDS exists in the cleanroom air, it will induce unwanted reactions with the surface of a Si wafer or a glass substrate to form a trimethylsiloxy bond (this is not adhesion but surface reaction), which results in persistent organic contamination. NH_3 and TMSiOH show a strong correlation in some cases, while they do not show any correlation at all in other cases. Figure 7.36 shows the relationship between NH_3 and TMSiOH.

HMDS generates NH_3 which induces particle generation and the T-top phenomenon on chemically amplified photoresist. Also HMDS forms the $(CH_3)_3$–Si–O-group on Si wafer surface and on glass surface which changes these surfaces to lipophilic and indirectly facilitates formation of a surface susceptible

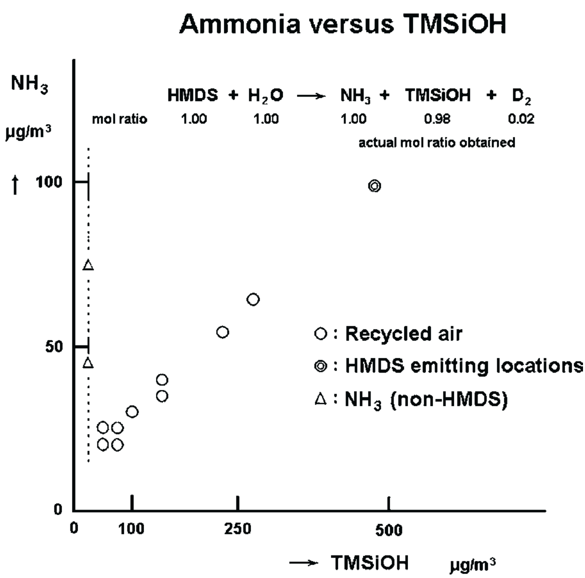


FIGURE 7.36 Scattering map of the relationship between NH_3 gas and TMSiOH based on analysis of cleanroom air contaminated by these compounds.

to adhesion of organic impurities. The organic compounds adhering to the surface tend to decompose, causing particle generation or problems in the film deposition process. Specifically, they may cause abnormal colors, haze and clouding on glass which adversely affects the optical properties and the display function. When a glass surface is irradiated with light such as UV radiation, the $(CH_3)_3$–Si–O-group is decomposed due to photoreaction and precipitates SiO_2. The SiO_2 precipitates may make the glass cloudy, generate particles, or cause other problems on the glass surface.

7.4.1.3.3 Amino Alcohol

Amines or amino alcohol are sometimes spiked in the water for rinsing and humidification as rust inhibitor. Since they are bases, they may also induce particle generation and T-top phenomenon.

Amines used in the lithography process, such as TMAH and *N*-methyl pyrollidone, are detected in the air or on a Si wafer. They may induce T-top phenomenon and degrade the color uniformity of the display. There are many other sources. Concrete used for buildings (especially when a building is newly constructed) constitutes a major source of NH_3 generation.[8]

7.4.1.4 Condensables: Siloxanes, Phthalates, Phosphates, and Other Organic Compounds

Table 7.14 shows the results of analysis for organic compounds in outdoor air. There are a few compounds to which the definition of condensables applies.[48,49,132] Outdoor air contains few condensables, while circulated air contains condensables in large volume. This means many of the condensables are generated in the cleanroom. Toluene, BHT, alkyl benzene, and condensables generated from urethane resin are often detected. Many of them are artificially synthesized. Construction materials of a cleanroom constitute the main sources for generation of condensables.

From the viewpoint of ultraclean technology, however, it is necessary to mention other minor sources as well. For example, human expiration, shoes, and garments of the operators are also sources of condensables.[74] Equipment such as the air conditioning system and process tools also generate condensables. Sealants, lubricants and paints used in the equipment are often found to have detrimental effects.[1]

Figure 7.37 shows GC/MS spectra of AMCs in outdoor air and AMCs in cleanroom air.[11,133–136] Cleanroom air, shown on the bottom chart, is found to contain more AMCs than outdoor air (upper chart) in terms of both type and volume. Figure 7.38 shows TIM mode GC/MS chart of cleanroom air and that from a Si wafer surface exposed to the cleanroom air for 24 hours. It is clearly evident that several condensables among those found in cleanroom air are selectively deposited on Si wafer surface in large amounts.

TABLE 7.14 Results of Analysis of Organic Compounds in Outdoor Air

Compound	Analysis Results μg/m^3	Determination Limit	
		μg/m^3	*ppb vol*
Furon 113	<0.1	0.1	0.01
Acetone	2.3	0.3	0.1
Methyl acetate	<0.07	0.07	0.02
n-Hexane	8.6	0.2	0.05
Methyl ethyl ketone (MEK)	0.72	0.01	0.003
Ethyl acetate	2.0	0.004	0.001
Chloroform	4.0	0.008	0.002
1,1,1-Trichloroethane	0.1	0.02	0.004
Carbon tetrachloride	0.1	0.006	0.001
1,2-Dichloro ethane	1.6	0.009	0.002
Benzene	2.8	0.04	0.01
Trichloroethylene	<0.08	0.08	0.02
1,2-Dichloro propane	<0.09	0.09	0.02
Methyl isobutyl ketone	0.3	0.06	0.01
Toluene	16.0	0.4	0.1
Butyl acetate	0.4	0.004	0.001
Tetrachloroethylene	0.3	0.002	<0.001
Chlorobenzene	0.8	0.02	0.005
m,p-Xylene	0.6	0.06	0.014
o-Xylene	0.5	0.04	0.008

Total hydrocarbons: 1.72 ppm vol.
Unsaturated hydrocarbons: 0.046 ppm C (ethylene: 5–60 ppm).
Non-methane hydrocarbons: 0.49 ppm C.

Table 7.15 compares the contaminants in the atmosphere and on Si wafer surface exposed in the atmosphere. The main contaminants in the column "Shiramizu and Kitajima[44]" are ester phthalates, while ester phosphates are the primary contaminants in the column "Camenzind and Kumar.[21]" Condensables generated from urethane and BHT are scarcely detected on the Si wafer surface. BHT is not only used as an antioxidant for polymers, but it is also spiked in various industrial materials. It is sometimes used in hair cosmetics. Of the various contaminants above, three contaminants are always detected in air and on

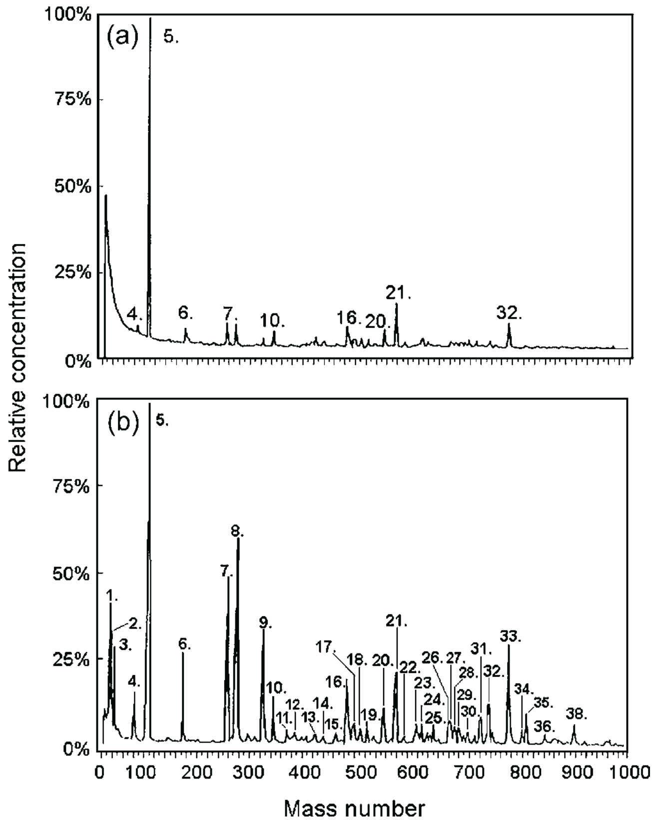


FIGURE 7.37 Comparison of GC/MS charts from (a) outside air and (b) cleanroom air (magnified 4 × in sensitivity).

silicon wafer surface that can cause contamination problems in the manufacturing processes. These contaminants, cyclosiloxanes, ester phthalates, and ester phosphates, are discussed below in greater detail.

The schematic relationship between contaminant concentration on Si wafer surface and the exposure time for various cleanroom contaminants is shown in Figure 7.39. When a Si wafer is exposed in a closed atmosphere, where the volume of air supplied is limited, such as a mini-environment (FOUP, Silicon wafer carrier), the relationship between the amount of AMC contamination on the Si wafer surface and the exposure time is shown in Figure 7.40.

7.4.1.4.1 Siloxanes

Silicone polymers constitute a major source of siloxanes. Silicone polymer is used in the sealant between the filter and the ceiling and as a sealant between wall boards. It is used for various applications to seal hard boards as well as soft tubes.[137]

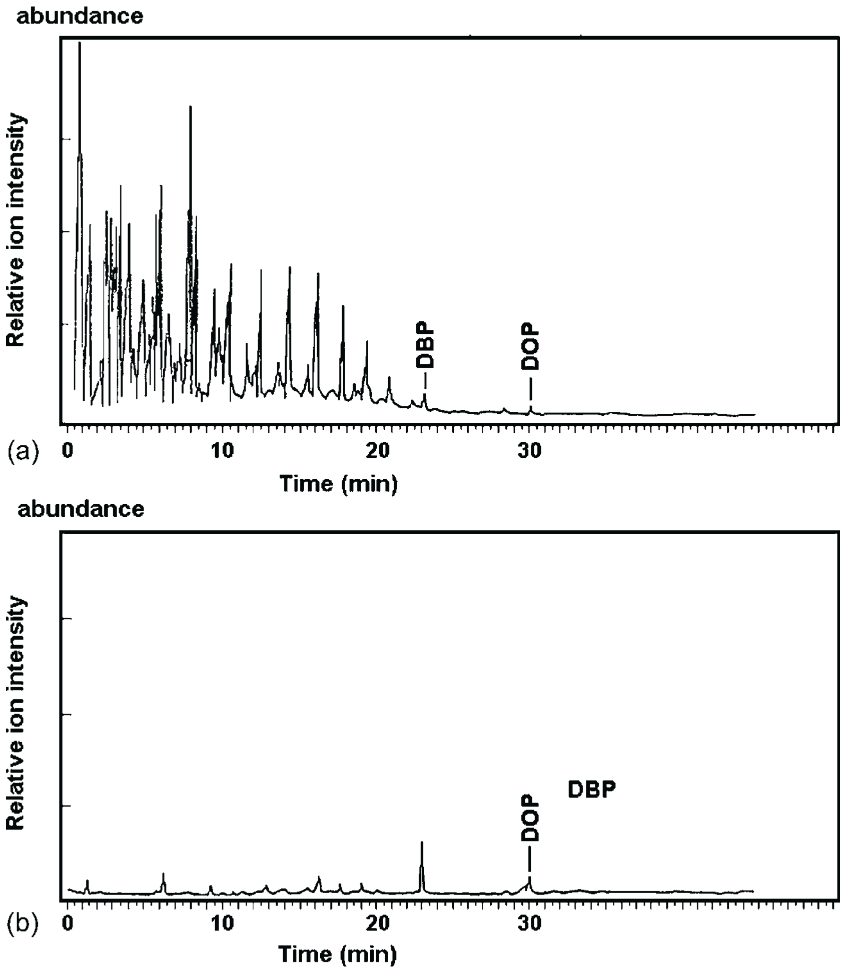


FIGURE 7.38 Total ion chromatograms of organic contaminants from (a) cleanroom air and (b) thermal desorption gas from Si wafer surface exposed for 24 hours to cleanroom air.

During the polymerization process, silicone polymer is easily mixed with a polymerization initiator and the monomer used as raw material. The D_{12} monomer featuring the lowest vapor pressure is more likely to remain in silicone than other monomers during the monomer removal step. The D_{12} (or D_6) monomer released from silicone, therefore, is preferentially detected in cleanroom air immediately after completion of cleanroom construction. The amount of D_{12} detected in cleanroom air gradually decreases as it is governed by amount of D_{12} (or D_6) remaining in silicone.[94] The trend in reduction of D_6, like other general organic AMCs, follows the coupled exponential model equation of Tamura et al.[138] In this silicone polymer, the polymerization initiator that

TABLE 7.15 Relationship Between the Concentration of AMCs in Air, on a Si Wafer Surface, and After Outgassing Test[21,44]

Item	Camenzind and Kumar[21]			Shiramizu and Kitajima[44]
			Required Level: Defect Level × 1/100	
Outgassing test	531 ppmW	0.2 ppmw	(5.3 ppmW)	
Air	900 ng/m^3	(0.34 ng/m^3)	9 ng/m^3	5.7 ng/m^3
Si wafer surface	13 ng/m^3	(0.005 ng/m^3)	(0.13 ng/m^3)	0.2 ng/cm^2
Problems	Yes	No	Not expected	Yes
Sticking probability	[1/200]	[1/200]	[1/200]	[1/100]
Exposure time	24 hours	24 hours	24 hours	24 hours

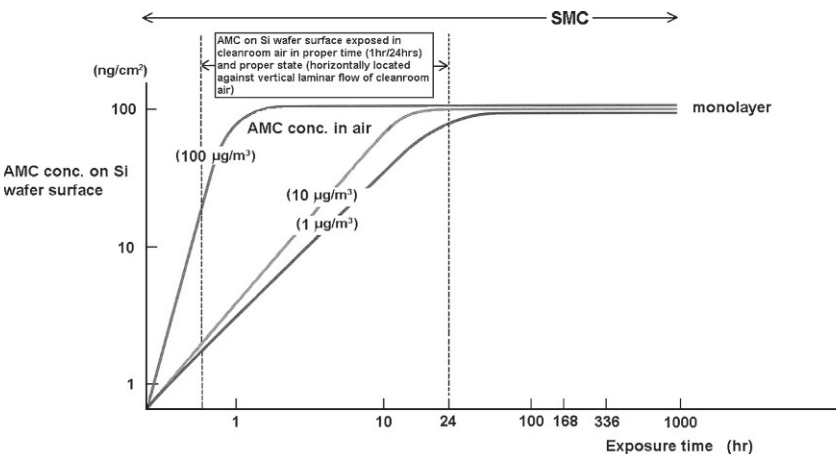


FIGURE 7.39 Schematic diagram of AMCs on a silicon surface exposed to cleanroom air in representative time (1hour/24 hours) and representative state (horizontally set against vertical laminar flow of cleanroom air).

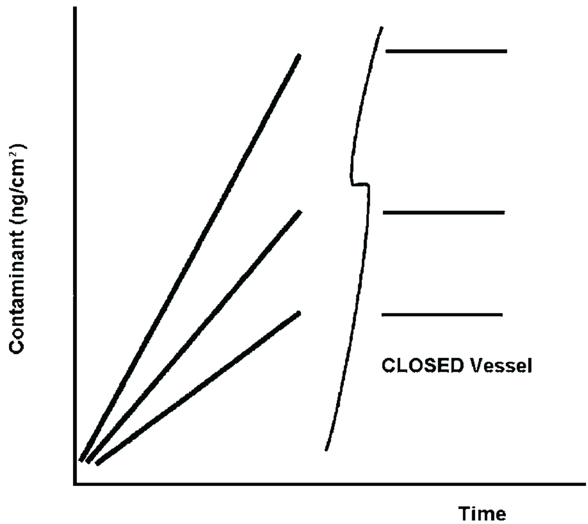


FIGURE 7.40 Typical representation of the concentration of surface contamination as a function of exposure time.

functions to cut the Si–O bond in the cyclosiloxane monomer remains in the polymer. This remaining catalyst starts to become active in severing Si–O bonds some time after completion of cleanroom construction, which triggers the generation of cyclosiloxanes such as D_3 and D_4. In fact, it has been reported that more D_3 and D_4 cyclosiloxanes are detected in cleanroom air when they are measured one year or several years after cleanroom construction than when they are measured immediately after completion of construction.[11,94,139–141] Cyclosiloxanes are adsorbed on the Si wafer surface and the glass surface which makes the surfaces lipophyllic. The lipophyllic surfaces are susceptible to adhesion of organic compounds, which will induce various contamination problems in the manufacturing process.

Organic compounds adhering to the surfaces are decomposed by light or heat, which may lead to particle generation on the surfaces. In order to obtain good information on this type of phenomena, it is critical to combine the results of cleanroom air analysis, Si wafer analysis and surface analysis data.

7.4.1.4.2 Ester phthalates

Ester phthalates are mainly used as plasticizers for polymers. They are also used as additives in the polymerization and formulation processes and as additives to paints and adhesive agents. This is because ester phthalates feature high boiling point and high viscosity, they are chemically stable, and they have an organic–inorganic ratio to affect solvent characteristics. There are few inexpensive substitutes. Ester phthalates are also used as standard for measuring open pore sizes for verification of particle removal filter performance.

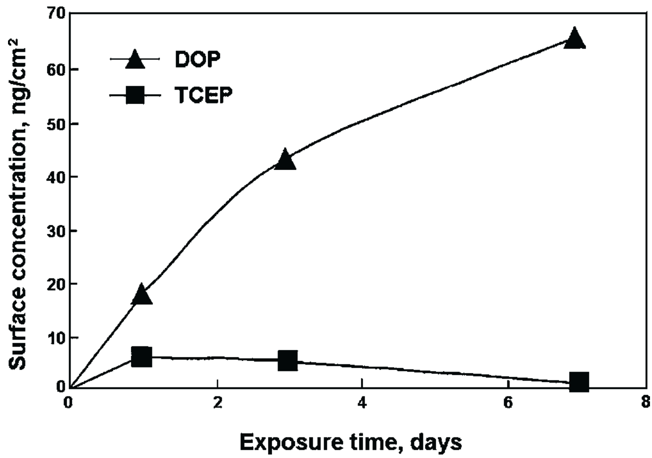


FIGURE 7.41 Change in concentration of DOP and TCEP on Si wafer surface as a function of time on exposure to cleanroom air (1–7 days).

Most commonly, the compounds detected on Si wafer include DBP, DOP, (DEP), dicyclohexyl phthalate (DCHP), di(*n*-octyl) phthalate (DOP), didecyl phthalate (DDP), and dinonyl phthalate (DNP). Among these phthalates, DOP is always detected in cleanroom air and it is reported to induce problems. There are a number of published papers on DOP since it is used frequently.[44,93] Since DOP has a high boiling point, its evaporation heat is high and its sticking tendency is also high. DOP remains persistently in the fruit-basket phenomenon. In other words, DOP adheres to the Si wafer by replacing phosphates and other organic compounds on the wafer surface, as shown in Figure 7.41 for DOP and TCEP.

Since DOP has a high boiling point, its evaporation temperature and the thermal decomposition temperature of its C–C aliphatic bond are close each other. Thermal decomposition takes place prior to evaporation and DOP has a long aliphatic chain. The radicals formed when the DOP C–C chain is thermally decomposed form new aryl rings and generate carbonaceous materials.[1,26] Ester phthalates adhering to a Si wafer are heated in subsequent manufacturing process steps and are partially carbonized to remain on Si wafer surface as the so-called carbonaceous materials. In turn, they react with neighboring atoms such as Si and C, and develop various defects that reduce the electric breakdown voltage of SiO_2.[28] Figure 7.42 shows the model for the thermal decomposition of DOP. By choosing a suitable atmosphere and the temperature conditions, it is possible to remove ester phthalates from Si wafer surface by forming highly evaporable and combustible carbonaceous materials.[142]

As described above, ester phthalates persistently remain on Si wafer surface and are likely to induce contamination problems. The degree of this persistency is revealed through comparison of data for different condensables and is

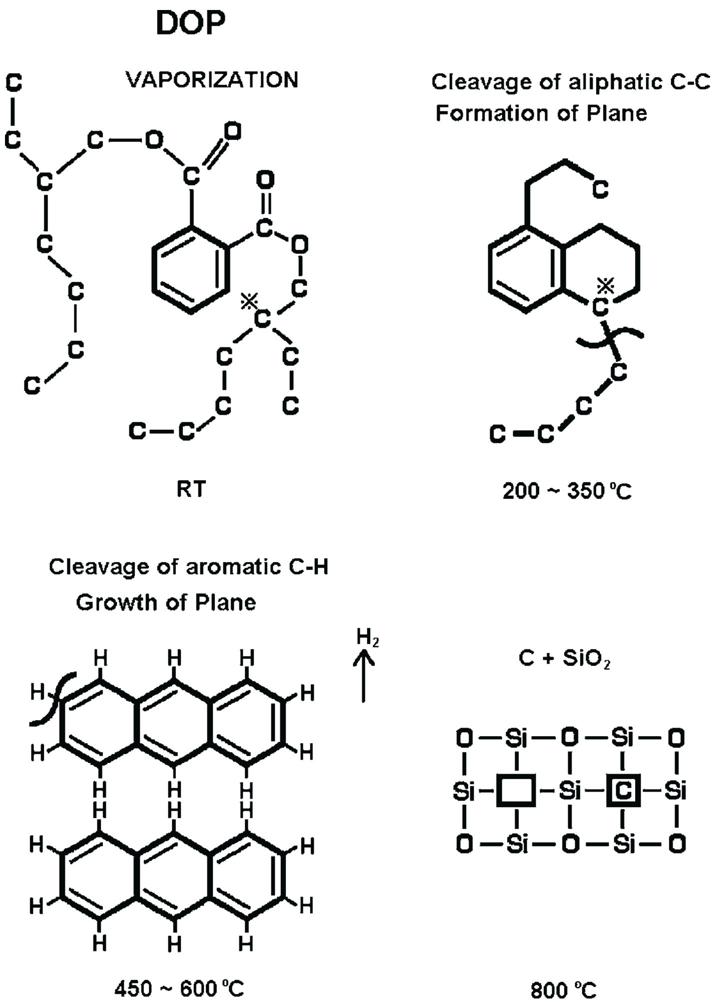


FIGURE 7.42 Schematic representation of the vaporization and thermal decomposition of DOP at (a) room temperature, (b) 200–350 °C, (c) 450–600 °C, and (d) 800 °C.

referred to semi-quantitatively as "staying tendency."[21] This is discussed in more detail in Sections 7.4.3.4.4 and 7.4.3.4.5.

7.4.1.4.3 Ester phosphates

Ester phosphates are mainly generated from cleanroom construction materials. Phosphates are often used as anti fire retardants that are added to HEPA gel such as sealant, foam and other construction materials. They are also used as plasticizers or stabilizers. Phosphates such as TEP, TBP, TCrP, TCEP, and TCPP are often detected in cleanroom air.

Phosphates are next to phthalates with respect to the boiling point, fruit-basket phenomenon, sticking probability, staying tendency, and thermal stability without evaporation. Like phthalates, phosphates cause those troubles that are common for organic compounds with high boiling points. They are more readily detected than other AMCs.

Phosphates feature higher staying tendency than phthalates since they contain other hetero atoms than C, H, and O. As has been reported in some cases, phosphates form acid points to induce formation of defects and precipitates on the surface. It has also been reported that phosphorus atoms exhibit undesirable behavior as a dopant.

For TCPP, Camenzind and Kumar[21] and Shiramizu and Kitajima[44] reported the relationship between the threshold level that causes problems and data on outgassing amount, concentration in the air, and concentration on Si wafer surface. By considering the data in a pair-wise manner together with the calculated sticking probability of TCPP, the required cleanliness level for problem-free operations has been defined (Table 7.15).

The "C-O-P" ester bond of ester phosphate undergoes thermal decomposition when heated to temperatures of 200 °C or higher. Under these conditions, phosphoric acid will remain on the surface of the substrate, although the ester structure is no longer preserved. This thermal action, which is difficult to eliminate, is shown in Figure 7.43. This phenomenon can be avoided by choosing a suitable atmosphere and heating temperature.

7.4.1.4.4 Dopants: boron and phosphorus compounds

What is unique about dopant contamination is that the effect on device physics is not proportional to the concentration of the contaminant. Usually boron and phosphorus are doped in Si wafers. Dopant contamination on Si wafer will enhance or neutralize the previously present dopant in the Si wafer. This is why the effect of dopant contamination is not linearly proportional to the dopant concentration in the air, or on the time of exposure to dopant contamination.

7.4.1.4.4.1 Sources of boron generation This section discusses what the analysis data indicate about sources of boron released to cleanroom air in the semiconductor manufacturing process. Boron is generated in the form of BF_3 gas, together with SiF_4 and PF_6 from the borophosphosilicate glass (BPSG)[a] passivation film during the HF etching process.[143,144] In some cases, boron is derived from source gases for the passivation film including B_2H_6, R_3B, PH_3 and R_3P ("R" represents an alkyl group or an alkoxy group in general, including CH_3–, C_2H_5–, and C_2H_5O).

A major contaminant that is most commonly generated is BF_3. Published reports suggest that BF_3 is generated when hydrogen fluoride gas used in semiconductor manufacturing process in a cleanroom is mixed into recycled air and it comes into contact with the HEPA filter material consisting of borosilicate

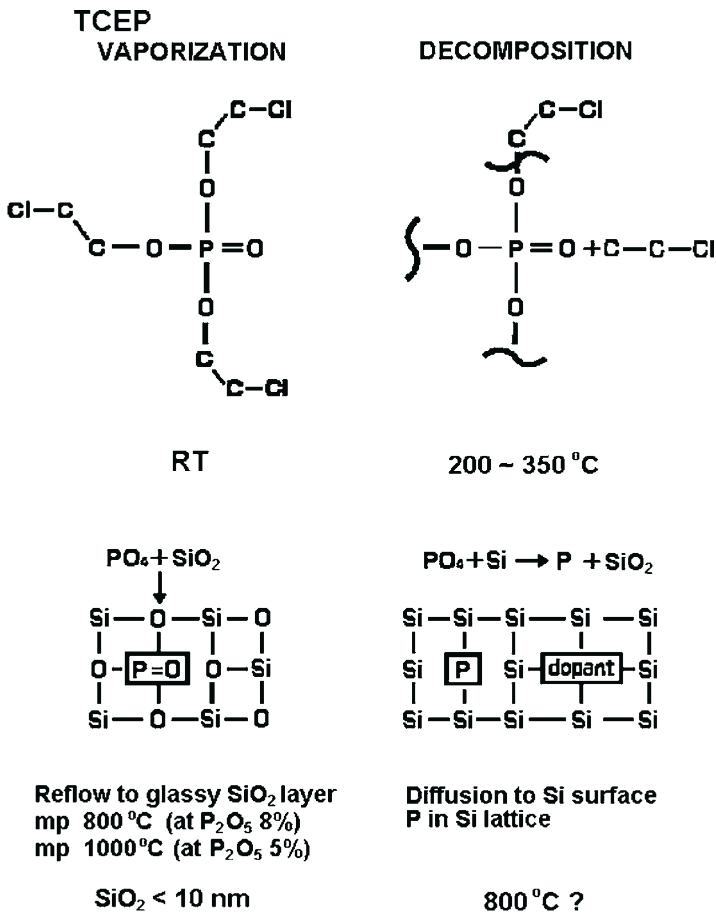


FIGURE 7.43 Schematic representation of the vaporization and decomposition phenomenon for ester phosphate at (a) room temperature, (b) 200–350 °C, (c) SiO_2 < 10 nm, and (d) 800 °C.

glass fiber.[145,146] Table 7.16 shows the results of analysis of boron and other metals in an absorbing solution in which cleanroom air containing AMCs was absorbed. Analysis of different metals is possible by using an acid absorbing solution spiked with nitric acid.

Table 7.17 shows the results of analysis of boron sampled from the air at different locations by means of the absorption-to-solution method and the collection-with-filter method. The results of both methods are similar and indicate the presence of boron compounds as particles, which are often observed in outdoor air. The boron concentration measured with the absorption-to-solution method is higher than that measured with the collection-with-filter method because some boron exists in a gaseous form. This is the case for cleanroom air.

TABLE 7.16 Comparison of Analysis Results Obtained with Various Absorption Solutions

Absorption Solution	Run	Metal (ng/m^3)				
		B	*Na*	*Al*	*Zn*	*Fe*
Pure water	1	37	140	4	33	3
	2	39	150	6	31	5
3% HCl	1	45	150	16	34	22
	2	43	160	18	30	24
3% HNO_3	1	43	150	20	33	23
	2	45	160	18	31	23

TABLE 7.17 Boron Concentration in Air at Various Locations

Sample	Site*	Boron (ng/m^3)			
		Solution Method	*Filter Method*	*Difference***	*Conc. Ratio: Gaseous Boron to Total Boron*
Outdoor air	A	3	3	0	0.00
	B	6	5	1	0.15
	C	9	9	0	0.00
	D	11	10	1	0.10
	E	22	20	2	0.10
	F	25	23	5	0.10
	G	32	31	1	0.30
	H	35	30	5	0.10
	I	130	125	4	0.04
	J	155	160	–5	0.00
Cleanroom	K	30	3	27	0.90
	L	180	8	172	0.96
	M	290	40	250	0.86

**A: Mountain valley; B: Field; C: Pasture; D: Town-1; E: Town-2; F: Town-3; G: Traffic; H: Plant; I: Seaside; J: Seaside; K: After HEPA filter-1; L: After HEPA filter-2; M: After HEPA filter-3.*
***Difference in boron concentration between the solution absorption method and the filter collection method.*

TABLE 7.18 Boron Concentration in Air Before and After HEPA Filtration and Chemical Filters

Air	Sampling Point	Boron (ng/m^3)	
		Room A	*Room B*
Outdoor air	Before HEPA filter	136	32
	After HEPA filter	5	1
Inlet air to cleanroom (95% recycled air + 5% outside air)	Before HEPA filter	5	5
	After HEPA filter	30	11
Inlet air to cleanroom (95% recycled air + 5% outside air)	Before chemical filter	30	11
	After chemical filter	<1	<1

Table 7.18 shows boron concentration measured upstream and downstream of a HEPA filter and a chemical filter. Boron in gaseous form increases downstream of the HEPA filter and decreases downstream of the chemical filter. Particulate boron in outdoor air decreases downstream of the HEPA filter. Boron in gaseous form in cleanroom air decreases downstream of the HEPA filter. Boron in gaseous form in cleanroom air, however, does not show an increase downstream when the HEPA filter is made from polymers which contain no boron. Figure 7.44 compares data from the absorption-to-solution method and those from the collection-with-filter method.

BF_2^+ (m/e = 48 and 49) in cleanroom air can be detected with API-MS.[146] Table 7.19 shows that the boron concentration in the air increases when the air contains alcohol[147] or moisture.[148] The existence of boron is not in the form of B_2F_6 due to corrosion of the glass with HF, rather it is present as $B(OH)_3$. The mechanism is the reaction of moisture or alcohol with B_2O_3 crystallized from BPSG as described below.

According to the solid–liquid phase diagram of the SiO_2–B_2O_3–P_2O_5 system (Figure 7.45) presented by Kern et al.[143,144] and Hummel et al.,[149,150] the boron, phosphorus, and silica glass is unstable in the region of very low-phosphorous composition where B_2O_3 is highly likely to be precipitated. This so-called "Vidrid" phenomenon is induced depending on the glass fiber manufacturing conditions.[1] It has been speculated that moisture or alcohol triggers evaporation of B_2O_3 precipitated on the surface into the air. The form of the contaminant generated in the air is different, depending on whether the HEPA filter fiber is uniformly glassy, or B_2O_3 precipitation takes place locally. Countermeasures for these effects need to be correspondingly

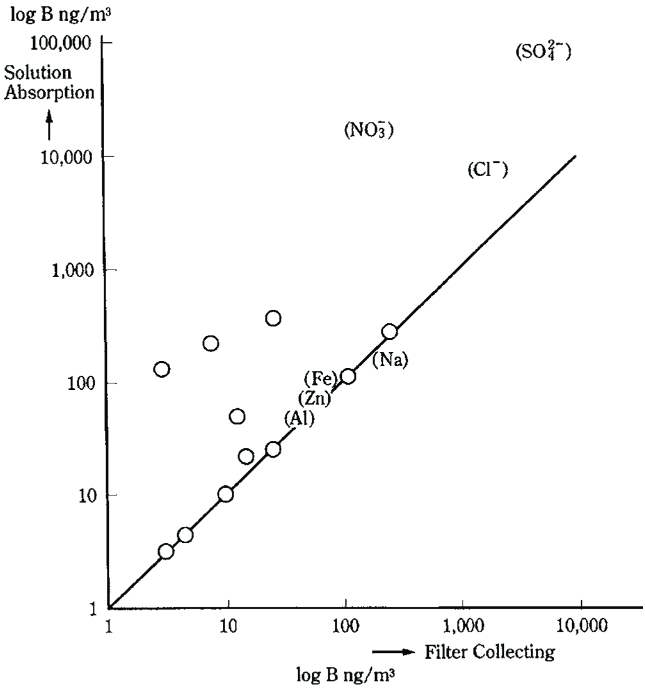


FIGURE 7.44 Scattering map of the analysis results for boron concentration from the solution absorption method compared with the filter collection method.

different. Among the boron compounds, BH_3, organic boron compounds, BF_3, BCl_3 and $B(OH)_3$ are expected to be detected. These compounds have higher vapor pressure in air than the metal, oxides and salts and they are highly likely to be present in the air.

As noted earlier, boron in cleanroom air adheres to the Si wafer surface and affects the electrical characteristics such as the threshold voltage of devices.[1] How the boron contamination described above is investigated and analyzed will help in the study of other metal AMCs in the future.

7.4.1.4.4.2 ***Sources of phosphorus generation*** Chemicals used in the patterning process and BPSG glass used for film deposition contain phosphorus compounds in large amounts. Therefore, in these process steps, PH_3 and $(RO)_3PO$ are sometimes detected as contaminants. They are often generated from phosphates which were described in Section 7.4.1.4.3.

Phosphorus compounds are often used as flame retardants in construction materials and as plasticizers. Phosphorous is particularly used in urethane paint, HEPA gel and foam.[93] Recently, another type of fire retardant sealant is being used that does not use ester phosphate.[151,152]

TABLE 7.19 Boron Concentration in Air Containing Moisture and Alcohol at Various Locations

Air	Sampling Point	Boron (ng/m^3)		Remarks
		Room A	*Room B*	
Outdoor air	Glass HEPA filter*	32	1	
	Glass HEPA filter*	35	5	
	Glass HEPA filter*	136	5	
	Glass HEPA filter*	155	1	
Inlet air to cleanroom	Glass HEPA filter*	5	30	HF: 300 ng/m^3
	Glass HEPA filter*	5	11	HF: 300 ng/m^3
	Glass HEPA filter*	10	91	Alcohol: >500 ng/m^3
	Glass HEPA filter*	3	15	Alcohol: >500 ng/m^3
	Polypropylene HEPA filter	3	3	(no boron source)
Inlet air to cleanroom	Chemical filter**	5	1	
	Chemical filter**	11	<1	
	Chemical filter**	32	<1	

*B_2O_3: 8%.
**Active charcoal filter.

7.4.1.4.5 Other contaminants

Gases of organic metals, metal hydrides, metal halides and metal hydroxides have higher vapor pressure than other compounds. They are sometimes detected as AMCs in air and they have a tendency to adhere to and react with the surface to develop defects or to generate particles.

7.4.2 Examples of Problems Caused by AMC Contamination

This section presents some examples of problems that are often attributed to AMC contamination.[1,44]

- Si wafer: particles, failure in lithography, TDDB failure, surface roughness, and poor thickness reproducibility.

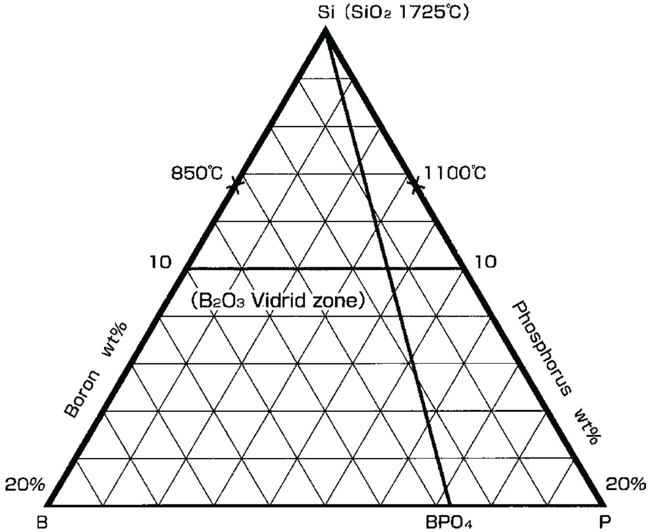


FIGURE 7.45 Diagram of the SiO_2–B_2O_3–P_2O_5 ternary system.

- LCD: surface roughness, abnormal color, blurring of color, and discoloration.[22,23,120,137,153,154]
- HDD: head crash.[24,25]

The remainder of Section 7.4.2 will discuss what problems occur frequently, what contaminants cause serious problems, and what effects can lead to misinterpretation of analysis data.

7.4.2.1 General

The adhesion of AMCs to a substrate surface generally results in some changes to the manufacturing process. It is necessary to consider AMC reactions such as evaporation and partial carbonization of AMCs, as well as the primary process reactions of the manufacturing process. Usually the reactions of AMCs are negligible in comparison with the primary process reaction. In some cases, however, these three reactions become competitive with each other, or they may be consecutive reactions in which all AMCs are evaporated before film deposition starts. Both the competitive reactions and the consecutive reactions will increase intra-and inter-lot variation.

7.4.2.2 Effects of Ammonia

Figures 7.6 and 7.7 showed XPS and TOF-SIMS spectra from a Si wafer surface on which particles of ammonium salts were detected.[77,85,86] Although siloxanes, phosphates, sulfites, and sulfates were detected by TOF-SIMS which can acquire information from the top surface, NH_4^+, NO_3, C–NH, and other

N_{1S} spectra were detected by XPS which can acquire information from a surface layer thickness of greater than 10 Å. Based on this data, amines and NH_3 adhere first to the surface, followed by organic compounds and acidic gases to form foreign substance particles.

NH_3 not only constitutes a primary cause of contamination, but it also may induce adhesion of other AMCs and generation of particles to indirectly cause further problems.

NH_3 not only generates particles of inorganic salts and organic salts, but it also reacts with the proton in optically amplified photoresist. This reaction consumes H^+ to make the top surface less photosensitive, causing the problem associated with the T-top phenomenon. This phenomenon, however, can be caused by other nitrogen compounds such as amines and amino alcohols. In some cases, problems may occur though the NH_3 monitor detects no excursion. Or in other cases, no problems are found even though the NH_3 monitor detects an excursion. It is critical to fully understand the measurement mechanism of the NH_3 monitor and mechanism of troubles to take place.

7.4.2.3 Effects of Siloxanes

The surface of optical glass and lenses often forms a haze when siloxanes adhering to the surface are converted to SiO_2 due to photoreaction with UV radiation which then crystallizes on the surface. As noted previously, the primary sources of siloxanes are cyclosiloxanes derived from silicone polymer (oligomer) in construction materials. HMDS used in the lithography process is also a siloxane.

7.4.2.4 Effects of HMDS

The characteristics of HMDS have been discussed in Section 7.4.1.3. In the air, HMDS is decomposed with moisture into NH_3 and trimethylsilanol (TMSiOH). The NH_3 monitor will indicate high values when the air is contaminated by HMDS. TMSiOH makes glass cloudy. Analysis of TMSiOH can reveal the relationship between air cleanliness and glass clouding.[1] What makes data interpretation complex is the fact that another contaminant cyclosiloxane that also makes glass cloudy. Since cyclosiloxanes are generated from construction materials, their concentrations hardly show a rapid change in a cleanroom. In some cases, however, cyclosiloxanes concentrations in a cleanroom are significantly different from those in another cleanroom. In order to make effective use of alarm information and take appropriate actions, it is critical to understand the functions of the contaminant monitor and the chemistry of clouding.

7.4.3 Chemistry of AMCs

Theoretical and experimental investigations on the chemistry of AMCs are discussed in this section.

7.4.3.1 Monolayer and Sub-Monolayer Contamination

By measuring the concentration of contaminants adhering to a Si wafer surface, several authors were able to obtain actual data to draw a monolayer adsorption curve (Figure 7.39). Figure 7.46 shows the variation of contaminant concentration on Si wafer, glass, and an aluminum plate as a function of time. This data confirms that a so-called "fruit-basket phenomenon" takes place as the amount of adsorbed contaminants gets close to the theoretical amount of monolayer coverage and the remaining "sites" decrease in number. The fruit-basket phenomenon occurs when organic compounds with lower boiling points rapidly adsorb on the wafer surfaces at first, but they tend to be gradually replaced by organic compounds with much higher boiling points. This phenomenon has a great impact on the reliability of a method to evaluate cleanroom construction materials. A construction material and a Si wafer are confined in a container together for a given period of time, after which the Si wafer is heated to evaluate the gases evaporated from its surface. This phenomenon was found to be useful for developing a new chemical filter to trap a specific contaminant.[1]

There are two major models to calculate the concentration of monolayer contaminants.

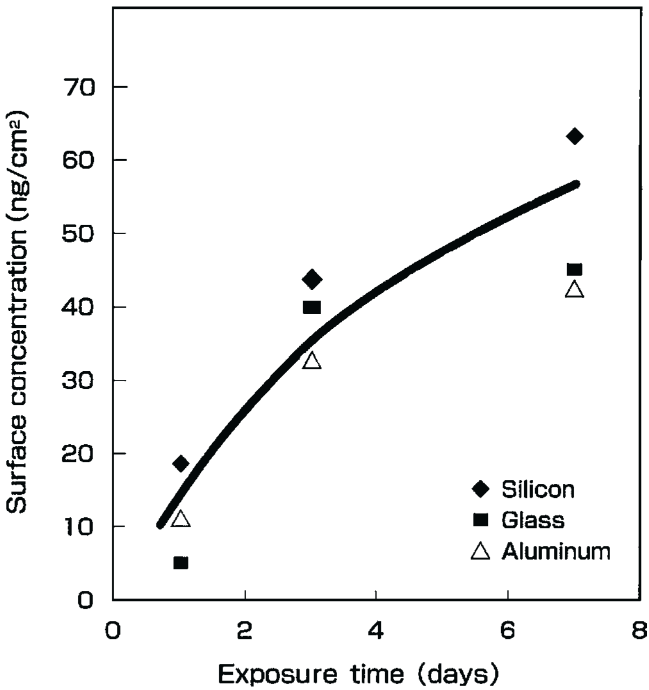


FIGURE 7.46 Relationship between exposure time and DOP concentration on a silicon wafer, glass substrate, and an aluminum surface.

LIVE GRAPH
Click here to view

1. *Calculation of silicon substrate from density, molecular (atomic) weight and Avogadro's number*. The surface concentration of Si atoms on Si wafer is calculated by using values of density, atomic weight and Avogadro's number. The calculated concentration is then replaced with the surface concentration of contaminants (Figures 7.47 and 7.48). When a metal is a contaminant, no significant difference is found even if the concentration is calculated with this method because of the Dulong–Petit law between atomic weight and density. Only the alkali metals deviate slightly from this model. When this model is applied to a monolayer of organic contaminants, DOP is 27 carbon atoms/cm^2 when Si atoms and carbon atoms on Si wafer are assumed to be in a 1:1 ratio.
2. *Langmui–Blodgett thin layer model for organic compounds.* The model described above is inappropriate for many of the organic contaminants since their specific gravity is around 1. The second model that is used to calculate monolayer concentration of organic compounds is developed based on the Langmuir–Blodgett thin layer model.[155] Figure 7.49 shows schematically the calculation method Si and organic contaminants are slightly different with respect to the number and distribution of the atoms per unit surface area. When calculating the concentration by using specific gravity, molecular weight (to be converted to carbon-atom-equivalent) and Avogadro's number and ignoring the distribution of Si atoms on the surface, the concentration of

1. Molecules/cm^3

$$\text{UNITS: } \frac{\text{molecules}}{\text{mole}} \times \frac{\text{g/cm}^3}{\text{g/mole}} = \text{molecules/cm}^3$$

$$A \times \frac{d}{mw} = \text{molecules/cm}^3$$

2. Molecules/cm^2

$$\left(A \times \frac{d}{mw}\right)^{\frac{2}{3}} = A^{\frac{2}{3}} \times \left(\frac{d}{mw}\right)^{\frac{2}{3}} \text{molecules/cm}^2$$

3. g/cm^2

$$\left(A \times \frac{d}{mw}\right)^{\frac{2}{3}} \times (\text{weight of a molecule})$$

$$= \left(A \times \frac{d}{mw}\right)^{\frac{2}{3}} \times \frac{M}{A} = d^{\frac{2}{3}} \left(\frac{M}{A}\right)^{\frac{1}{3}} \text{g/cm}^2$$

where:

d = density (g/cm^3)
M = molecular weight
A = Avogadro's number, 6.023E23

FIGURE 7.47 Calculation of monolayer surface coverage in different units.

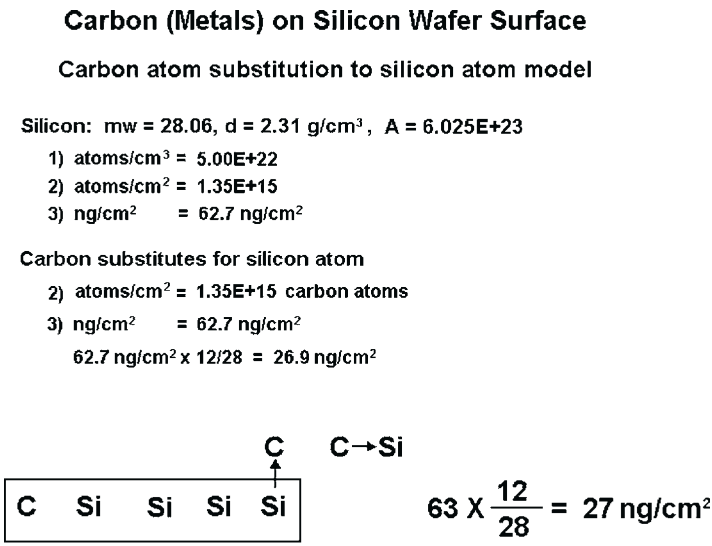


FIGURE 7.48 Model of monolayer surface coverage for the case of substitution of a carbon atom for a silicon atom.

phthalates turns out to be 1.3E14 DOP molecules/cm^2, 88.2 DOP ng/cm^2 and 65 C ng/cm^2. The measured values of surface contaminant concentration that reach a constant value with exposure time are always in the range of 60–100 ng/cm^2 (Figure 7.49). Figure 7.50 compares the concentration on the surface per unit surface area for several metals and organic compounds.

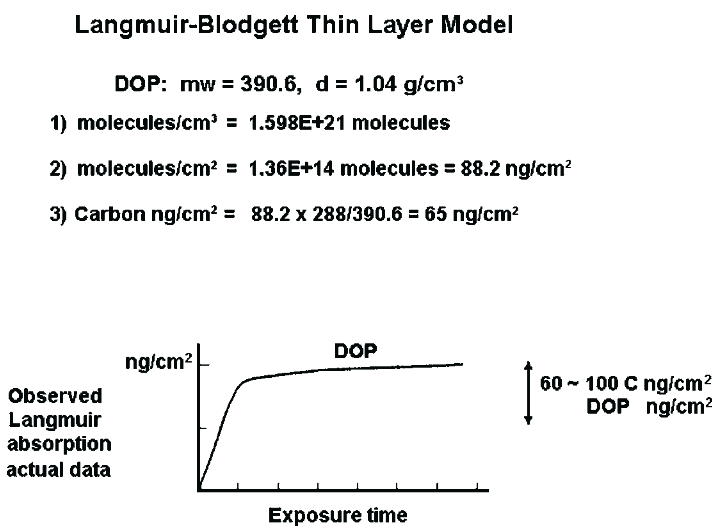


FIGURE 7.49 Model of monolayer surface coverage for the Langmuir-Blodgett thin film model of organic compounds.

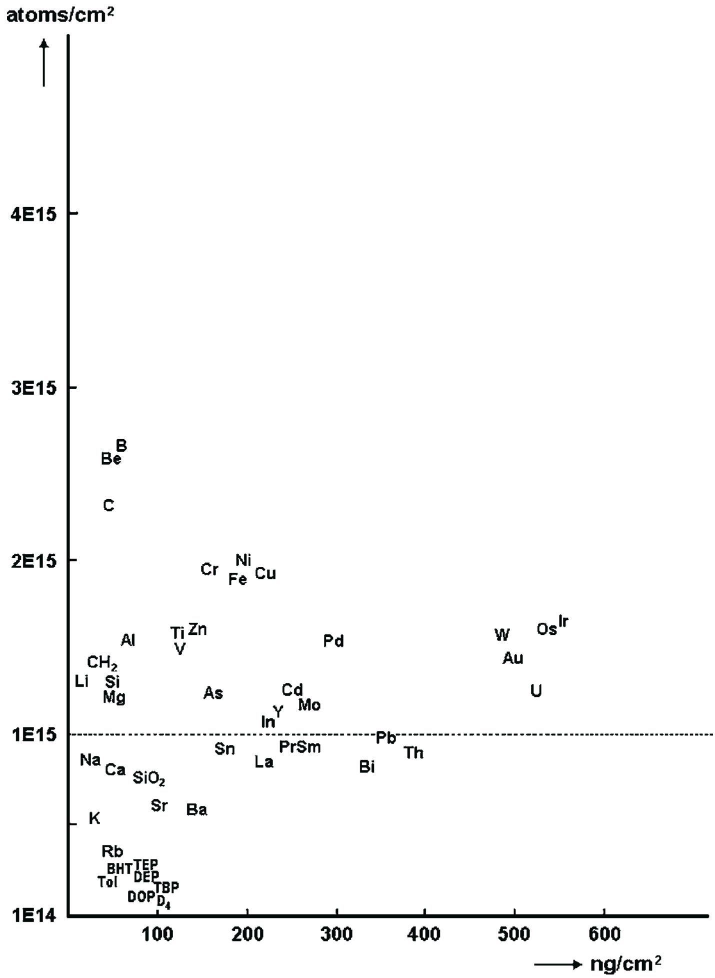


FIGURE 7.50 Concentration of contaminants per unit surface area for various metals and organic compounds.

7.4.3.2 Clausius–Clapeyron Relation

A new method based on Clausius–Clapeyron equation has been developed to evaluate the amount of gases generated during heating. The new method is highly reproducible and accurately represents real situation. Figure 7.51 shows chromatogram data on gas evolution from a polypropylene sheet. The amount of gases generated as a function of temperature is shown in Figure 7.52.

The Clausius–Clapeyron equation considers vaporization of organic compounds, including polymer, the glass transition temperature T_g, and the effect of moisture or steam (>100 °C). It has been shown that the amount of gases

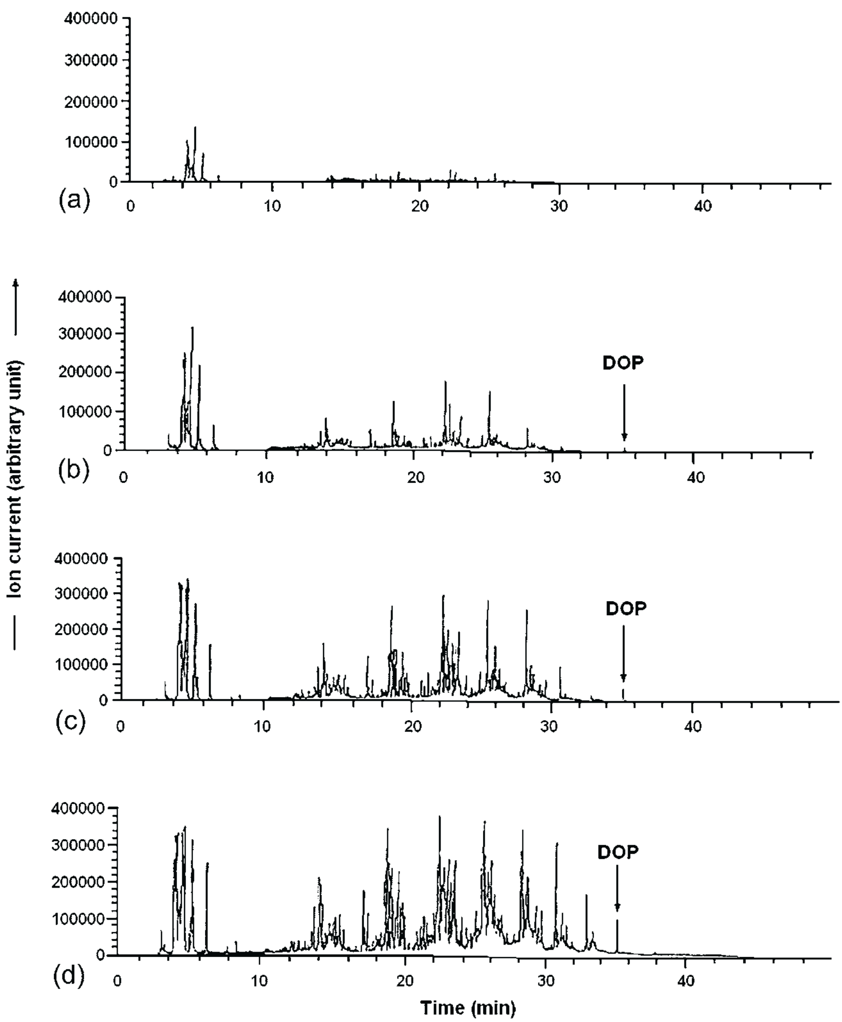


FIGURE 7.51 GC/MS chromatograms of gas from a polypropylene sheet sample at (a) 60 °C, (b) 80 °C, (c) 100 °C, and (d) 120 °C (outgassing time: 1 hour. Gas flow rate: 300 ml/min).

generated can be estimated at any arbitrary temperature when the Clapeyron–Clausius equation is applicable and the gas generation mechanism is constant within the region where linearity is observed, and that the slope is almost the same as that of the vapor pressure–temperature curve for each compound. Figure 7.53 shows the applicable form of the equation. It has been found for many contaminants that the amounts of gas generated at 150, 120, 100, and 80 °C are multiples of the amount of gas generated at room temperature. Figure 7.54 compares the amount of outgassing of polypropylene at room

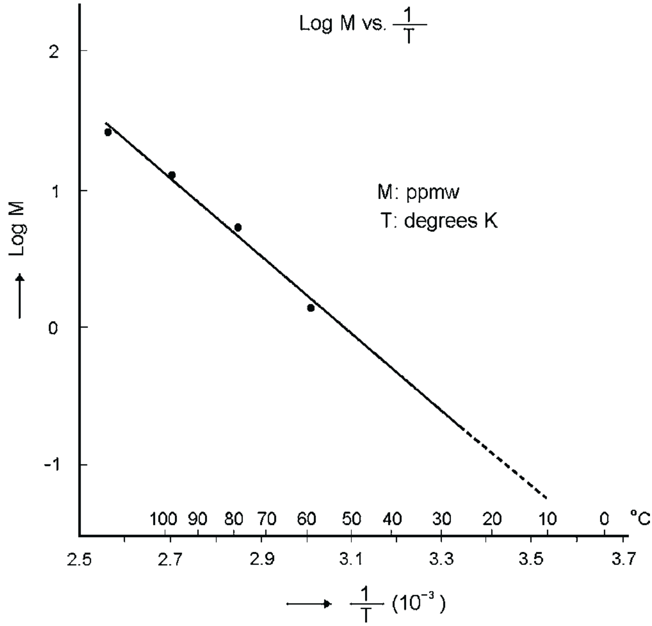


FIGURE 7.52 Relationship between the logarithm of the concentration in μm/g of outgassing contaminants and the reciprocal outgassing temperature in degrees K.

Clausius - Clapeyron Equation

$$\frac{dp}{dT} = \frac{\Delta H}{T * \Delta V}$$

$$\Rightarrow \ln p = -\frac{\Delta H}{R * T} + C$$

$$\Rightarrow \ln \frac{p_2}{p_1} = \frac{\Delta H}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\ln p = k * \Delta H/R \times 1/T$$

FIGURE 7.53 Representation of the Clausius–Clapeyron equation (p = pressure of the phase; T = absolute temperature; ΔH = molar enthalpy of the phase transition; ΔV = difference in the molar volumes of the two phases 1 and 2; c = integration constant; R = gas constant).

temperature estimated from the measured amount at elevated temperature, and the measured outgassing concentration at room temperature. Since the amount of outgassing from a contaminant depends significantly on its vapor pressure, this new method is found to be effective in measuring those contaminants that are lie between the contaminants which can decrease in amount in a day and those which scarcely decrease even after three years.[11,133]

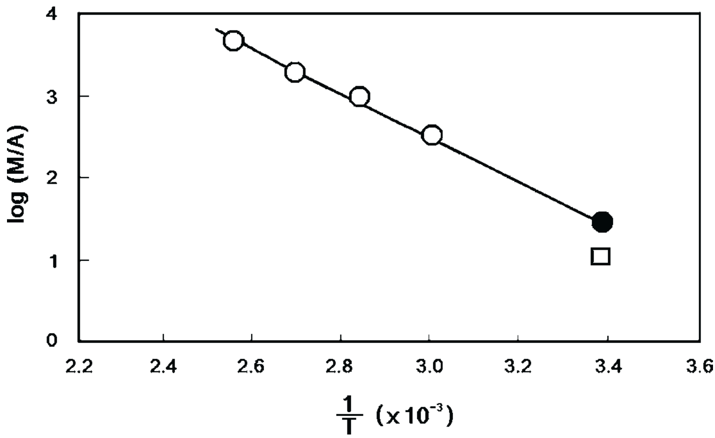


FIGURE 7.54 Outgassing amount from a polypropylene sample at room temperature: comparison of the measured value (circles) with the calculated value (square symbol).

Figure 7.55 shows the vapor pressure of organic compounds as a function of reciprocal absolute temperature. The outgassing data from polypropylene measured with the JACA 34–1999 method[100] have been included in this figure. The temperature dependence of the amount of gas evolved is in good agreement with the vapor pressure dependence.[103,155–157]

7.4.3.3 Outgassing Phenomenon: Two-Phase Exponential Model of Vaporization and Diffusion of Contaminants

All the JACA methods[100] emphasize taking measurements under those conditions which enable evaporation to take place at an equilibrium state between the gas and the surface. A linear relation can be found between the measured amounts and reciprocal of the temperature that triggers surface vaporization and gas release ($1/T$). By using this linear relation, it is possible to estimate the amount of gas generated at any arbitrary temperature.[94,158,159] The same sample was repeatedly measured with this method to verify its reproducibility. The method is found to be highly reproducible. Also, since this method takes measurements in an equilibrium state between the gas and the surface, the measured values are truly represent the real physical situation.

The two methods described in Section 7.3.3 will be discussed here. The IEST dynamic headspace screening test method[25,99] is capable of evaporating all AMCs in a sample and measure them promptly. In other words, the IEST method features high throughput. Therefore, this method is effective when it is necessary to select candidate materials from a large number of materials.

The JACA methods enable us to use a measured value at a certain temperature as a reference basis to estimate the amount of gas evaporated within a unit time at an arbitrary temperature. For example, compared with amount of gas

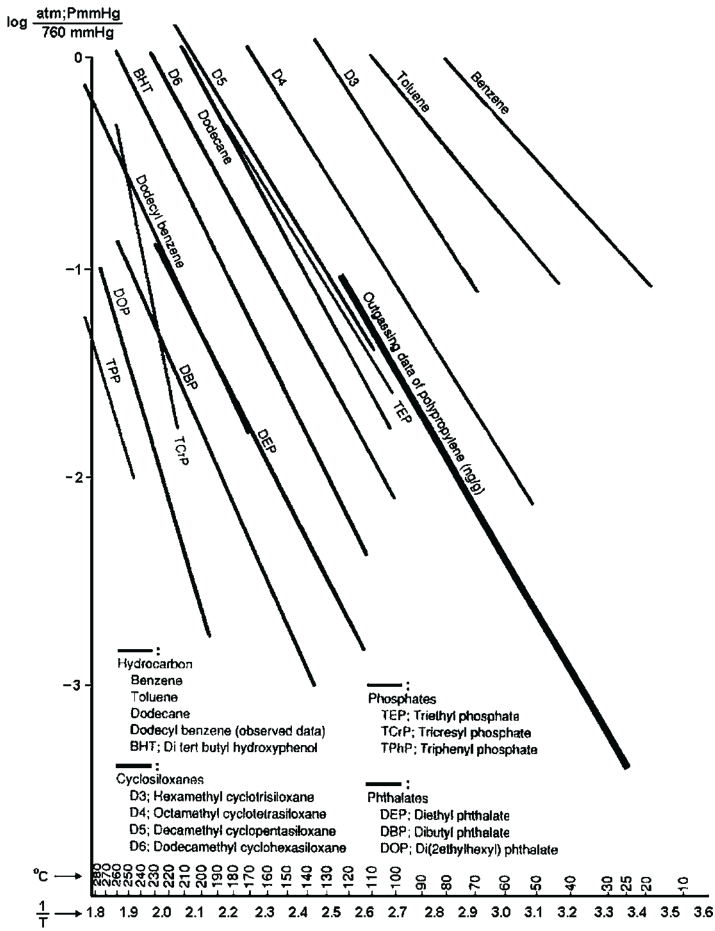


FIGURE 7.55 Vapor pressure of relevant organic compounds as a function of reciprocal temperature together with data on outgassing amounts (ng/g) from polypropylene.

generated at room temperature, the amount of gas generated is 900 times greater at 150 °C, 200 times greater at 120 °C, 140 times greater at 100 °C, and 65 times greater at 80 °C. Takeda et al.[103] have proposed a method to calculate the amount of gas evaporated at room temperature (ng/m^2·hour) by using a nomogram (Figure 7.56), based on measurement data and measurement conditions such as temperature, heating time and surface area of the sample. They reported that the calculated values are in good agreement with measured values.

The sample in the IEST method is a very small pin type with a comparatively large surface. The temperature elevation is rapid. The evaporation of the contaminant from the sample surface is the rate-determining step. Diffusion from the interior to the surface of the sample is not a rate-determining step. All contaminants on the sample surface are quickly evaporated. On the other hand, the

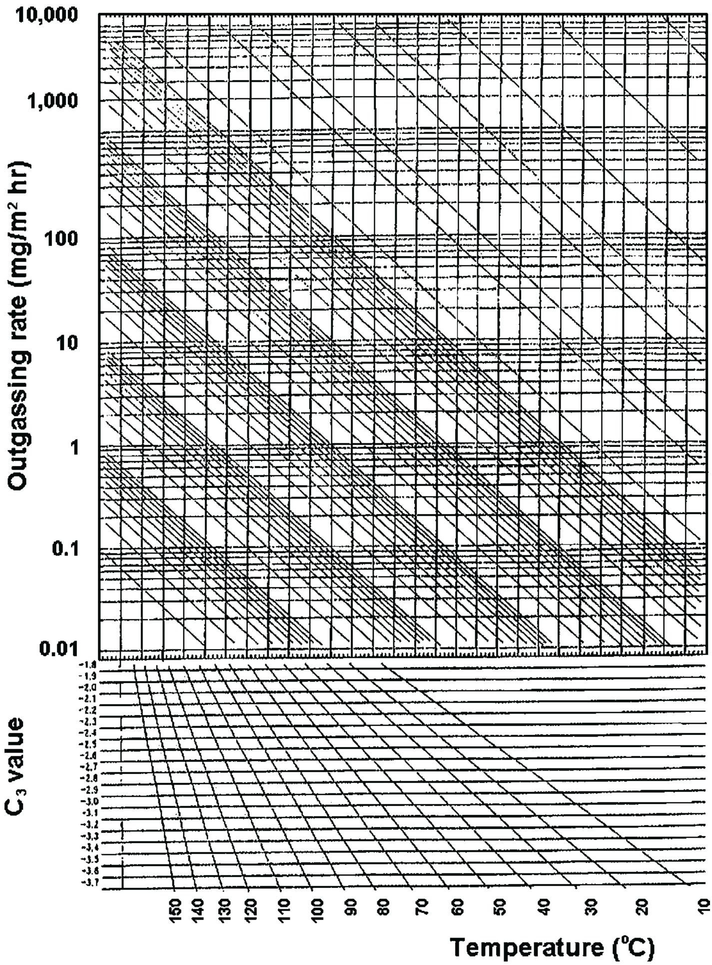


FIGURE 7.56 Nomogram for prediction of the outgassing rate of organic AMC compounds.

JACA methods enable evaporation to take place at a gas–surface equilibrium state where contaminants transfer from the interior to the surface of the sample which is the rate-determining step.

Tamura et al.[125,160] reported reaction rate equations for the two methods by applying the two-phase exponential model of Chang and Krebs.[161] They assumed two reaction rate steps, evaporation from the surface and diffusion to the surface, and formulated a model using a two-phase exponential equation. They reported that the results from this model are in a good agreement with the measured values.

This report clearly defines the positions of the IEST and JACA methods. The measurement data by the JACA methods enable us to estimate the amount of gas evaporated within a unit time, to calculate the amount of gas evaporated over time, and to calculate the AMC concentration in the air at 7 days later.[106,160,162]

To ascertain the relationship between the two models above, it is instructive to consider a hybrid model. By modifying the sample size and shape, and selecting the proper heating conditions, a hybrid model has been developed and experimental data have been obtained to validate the model. The data show both reaction steps surface, evaporation and diffusion to the surface, contribute to the rate-determining step. Using this data, both evaporation and diffusion rates can be estimated and compared with each other.

7.4.3.4 Sticking Probability, Sticking Coefficient, and Staying Tendency

The authors present the actual contamination level in a cleanroom which was regularly measured over several years from start of cleanroom operation. This section also describes the relationship of AMC concentrations at the contamination source, in the air and on Si wafer surface, and their relationship to the problems caused in the manufacturing process. Figure 7.57 shows TIM mode chromatography data for contaminants at different exposure times. The variation of chemical filter performance over a period of several years is discussed in Section 7.5.3.

7.4.3.4.1 Sticking probability[163]

Sticking probability as an indicator of susceptibility to contamination was found to be useful to represent the relationship between AMC concentration in the air and that on Si wafer surface which was exposed to the air.[164,165] Suzuki and Maki's equation[163] used for particle contamination can be applied and the sticking probability can be calculated only from the data obtained. Qualitatively, the sticking probability shows the relative intensity of AMC-to-substrate surface adhesion. Table 7.20 shows the calculation results. Figure 7.58 shows relationship between AMCs concentration in cleanroom air and that on Si wafer which was exposed to the cleanroom air for 24 hours.

Sticking probability is particularly useful when discussing how easily AMCs adhere to substrate surface and how AMCs relate to the fruit-basket phenomenon, compared with particles in cleanroom air. The sticking probability of a particle is 1/400, while the sticking capability of DOP and of phosphate is 1/100 and 1/200, respectively.[133] Figure 7.59 shows DOP concentration in the air on the horizontal axis and that on Si wafer surface on the vertical axis for different exposure time and sticking probability. This figure demonstrates that it is possible to predict the concentration of AMCs on silicon wafer surface in air for a given exposure time. It is also possible to predict the required cleanliness level for air and the exposure conditions to maintain the substrate surface cleanliness level required for each condition. Figure 7.60 shows the levels for DOP and siloxanes.

Momoji et al.[65] reported the sticking probability of acidic and basic contaminants on the surface of a Si wafer using Ion Chromatography and Capillary Electrophoresis Mass Spectrometry. The data are shown in Table 7.21 and Figure 7.61.

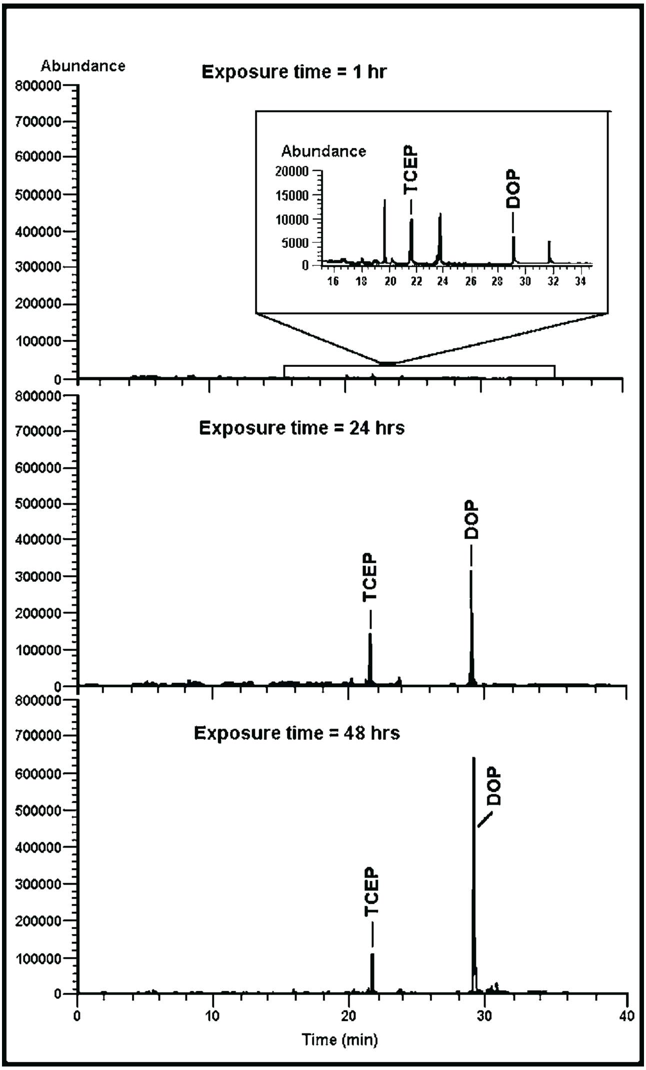


FIGURE 7.57 Total ion chromatograms of the surface contaminants on a Si wafer exposed in cleanroom air.

7.4.3.4.2 Sticking Coefficient

A recent model of the sticking coefficient uses the gas diffusion coefficient in its formulation.[165,166] This model should be useful for quantitative prediction in a sealed space with no air flow where diffusion is the rate-determining step. At present, this model is often applied to a mini-environment which is totally different from a cleanroom environment in terms of the rate and patterns of air flow.

TABLE 7.20 Sticking Probabilities for Various Contaminants

Contaminant	Sticking Probability
Di(2-ethyl hexyl) phthalate	0.010000
Trichloro ethyl phosphate	0.005000
Cyclosiloxanes	0.002500
Aromatic hydrocarbons	0.000017

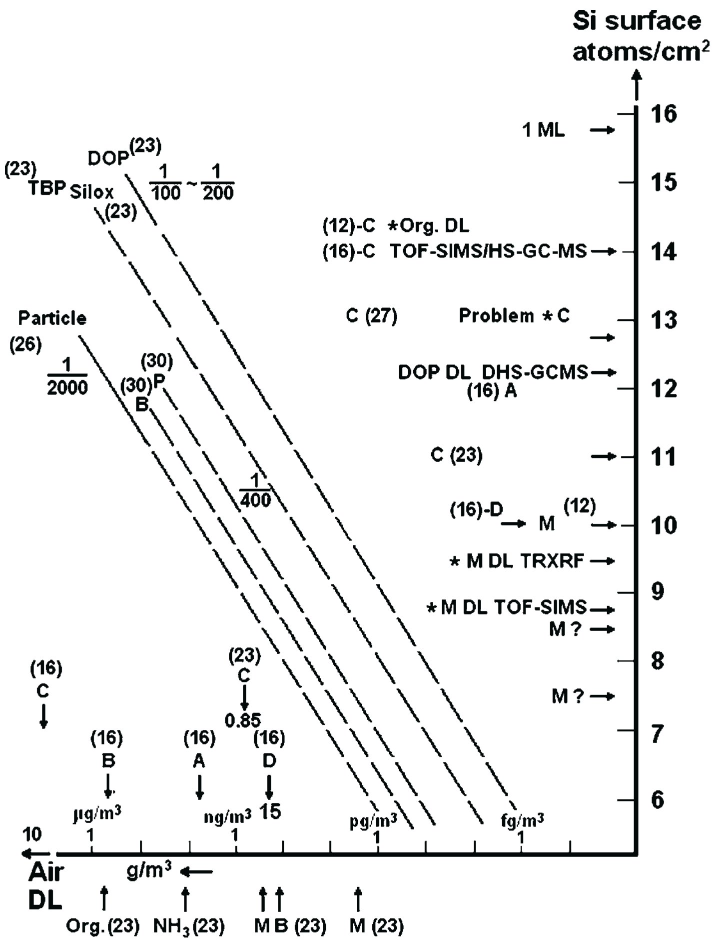


FIGURE 7.58 Relationship of the concentration of contaminants between air and a Si wafer surface (calculated for 24 hours exposure).

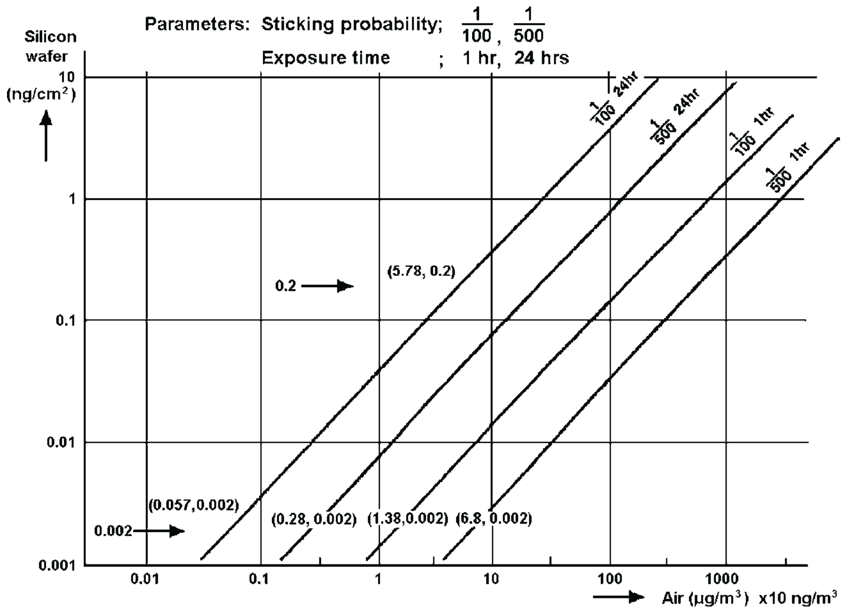


FIGURE 7.59 Concentration of DOP on a Si wafer surface (ng/cm^2) and in air (ng/m^3) for varying exposure times and different sticking probabilities.

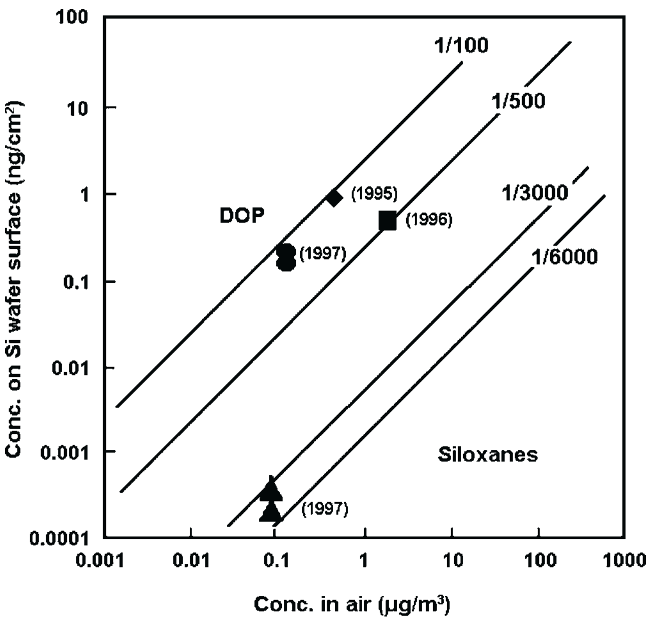


FIGURE 7.60 Relationship between the concentration of organic compounds in cleanroom air and on Si wafer surface necessary to maintain a specified cleanliness level on 1 hour exposure.

TABLE 7.21 The Sticking Probability of Ionic Species on a Si Wafer

Species	Sticking Probability	
	Condition 1 (without amine)	*Condition 2 (with amine)*
Cl^-	6×10^{-5}	3×10^{-3}
NO_3^-	6×10^{-5}	4×10^{-4}
SO_4^{2-}	3×10^{-5}	$<10^{-7}$
NH_4^+	3×10^{-5}	no sticking
Alkanol amine	does not exist in air	5×10^{-3}

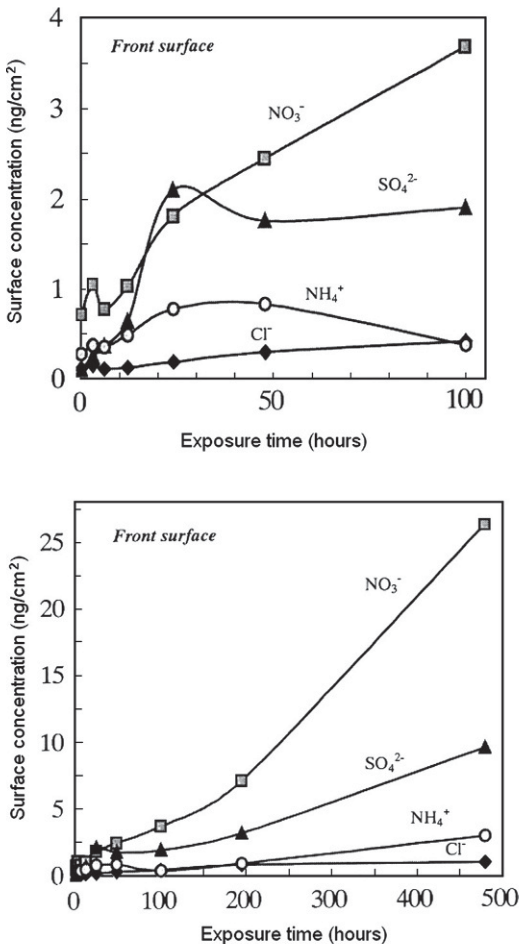


FIGURE 7.61 Relationship between exposure time and concentration of ionic species on the front surface of a Si wafer for (a) exposure time from 0 to 100 hours (upper figure) and (b) exposure time from 0 to 480 hours (lower figure).

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The sticking probability is the basis for a contaminant model dominated by the flow rate of the air such as particles, while sticking coefficient is the basis for a model of contaminants dominated by gas diffusion. Namiki et al.[166–168] have developed a model to integrate these two parameters. Using the model, they calculated the migration rate, average migration distance per unit time, and the deposition rate of gaseous molecules and particles, as shown in Figure 7.62. Three regions, namely a turbulent region, turbulent boundary layer, and a viscous bottom layer, are identified to express the adsorption and the desorption mechanisms on the surface. This model is in good agreement with actual data.

By considering the interaction with water molecules in the air, Kagi et al.[168,169] have obtained the desorption coefficient of DOP from the values measured with API-MS, and calculated the amount of adsorbed DOP. They reported that their calculated results are in good agreement with the measured values. Figure 7.63 compares the calculated DOP concentration with the measured DOP concentration on Si wafer surface.

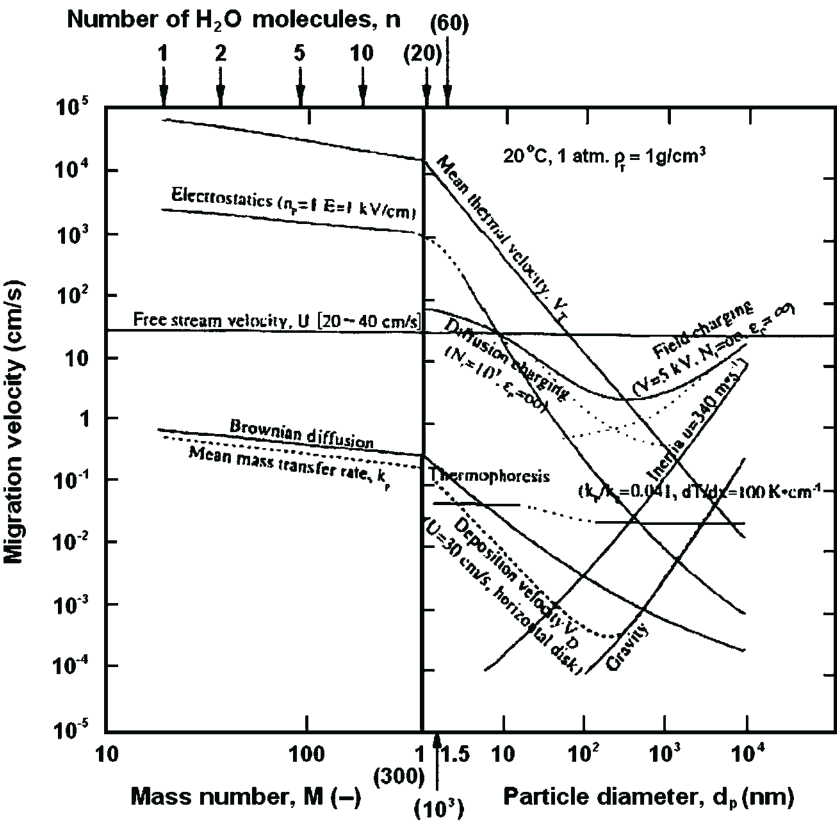

FIGURE 7.62 Migration velocity and average migration length for unit time and deposition rate for particles and gaseous molecular contaminants.

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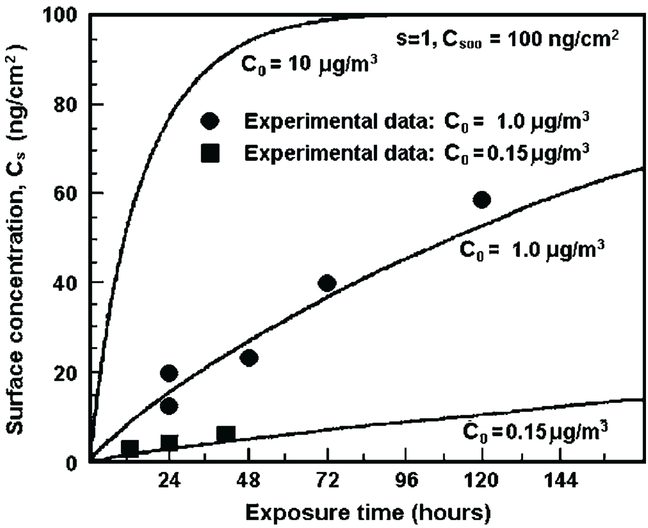

FIGURE 7.63 Comparison of the calculated concentration with the measured values for DOP concentration on a Si wafer surface according to the Kagi model.[168,169]

Using this numerical model, Tamura et al.[138] reported that an explanation using sticking probability would be more applicable to those cases where the rate of air flow was high since the flow rate affects the amount of AMCs adsorbed on the surface.

In addition to the above, Ishiwari et al.,[69–71] Godo et al.,[170–173] Habuka et al.,[174] and Buegler et al.[98] reported numerical expressions to express the relationship between AMC concentrations in air and on Si wafer surface. In many cases, the adsorption of AMCs onto the substrate was explained by the van der Waals adsorption mechanism. Godo et al.,[170–173] have attempted to explain the adsorption mechanism using a molecular structure model. However, Bent[175] has reported that there is an interaction between the substrate surface and compounds, such as ester phthalates, which have "the conjugated double bond" in the chemical structure.

The C–H stretching vibration is different between the monolayer and the multilayer of butadiene on the surface of a silicon wafer. (The stretching vibration of C–H: 3080 cm^{-1} at the multilayer comes to be 2994 cm^{-1} at the monolayer). The analysis method is "Multieflection FT-IR" which Nishiyama and Ohmi have reported.[84] The results of FT-IR have been reported for many years on the interaction between the metal and the compounds including a "conjugated double bond" in the chemical structure. The interaction between acrylonitrile and alkyl aluminum/titanium chloride was reported by Yasui et al.,[176] and the interaction between methyl methacrylate, methacryonitrile and $LiALH_4$ was reported by Fujimoto et al.,[177] (The stretching vibration of the triple bond of C≡N of methacrylonitrile: C≡N; 2230 cm^{-1} in solvent becomes 2267 cm^{-1} as adduct with $LiAlBuEt_3$. The stretching vibration of the double bond of C=O of methyl methacrylate: C=O, 1724 cm^{-1} becomes 1732 cm^{-1} as adduct with $LiAlH_4$). These discoveries contributed to the high tactility polymer consumption. When considering a mechanism for AMC adsorption onto the surface of the substrate, it should be taken into consideration that there is a chemical interaction between two non-transition metals, which have no d-orbital electrons, and the organic compounds, which have a conjugate double bond. The shift in stretching vibration of C–H, C=O and C≡N, which were reported in the results from FT-IR analysis, is thought to be a demonstration of the interaction phenomena between two metal atoms and a conjugated double bond. Consideration of this interaction was described by Buriak,[178] Cypryk and Apeloig,[179] and Schwarz and Hamers.[180] Schwarz and Hamers reported that there was a shift to 1985 cm^{-1} in the C≡N stretching vibration on the surface of a silicon wafer, while a stretching vibration of 2230 cm^{-1} was detected in a liquid medium. In addition, the mechanism and bonding configuration between two Si atoms and acrylonitrile was also considered.

The research into AMC adsorption onto the surface of a substrate has progressed since both physical adsorption, according to van der Waals adsorption mechanism, and chemical adsorption, are taken into account. It is expected that the model of interaction between the substrate surface and the AMCs will become more accurate as the research progresses into the areas of adsorption probability, staying tendency, fruit-basket phenomenon, and other interactions.

7.4.3.4.3 Staying Tendency

The concept of staying tendency has been proposed[25] to express the relationship between concentration of AMCs on Si wafer surface immediately after exposure in air, and the amount of residual AMCs (not evaporated) on Si wafer surface which is converted to carbonaceous materials after thermal treatment. As noted previously, the carbonaceous material can cause problems in the subsequent manufacturing process.

7.4.3.4.4 Fruit-Basket Phenomenon

Hayashi et al. have reported on the fruit-basket phenomenon.[181,182] It is also referred to as “musical seat games” phenomenon.

This is the phenomenon where AMCs previously adhering to Si wafer are replaced by those which reach the Si surface later but are much more adhesive. Organic compounds with lower boiling points rapidly adsorb on the wafer surfaces at first, but they tend to be gradually replaced by much higher boiling points organic compounds. Figure 7.64 illustrates the fruit-basket phenomenon. The lower diagram shows a model based on the assumption that each AMC adheres separately. In reality, however, multiple AMCs coexist as shown in the upper diagram, and the most adhesive AMCs will ultimately adhere to the Si surface.

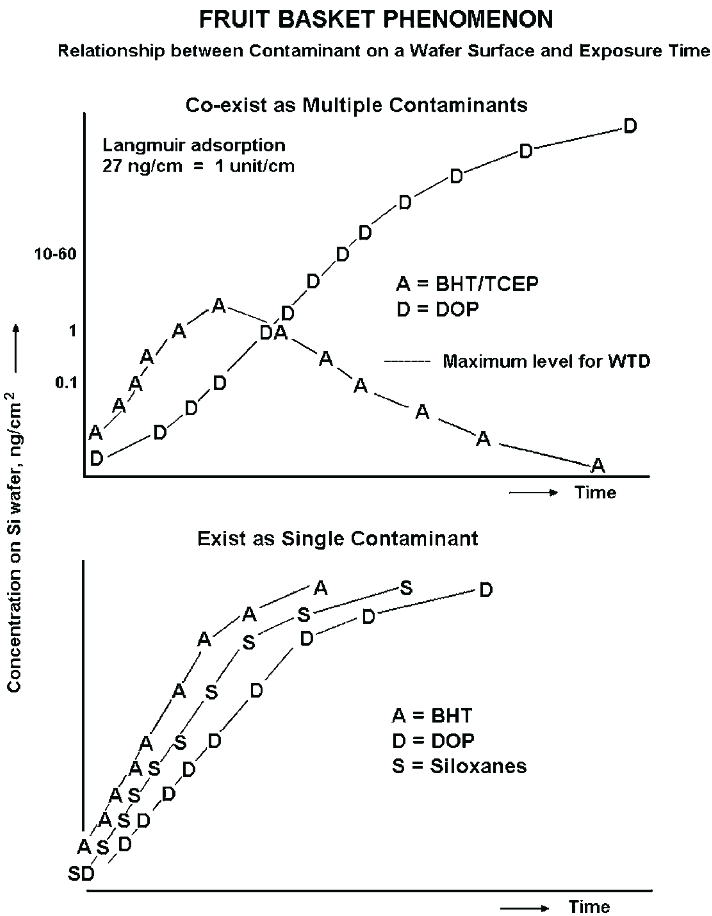

FIGURE 7.64 Schematic model of the fruit-basket phenomenon. (a) Co-exists as multiple contaminants. (b) Exists as single contaminant.

As shown in Figures 7.41 and 7.65, the adhering ester phosphates are progressively replaced with ester phthalates.[133] For the first 24 hours, ester phosphates, just like DOP, increase in proportion to time on Si wafer. The amount of ester phosphates exceeds 5 μg/cm^2 after 24 hours (Figure 7.41). At this stage, 5–10% of the "available seats" are occupied if it is assumed that the maximum amount of ester phosphates that adhere is 60–100 μg/cm^2. As the number of vacant seats becomes smaller, the fruit-basket phenomenon starts to take place. Ester phosphates begin to gradually decrease after showing the maximum concentration value after one or two day exposure. When multiple AMCs coexist, the fruit-basket phenomenon takes place. Ester phthalates follow a similar S-shaped curve, and other AMCs reach peak values once and then keep decreasing.

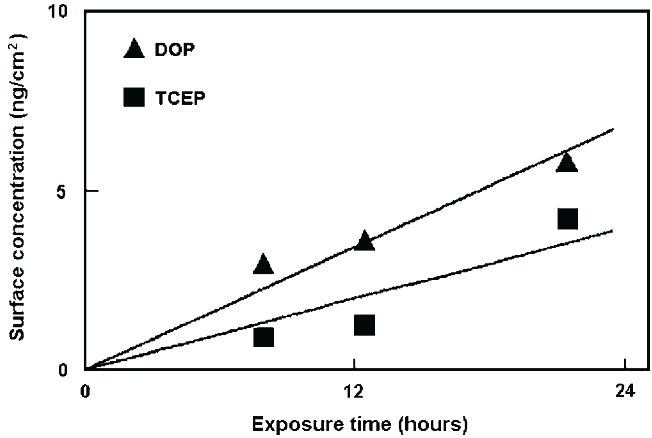

FIGURE 7.65 Change in concentration of DOP and TCEP on a Si wafer surface as a function of exposure time (24 hours) to cleanroom air.

Wafer thermal desorption GC/MS (WTD-GC/MS) is a method to measure the concentration of AMCs on Si wafer exposed to cleanroom air for the purpose of evaluating AMC concentration in cleanroom air. When this method is used, it is necessary to select the exposure condition (time) to allow the concentration of AMCs on the Si wafer to increase proportionally with time. For the case shown in Figure 7.64, the concentration of ester phosphates decreases due to the fruit-basket phenomenon since the amount of DOP exceeds 1–5 ng/cm^2. Thus, in this case, the concentration of ester phosphates on Si wafer measured within the first 24 hours must be used. As shown in this figure, the fruit-basket phenomenon takes place when the amount of DOP exceeds 1–5 ng/cm^2. In other words, the fruit-basket phenomenon does not take place when the amount of DOP is less than 1 ng/cm^2. As has been reported, the DOP concentration on Si wafer reaches a constant value with time at a concentration is 60–100 DOP ng/cm^2 which is assumed to be a monolayer. If it is assumed that a monolayer of DOP is 100 DOP ng/cm^2, the fruit-basket phenomenon starts when 1/100 of the total available seats are occupied by DOP.

7.4.3.4.5 Carbonization Phenomenon

The present authors have investigated the carbonization behavior of contaminants and the methods to prevent problems by evaluating the thermal behavior of DOP. A measurement method was developed to observe the thermal behavior of DOP in a reducing/oxidizing environment. The new measurement method has enabled us to reveal various phenomena and their mechanisms which has contributed to the resolution of problems and to the improvement of the cleanliness of the manufacturing environment.[1,88,183]

DOP is often detected especially on Si wafer surface exposed to cleanroom air in which numerous organic compounds exist. DOP also draws attention as it causes device problems such as the reduction of electric breakdown voltage of oxides. Hence, the thermal behavior of DOP was studied in comparison with that of ester phosphates and siloxanes both of which feature high sticking probability.

Evaporation is a common phenomenon which takes place primarily when organic compounds are heated. If it were only evaporation that occurred when DOP was heated, DOP would be removed easily. In reality, however, DOP-induced problems are always observed. Based on this fact, it has been hypothesized that the heating process triggers not only DOP evaporation, but also other phenomena such as DOP carbonization.[184,185] Changes in the thermal behavior of organic compounds were examined by means of thermogravimetry (TG) and differential thermal analysis (endotherm/exotherm calorimetry) (DTA).

The behavior of various organic compounds in air flow and in nitrogen gas flow was investigated by thermogravimetry (TG) and differential thermal analysis (DTA). Figures 7.66–7.68 show the measurement results and Table 7.22 summarizes these results. All contaminants except DOP and TBP in the air keep absorbing heat until they reach temperatures where there is no mass, indicating that these contaminants are evaporated. Like other organic compounds, DOP absorbs heat in nitrogen gas flow. In air, however, DOP generates heat, which indicates that DOP burns. For TBP in the air, a precise observation of its DTA curve reveals that the heat generation curves overlaps with the evaporation curve. The DOP weight is totally eliminated in air at a temperature of 300 °C or higher. However, it appears that DOP thermal decomposition might be taking place around 200 °C and that TBP also tends to be thermally decomposed to some extent.[1]

These phenomena are helpful when performing contaminant analysis and investigating contaminant behavior by interpreting relevant measurement data. Those organic compounds with high sticking probability adhere to the Si wafer surface, and they go through various thermal treatments in the subsequent manufacturing process steps. Investigation of their vapor pressure suggests that evaporation is the primary reaction when they are heated: they are evaporated and released from the Si wafer surface. When temperature is increased rapidly, however, thermal decomposition and oxidation in the air, as well as primary evaporation reaction, may take place. It is speculated that due to these secondary reactions, organic compounds are partially carbonized and remain on the Si wafer surface. Measurement data taken by means of the TG and DTA are very useful to assess the staying tendency of each organic compound.[25]

7.4.3.5 Investigation of Contamination Based on the "Organic Conceptual Diagram"[186–189]

This section describes the application of the Organic Conceptual Diagram to AMCs in cleanroom air. In the semiconductor manufacturing process, there are problems induced by AMCs which adhere to the surface of substrates such

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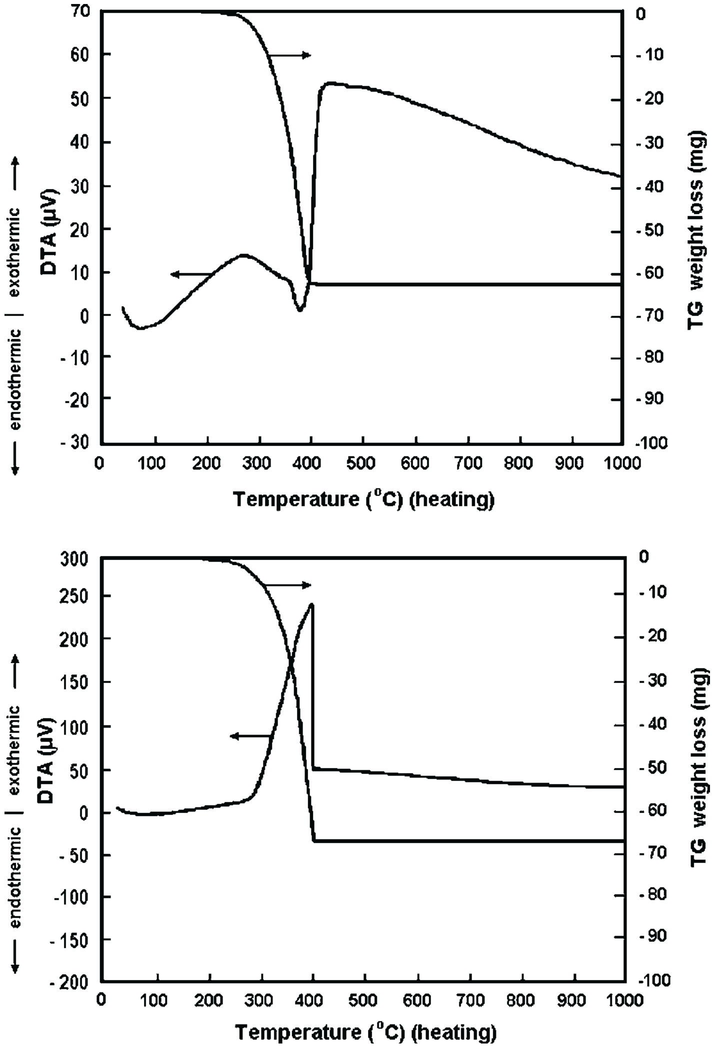

FIGURE 7.66 Thermogravimetry and differential thermal analysis data of DOP for (a) N_2 (upper figure) and (b) air (lower figure).

as a Si wafer and glass. There is a wide range of relevant contaminants, including acids and bases and organic compounds. In order to sort these numerous AMCs, the Organic Conceptual Diagram can be applied.[172–175] Figure 7.69 shows the diagram for AMCs. The key features are highlighted below.

- Basic contaminants react with protons of photoresist to interfere with optical amplification, which triggers T-top phenomenon due to lack of photosensitization. These basic contaminants are located in the strong inorganic region of the diagram.

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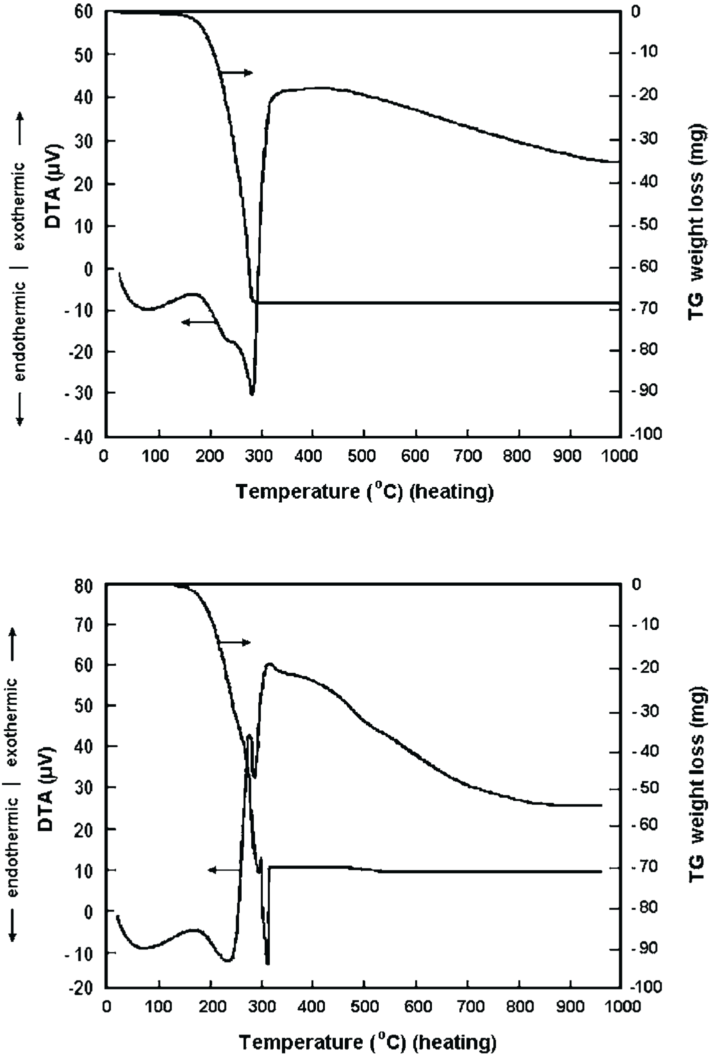


FIGURE 7.67 Thermogravimetry and differential thermal analysis data for TBP in (a) N_2 (upper figure) and (b) air (lower figure).

- Ester phosphates and ester phthalates adhere to Si wafer surface, and are partially carbonized on heating to generate defects in oxides which result in degradation of the electric breakdown voltage. Both ester phosphates and ester phthalates are located in the strong organic region above the sensitization limit line. Cyclosiloxanes, after adhering to Si wafer, are sensitized with UV and photolyzed allowing silicon dioxide to be precipitated which hazes

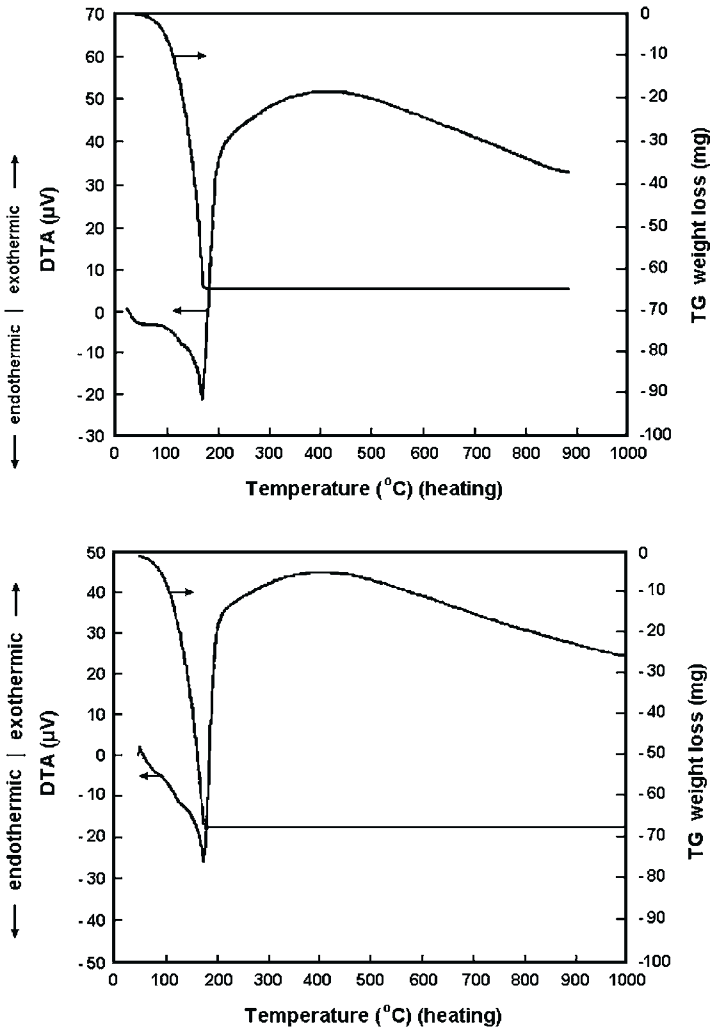


FIGURE 7.68 Thermogravimetry and differential thermal analysis data for cyclosiloxane tetramer (D4) in (a) N_2 (upper figure) and (b) air (lower figure).

or clouds the lens and mirror. The cyclosiloxanes are located just between the organic region and the inorganic region. Some relationships are also observed between the type of problems and the location of AMCs on the diagram.

- The surface of a Si wafer of Si–H type, after being treated with HF is extremely organic and water-repellant. On this type of surface, located very close to and almost parallel with the organic axis, ester phthalates do not adhere so much. Native oxides are found to contain both organic and

TABLE 7.22 Results of TG and DTA for Various Contaminants

Compound	Atmosphere		Temperature (°C) for 99.5% Weight Loss
	N_2	*Air*	
DOP	Endothermic	Exothermic	378
TBP	Endothermic	Endothermic and exothermic	282
OMCTS	Endothermic	Endothermic	172

inorganic portion as determined by contact angle measurements of a water droplet. This is probably because a variety of surfaces including inorganic oxides, Si substrates, and organic Si–H films coexist. Ester phthalates, cyclosiloxanes, acid contaminants, and base contaminants are found to adhere to the surface with these native oxides. The Organic Conceptual Diagram is very effective in exploring appropriate synthetic high molecular weight dyes. Similarly, the relationship between contact angle of a water droplet and the Organic Conceptual Diagram of AMCs can be more clearly defined as more data are accumulated.

- When a new contaminant is detected, by locating it on the Organic Conceptual Diagram, its behavior could be described by similarity with AMCs of known behavior located nearby on the diagram.
- DTA curves indicate that both heat absorption due to evaporation and heat generation due to thermal decomposition take place when AMCs are heated in an inert gas atmosphere. It is expected that the staying tendency of residues after carbonization of organic compounds can also be estimated when contours are drawn from available data. An example of a schematic diagram is shown in Figure 7.70 to define the sticking probability and the staying tendency by means of the contours on the Organic Conceptual Diagram.
- The Organic Conceptual Diagram is useful when investigating compounds which are suspected to cause troubles. Specifically, the Diagram is informative when estimating the mechanism and the likelihood of a potential problem and when determining the priority of experiments.
- Recently, high-performance personal computers have become available to present the *n*th-order figure and easily obtain a map along a desired cross section plane, for example, the distribution of u on the x–z plane at certain x and y values from a (x, y, z, u) tetra-dimension figure. It becomes possible to make more effective use of The Organic Conceptual Diagram.
- Compared with the Organic Conceptual Diagram, quantitative structure activity relationship (QSAR), one of the computer-based quantitative

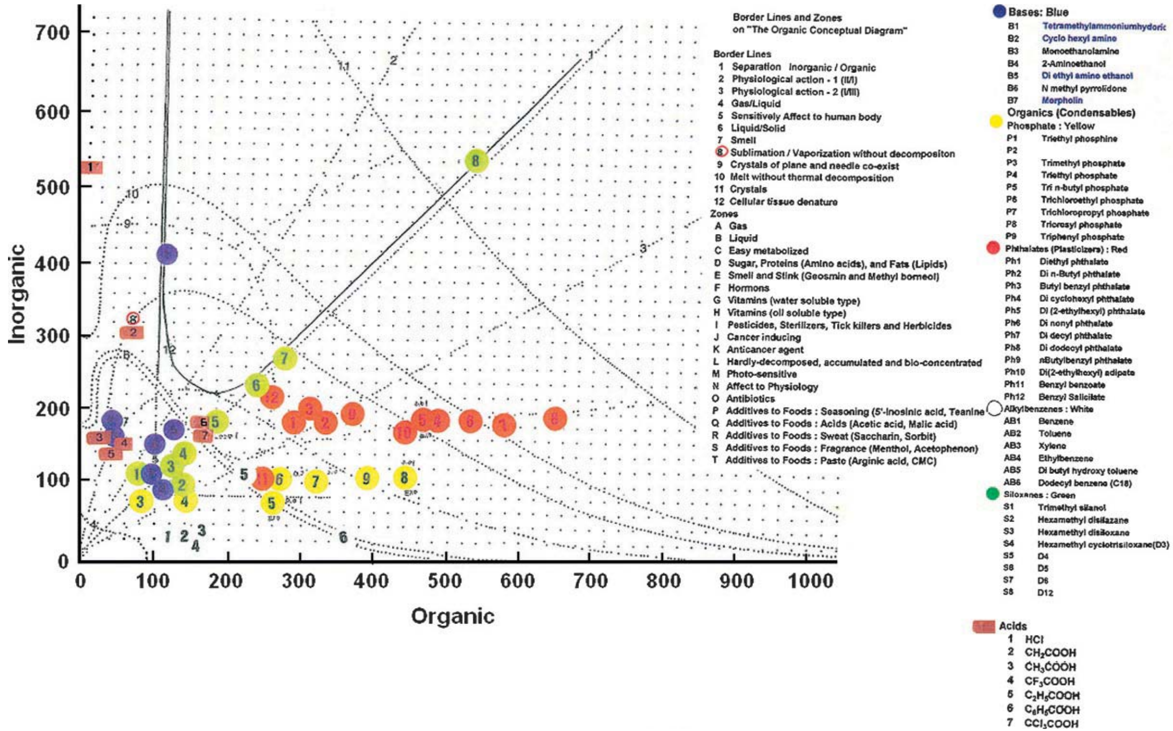


FIGURE 7.69 The organic conceptual diagram for AMCs.

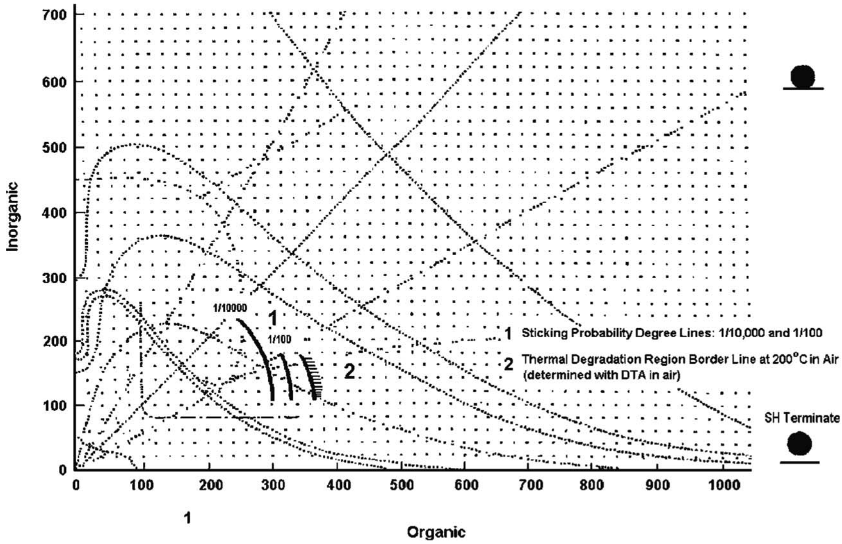


FIGURE 7.70 Schematic representation of sticking probability and staying tendency on the organic conceptual diagram for AMCs.

chemistry techniques, still has a limited database. In addition, it uses hydrogen bound as an important calculation factor. Therefore, at present the Organic Conceptual Diagram, which has a large experimental database, is more useful to the study of AMCs.

7.5 EXAMPLES OF APPLICATION OF KNOWLEDGE AND TECHNOLOGY

This section discusses the sources and excursion of AMCs and countermeasures against contamination.

7.5.1 Excursion of AMCs

7.5.1.1 Data Analysis

Problems cannot be solved merely by revealing the state of contamination on a substrate surface. It is necessary to identify the contaminant sources from upstream cleanroom air, outdoor air, and cleanroom construction materials (building materials, tools installed in the cleanroom, paint, cleanroom garments, shoes, people, chemicals, and water). It is very common to identify contaminant sources by combining and comparing different data taken from these potential sources. So far a database of more than 100,000 contaminants has been built. Still, a new contaminant source is added to the database from time to time, though it is less frequent than before.

7.5.1.2 Investigation of Contaminant Sources in Cleanroom

Methods have been established to identify contaminant sources in an operating cleanroom.[110,190,191] Figure 7.71 shows a method for easily sampling contaminants from ceilings, walls, and any other locations within a cleanroom.[191] Using this sampling method, Itoh et al.[191] have shown that the AMC concentration in the cleanroom was significantly reduced by identifying the major contaminant source in the walls of a cleanroom and substituting it with a low outgassing material (Figure 7.72).

Figure 7.73 shows a double cylinder sampling system.[110,190] By using this system, taking measurements and applying a modeling equation, it is possible to estimate the variation of AMC concentration in the air. Figure 7.74 shows the variation of AMC concentration in cleanroom over several years after construction of the cleanroom was completed. Immediately after completion of construction, the concentration of AMCs showed a considerable drop on a daily basis. A few months of operation after construction, the AMC concentration declined slowly. It has been suggested that evaporation of the AMCs adhering to the surface are a major driving force for the decrease in concentration immediately following construction, and that diffusion and transfer of AMCs to the surface of the materials is the major driving force several months after construction. In other words, two behaviors of AMCs, namely evaporation from surface and diffusion from the interior to the surface for replenishment, are relevant.

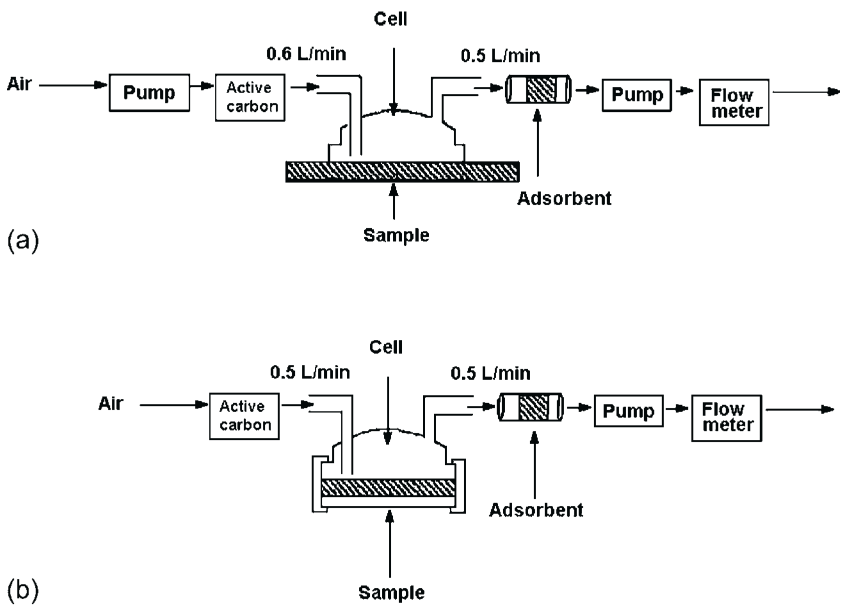


FIGURE 7.71 Schematic of the outgassing methods for evaluating a cleanroom surface after construction by (a) field outgassing method and (b) laboratory outgassing method.

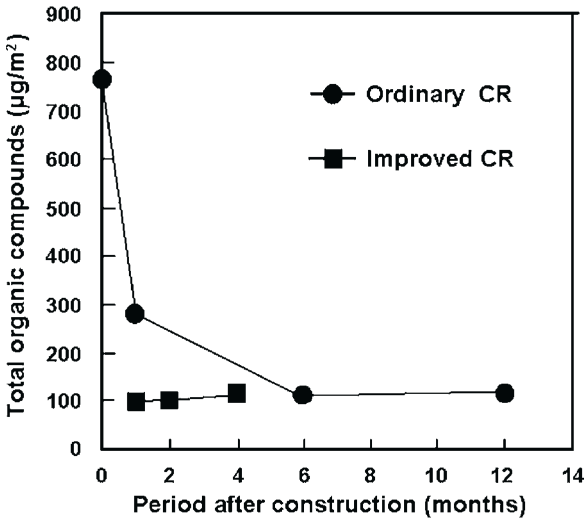


FIGURE 7.72 Relationship between the amount of total organic compounds in cleanroom air and the period after construction for (a) an ordinary cleanroom and (b) an improved (wall substituted) cleanroom.

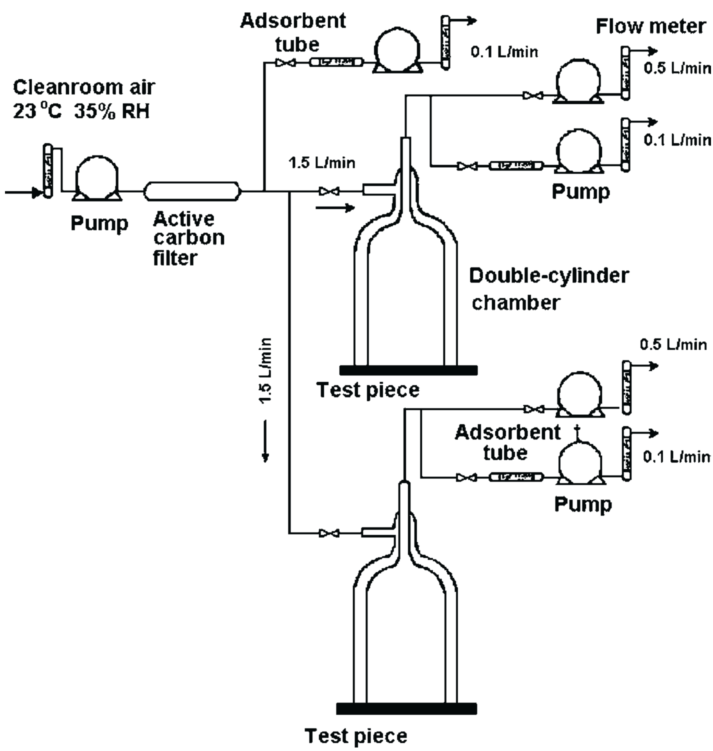


FIGURE 7.73 Schematic diagram of the double-cylinder chamber test method.

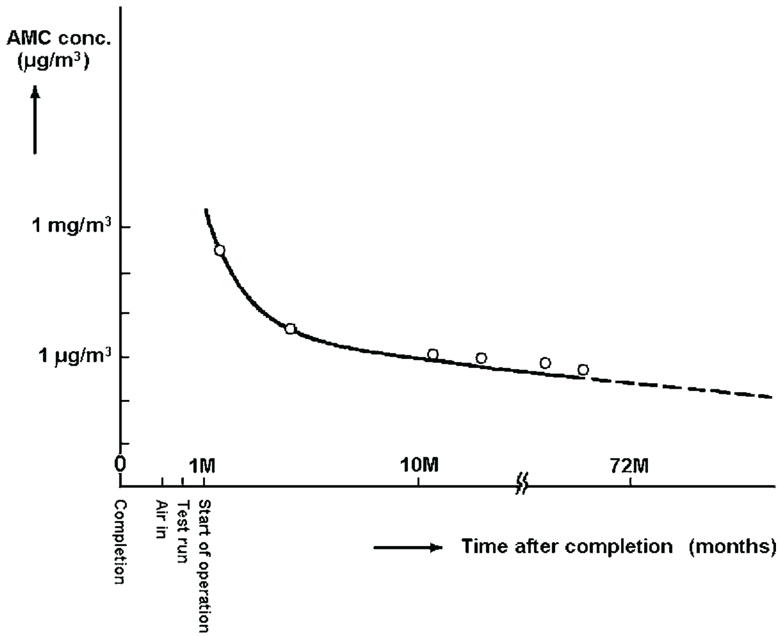


FIGURE 7.74 Schematic model of the concentration or organic contaminants in cleanroom air as a function of time after completion of construction.

This is very similar to the phenomenon of the outgassing evaluation method. Fujii et al.[166,192,193] reported that the data obtained by measuring samples taken with the double pipe sampling system are in a good agreement with theoretical values expressed with a double exponential equation based on the assumption of two reaction steps, namely evaporation from surface and diffusion to the surface. It is possible to estimate the concentration at a few months or at several years later.

7.5.2 Selection of Construction Materials with Fewer AMC Contaminant Sources

Kobayashi et al.[194,195] reported in 1996 that they had succeeded in constructing a cleanroom with very high cleanliness since they had selected low-outgassing cleanroom construction materials using the outgassing test method. Since then the application of this method has become common and many of the newly constructed cleanrooms feature very high cleanliness. This method has also been applied to mini-environments.[104,105] Recently, published evaluation reports for mini-environments have been increasing.[124,125,193,196–199] Also, it has now become possible to fabricate Si wafer carriers with low concentration of contaminants by selecting low-outgassing plastics. The number of published reports related to newly detected contaminants has recently tapered off, while published reports on emission sources are still increasing.[124,125,193,198–209]

Section 7.3.3 discussed this issue in detail. Section 7.5.6.4 describes how the low levels of AMCs have been achieved in a cleanroom.

7.5.3 Chemical Filter for Air Cleaning

7.5.3.1 Change of Total Organic Concentration with Time

Figure 7.75 shows the concentration of total organic compounds in cleanroom air as a function of the years of cleanroom operation. As can be seen, with only an ULPA filter and no chemical filter, the total organic concentration decreased rapidly in the first 6 months of cleanroom operation, though the initial concentration was about as high as 800 μg/m^3. Since then, the total organic concentration has been maintained constant at about 100 μg/m^3 for 3 years.

The total organic concentration in outdoor air was about 10 μg/m^3. At the start of cleanroom operation and after one month of operation, the total organic concentration measured at the chemical filter and at the ULPA filter was far lower than that measured under ULPA filter alone with no chemical filter installed. After 6 months operation, the total organic concentration was found to gradually increase. After 2 years of operation, the total organic concentration measured at the chemical filter was almost the same as that measured under the ULPA filter.

7.5.3.2 Change of Siloxanes Concentration in Cleanroom Air

A variety of organic compounds are detected in cleanroom air. Among these organic compounds, siloxanes are known to be gases derived from silicone

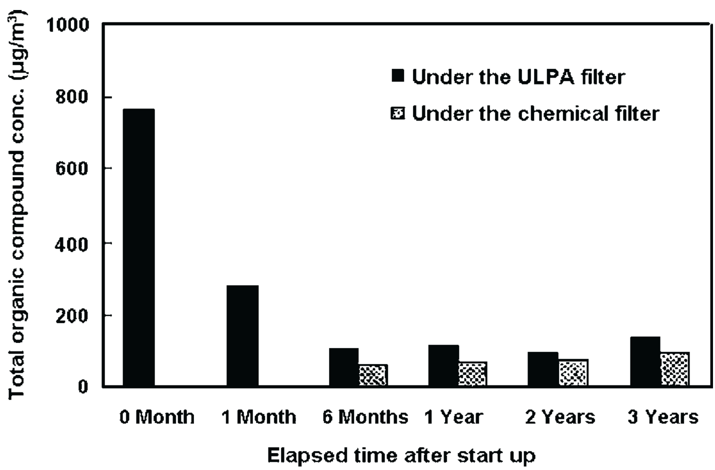


FIGURE 7.75 Time-dependent fluctuation of the concentration of organic AMCs in cleanroom air during the 3 years since the start of operation.

sealant. Silicone sealant is used in large quantities in cleanrooms. At the start of cleanroom operation, gases derived from monomer low molecular weight cyclosiloxanes contained in the sealant are generated in very large amounts. When the polymerization catalyst (initiator) added during silicone polymer production remains in the sealant, it is suggested that the catalyst gradually decomposes the polymer to generate low molecular weight siloxanes which constitute a major contaminant source. The degradation products mainly include octamethylcyclotrisiloxane (D_3) which is a trimer of cyclosiloxane, and decamethylcyclotetrasiloxane (D_4) which is a tetramer of cyclosiloxane. The low molecular weight siloxanes detected as degradation products are released for a relatively long time. For siloxanes (cyclic trimer to hexamer), Figure 7.76 shows the change in their contamination levels and their concentrations measured under chemical filters as a function of time of cleanroom operation, and Figure 7.77 shows changes in their concentrations measured under ULPA filters. Clearly, the concentration of siloxanes remains at a certain level for years though it initially shows a gradual decrease. Table 7.23 shows the total organic concentration as well as concentration of specific siloxanes such as cyclosiloxanes (D_3 to D_6), as well as the phthalates DBP and DOP, measured under ULPA filters and chemical filters after three years of cleanroom operation. The total organic concentrations measured under the two types of filters are found to be of the same order. This indicates that the removal efficiency of the chemical filter for total hydrocarbons is relatively low. With respect to the concentration of low molecular weight cyclosiloxanes, however, the removal efficiency of the chemical filter is still observed after 3 years of operation in reducing the concentration of siloxanes to less than a 10th of the value with a standard ULPA filter.[133]

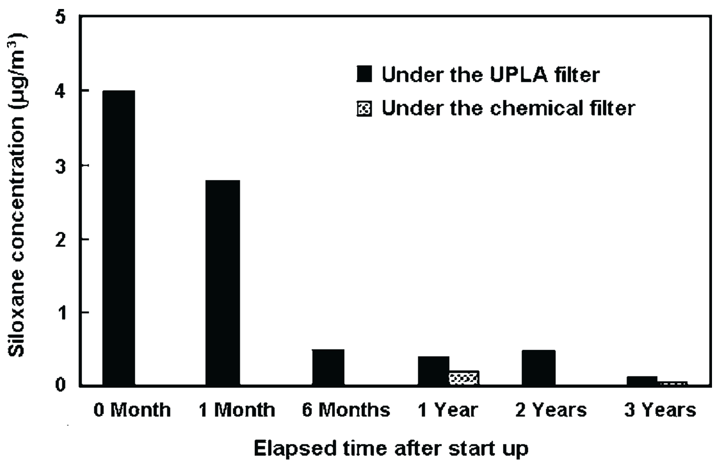


FIGURE 7.76 Effect of chemical filter on the time-dependent changes in the concentration of siloxanes in cleanroom air.

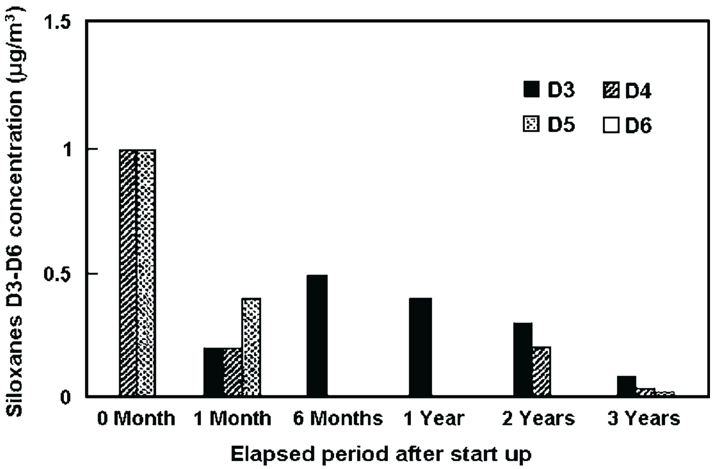


FIGURE 7.77 Time-dependent change in concentration of siloxanes in cleanroom air during the 3 years since the start of operation with ULPA filtration only.

TABLE 7.23 Air Contaminant Concentration After the ULPA Filter and After Chemical Filter Combination After 3 years of Cleanroom Operation

Analyte	Standard ULPA Filter (μg/m^3)	Chemical Filter + ULPA Filter (μg/m^3)
Total organic compounds	140	95
DOP	0.15	0.02
DBP	0.03	<0.01
Siloxanes (D3–D6)	0.12	0.04

7.5.3.3 Effect of Chemical Filter

For total organic concentration, removal efficiency of chemical filter declines gradually over time and is very limited after three years of cleanroom operation. For ester phthalates and cyclosiloxanes, such as DOP and DBP which cause serious problems in semiconductor manufacturing process, the concentrations measured under chemical filter are lower by about one order of magnitude than the concentrations measured under a standard ULPA filter. This suggests that a chemical filter still maintains its removal capacity after three years of use. A standard evaluation method for chemical filters has been published.[139,208–213] The principle of the evaluation method is shown in Figure 7.78.[213]

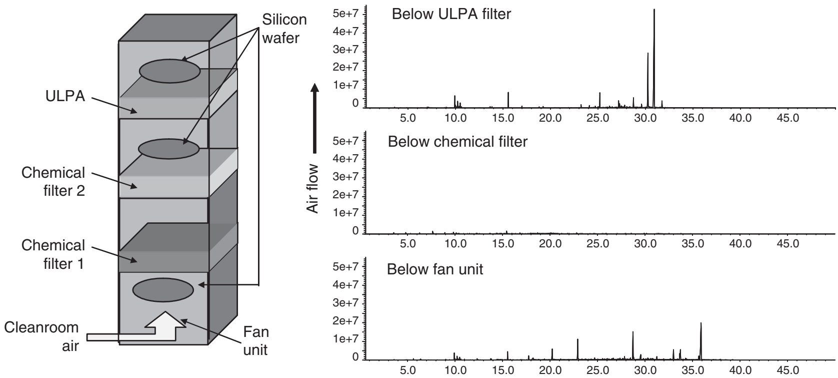


FIGURE 7.78 Standard evaluation method for a chemical filter.

7.5.3.4 Published Reports on Removal Mechanisms

Several reports have been published that describe chemical filters to remove trace AMCs.[94,214–216] Recently, reports on removal mechanisms have also been published.

Sometimes $NH_4H_2PO_4$ is detected as a solid on Si wafer surface. In ambient conditions at a temperature of 23 °C, relative humidity (RH) of 40% and NH_3 concentration of 10 ng/m^3, $NH_4H_2PO_4$ is a stable compound with respect to its equilibrium vapor pressure. It is assumed that $(NH_4)_2HPO_4$ and $(NH_4)_3PO_4$ exist in equilibrium when NH_3 concentration in the air is much higher.[217,218]

Chemical filters for NH_3 are sometimes impregnated with P_2O_5 which serves as an absorption site. The chemical state of ammonium phosphate salts which are in equilibrium at 40% RH and ammonia in air are (NH_4) H_2PO_4 at 10 ng/m^3 and $(NH_4)_3PO_4$ at 1000 ng/m^3, respectively. In other words, the chemical state of active sites is a function of the equilibrium concentration in air.

The rate of removal efficiency (%) was calculated by using the ratio between NH_3 concentration measured downstream of the filter and that measured upstream of the filter in an experiment performed in an atmosphere in which NH_3 concentration is so high (much higher than 10 ng/m^3) that it is in equilibrium with $(NH_4)_3PO_4$ downstream of the filter. It is necessary to check whether the same rate is obtained when NH_3 concentration in ambience is so low (10 ng/m^3) that its ultimate concentration becomes equilibrium with $NH_4H_2PO_4$. It should be noted that the form of the stable compound in an equilibrium state depends on NH_3 concentration downstream of the filter. It is necessary to analyze the data obtained from experiments which employ NH_3 concentration close to the actual operating condition.

The common logic employed in chemical filter evaluation is as follows: "The experimental data are obtained to indicate that NH_3 concentration of 1 mg/m^3 measured upstream of the filter is reduced to 1 μg/m^3 downstream of the filter. The removal efficiency is 99.9%. Therefore, this filter is capable of reducing NH_3 concentration from 10 μg/m^3 at its upstream location to 10 ng/m^3 at its downstream location."

In this type of experiment, the concentration of a target AMC is measured in an environment where the AMC exists by itself at high concentration. In reality, however, various types of AMCs coexist in the environment and the concentration of the target AMCs is relatively low. The adsorption equilibrium state of AMCs at low concentration may be different from that of AMCs at high concentration. Also, the fruit-basket phenomenon may take place among various coexisting AMCs. The removal efficiency obtained in experiments where the target AMC exists at high concentration cannot necessarily be applied as is to low concentration AMCs. It is not necessarily correct to refer only to removal efficiency obtained in experiments to decide whether the cleanliness requirement (20 ppt in ITRS 2002, 10 ppt in 2010) is achievable. In other words, researchers need to look at AMC concentration measured in experiments in which the ambient AMC concentration is close to the AMC concentration in actual cleanroom air.

7.5.3.5 Other Issues

Activities to standardize chemical filter evaluation methods are under way,[12] and some of them are reported in JACA. In the future, it will be important to focus on the following aspects.

- Critical evaluation parameters (BET, micropores, absorption rate, removal efficiency of high-concentration contaminant and the like) for chemical filters to remove water and high-concentration odors
- Macropores (200–500 Å) for diffusion and AMC concentration downstream of the filter to evaluate catalytic carriers (surfactants)

Recent reports suggest that a ceramic filter is effective in removing some AMCs, including phthalates, phosphates, and siloxanes.[219–221] When the ultimate concentration required in the filter after removal of AMCs is very low, the fruit-basket phenomenon is expected to take place in the filter. Once this phenomenon occurs, it becomes necessary, in the case of adsorption and desorption of high molecular weight AMCs, to consider a model in which diffusion is a rate-determining step as in the case of catalytic chemistry. Not only parameters relating to adsorption, such as BET, surface area, and micropores, but also the number of macropores which determine ease of diffusion and migration must be evaluated. For a ceramic filter, it should be technically advantageous if it is designed to allow the macropore diameter and the number of pores to be changed appropriately, as well as to allow the quality of adsorbing points to be selected and adjusted.

7.5.4 Removal from the Substrate Surface (Carbon Chemistry)

7.5.4.1 Cleaning Technology using UV Photoelectron with Catalyst

Fujii et al.[222–227] have reported methods to decompose organic compounds by means of UV photoelectrons, UV and catalyst sources, or other similar excitation sources. Some of these methods have been introduced into actual manufacturing process.[228–231] A silicon wafer carrier equipped with this removal system has been reported.[232,233] This removal system combined with other methods has been successfully applied to the manufacturing process and to mini-environments.[234]

While the target contaminant decomposes by UV photoelectron/catalysis, the detection and structural analysis of the secondary contaminants which cause the secondary reactions to occur are reported. While certain compound combinations are cleaved, there is the possibility that these rejoin to generate another compound. Not all carbon compounds decompose to form CO_2. Some of these include the following: aliphatic C–C bond with a binding energy of 83 kcal/mol; C=C bond which has a higher binding energy; and aryl C=C bond which constitutes a benzene ring. From now onwards, further research will be to investigate which bond combinations decompose easily.[235] Table 7.24 shows the binding energies of the C–C, C–N, C–H, and C–O bonds for several carbon compounds.

7.5.4.2 Other Cleaning Technology

It has been reported that organic compounds are reduced by means of catalysts. There are several reports on organic compound decomposition by means of ceramic catalyst, decomposition on TiO_2-plated surface, and non-thermal equilibrium plasma.[236–240]

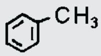
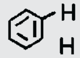
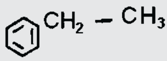
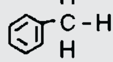
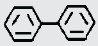
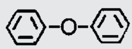
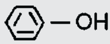
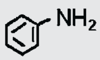
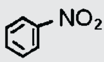
7.5.5 Monitoring Systems

NH_3: A method has been proposed to absorb NH_3 into aqueous solution with a double-pipe wetted-wall technique and to quantify the NH_3 with ion chromatography. A multiple-point sampling technique has been commercialized. In 2001, more than 100 systems were in actual use in Japanese semiconductor fabs. Methods with a determination limit of 20 ng/m^3 have become common.[241–243] STRJ 1999[45] calls for in-line monitoring systems to feature the following sensitivity: 800 ng/m^3 for NH_3, 80 ng/m^3 for amines and 214 ng/m^3 for organic compounds. The above-mentioned NH_3 monitoring system meets this requirement.

NO_X: Monitoring systems employing chemical-luminescence and absorbance techniques are commercialized.[244,245]

IMS: IMS is used to monitor specific organic compounds. The basic principles of IMS are described by Budde and Holzapfel.[51–53,246] Illuzzi and Landoni have applied IMS to monitoring AMCs in an actual fab cleanroom.[247,248]

TABLE 7.24 Bond energies for cleavage of various carbon bonds

Bond energy		kcal/mol	Bond energy		kcal/mol
C–C	$HC \equiv CH$	230	C–H	$-\overset{H}{\underset{H}{C}}-H$	98
	$H_2C = CH_2$	158		>C – H	90
	$H_3C - CH_3$	83		>CH–H	94
		87			104
		63			78
		87	C–O	C_2H_5-OH	90
	$H_7C_3-C_3H_7$	76			90
	$CH_2CH_2CH_2-C_3H_7$	54.5			107
C–N		94		$H_3C-\overset{O}{\overset{\|}{C}}-O-C_2H_5$	87.1
		62			
	$C_2H_5-NH_2$	78			
	$C_2H_5-NO_2$	52			

SAW: Bower[249] has proposed the use of SAW to mainly monitor condensables. Recently, applications of SAW to AMC monitoring in actual cleanrooms are increasing.[250–253] Figure 7.79 shows a typical SAW signal as a function of exposure time.

API-MS: Several authors have proposed the use of API-MS for monitoring AMCs.[254–257]

7.5.6 Recent Developments

Recent developments in AMC technology are discussed in this section.

7.5.6.1 Cleanliness Requirements for Si Wafer

7.5.6.1.1 Cleanliness

The cleanliness requirements for Si wafer are shown in Tables 7.1 and 7.2 taken from the ITRS Roadmap[b,18.] Table 7.2 indicates that the cleanliness required in

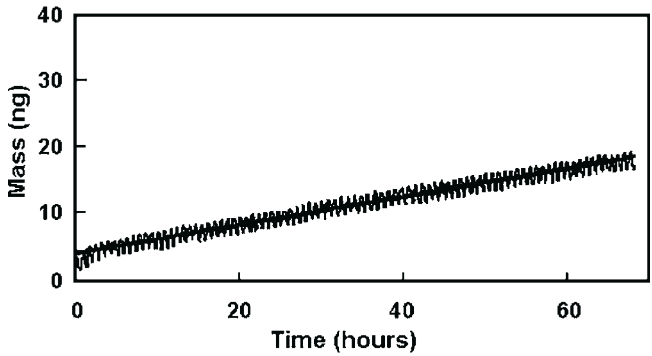


FIGURE 7.79 Typical result of SAW sensor monitoring showing the SAW mass as a function of exposure time.

2002 is 5.3×10^{13} C atoms/cm^2. For a monolayer model, this cleanliness level is equivalent to 2.5 ng/cm^2. For a 300-mm Si wafer, cleanliness of less than 1750 ng must be achieved.

7.5.6.1.2 Analysis

To analyze this level of cleanliness, analytical methods need to quantify a level of 100 ng to one order of magnitude lower than the required level. For a witness plate Si wafer of 1 cm $\times$ 3 cm, however, it is necessary to analyze on the order of 1 ng (0.3 ng/cm^2).

The STRJ Roadmap[45] requires a measurement method that has a detection limit for organic compounds on Si wafer surface of 1.4 E atoms/cm^2 (7 pg/cm^2) in 2002. It is possible to quantify sub-nanogram level contamination on a 300 mm Si wafer by controlling blank values at a low level. Cleanliness of 1 ng on a 300 mm Si wafer is equivalent to 1.5 pg/cm^2, which meets the required determination limit.

7.5.6.1.3 Required Cleanliness

The approach to establish a required cleanliness level described by Kinkead et al. in SEMATECH No. 95052812ATR[2] is that "a cleanliness level is to be 1/100 of the minimum concentration (in the air/on Si wafer surface) that was reported to cause detrimental effects." In accordance with this approach, the required cleanliness level of either DOP or TCPP should be 0.02 and 0.13 ng/cm^2 on Si wafer surface, respectively. The determination limit of the conventional analytical methods for 200 mm wafer is 0.003 ng/cm^2. This level of cleanliness can be evaluated if a wafer is used as a sample.

7.5.6.1.4 Air

Table 7.25 lists the cleanliness levels from the ITRS roadmaps for 1997 and 1999. From 2002 to 2011, the required levels of cleanliness for AMCs such

TABLE 7.25 AMCs in the ITRS Roadmap for 1994, 1997, and 1999[18]

Year	1995	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2008	2009	2011	2012	2014
Technology Node	*350 nm*		*250 nm*			*180 nm*	*130 nm*			*100 nm*		*70 nm*		*50 nm*		*35 nm*
1994 Edition																
VOC ($\mu g/m^3$) (ca. ppbM)	100		30			10										
Ionics ($\mu g/m^3$)	1		0.3			0.01										
THC (ppb)	10		3			1										
Metals (ppb)	0.1		0.03			0.01										
1997 Edition																
Critical particle size (nm)		125		90		75		65			50		35		25	
Particles > critical size (nm)		27		12		8		5			2		1		1	
AMC's (ppt M)																
Lithography—Bases (as amine)		1000		1000		1000		1000			1000		1.0×10^3		1000	
Gate—Metals (as Cu, $E = 2 \times 10^{-5}$)		0.7		0.3		0.3		0.2			0.1		0.07		<0.07	
Gate—Organics (as MW ≥ 250, $E = 1 \times 10^{-3}$)		300		200		200		100			100		70		50	

Salicidation contact—Acids (as Cl^-, $E = 1 \times 10^{-5}$)		10		10		10		10			10		10		10	
Salicidation contact—Bases (as Na^+, $E = 1 \times 10^{-6}$)		80		40		30		20			10		4		<4	
1999 Edition																
Critical particle size (nm)				90	90	75	65	60	55	50		35		25		18
Particles > critical size (nm)				12	10	8	5	4	3	2		1		1		1
AMC's (ppt M)																
Lithography—Bases (as amine)				1000	1000	750	750	750	750	750		<750		<750		<750
Gate—Metals (as Cu, $E = 2 \times 10^{-5}$)				0.3	0.3	0.25	0.2	0.2	0.15	0.1		<0.07		<0.07		<0.07
Gate—Organics (as MW ≥ 250, $E = 1 \times 10^{-3}$)				200	170	130	100	90	80	70		70		50		<50
Gate—Organics (as CH_2)				3600	3000	2400	1800	1620	1440	1260		1260		900		<900
Salicidation contact—Acids (as Cl^-, $E = 1 \times 10^{-5}$)				10	10	10	10	10	10	10		10		10		10
Salicidation contact—Bases (as Na^+, $E = 1 \times 10^{-6}$)				40	32	24	20	16	12	10		4		<4		<4
Dopants P or B				<10	<10	<10	<10	<10	<10	<10		<10		<10		<10

as metals, organics and bases will be gradually lowered, while those for acids and dopants will remain unchanged.

7.5.6.2 Trends in Standardization

7.5.6.2.1 Representation of Cleanliness

SEMI F 21-1102[258] designates the classification of AMCs. For each type of AMC, the class designation specifies the maximum allowable concentration of that contaminant for the cleanroom to meet the requirements for that specific designation. The concentration is in parts per trillion (ppt) calculated on a molar basis. Both ITRS[18] and SEMATECH Technology Transfer No. 95052812A-TR[2] have adopted this representation. JACA No.35A-2003[90] has adopted ng/m^3. For cleanliness, exponential values expressed with g/m^3 have been adopted. Table 7.15 shows the JACA cleanliness levels. The ISO Technical Committee 209 is also working on standardization of AMCs.

7.5.6.2.2 Analytical Methods

- Si wafer surface: JACA,[73] ASTM,[259,c] and other organizations have published analytical methods for inorganic and organic compounds.
- Air: JACA has established the analytical methods.[73,90] ISO Technical Committee 209 WG8 is also working on standardization of the methods.
- Outgassing evaluation. SEMI,[116,259–261] IEST,[99] JACA,[100] and ECMA[262] have published standards on analytical methods.

7.5.6.3 New Analytical Techniques

IMS-GC/MS[263] and Empore™ disk in column GC/MS[264–267] are drawing attention as methods to analyze trace AMCs. Methods to analyze phthalates and phosphates at sub-ng/m^3 in the air have been reported.

GC/MS is becoming popular for analysis and investigation of the properties of low-concentration acids and bases.[63–65]

7.5.6.4 Achieving Cleanroom Cleanliness at Low AMCs Level (DOP 0.1 ng/m^3)

This chapter has described the experience and the technology utilized in order to reduce AMCs for experimental investigation of endocrine disruptors in the cleanroom constructed at the National Institute for Environmental Studies in Japan. After construction of the cleanroom was completed using low AMC content materials, the AMC concentration of the cleanroom air was evaluated. The amount of DOP and DBP was found to be below 0.1 and 0.8 ng/m^3, respectively. These results satisfy the present demand for an environment for the experimental investigations of endocrine disruptors and also for the manufacture of semiconductors. Our approach for improving cleanroom environments was proven to be suitable, as shown by the results in Table 7.26.

TABLE 7.26 Results of AMC Concentration in for a Cleanroom Constructed from Low AMC-containing Materials

Compound	Measured Value (μg/m^3)	Determination Limit (μg/m^3)
D3	0.2	0.2
D4	<0.1	0.1
D5	<0.1	0.1
D6	<0.1	0.1
DBP	0.8	0.3
DOP	<0.1	0.1
TBP	<0.1	0.1
TCEP	<0.1	0.1
TCPP	<0.1	0.1

7.6 FUTURE DIRECTIONS

7.6.1 Technology for Cleanliness of AMCs

Technology developments for AMCs will be focused in two primary directions. One direction is in the area of ultraclean technology, while the focus of the other direction will be on "clean enough."

- At the frontier of technology development, new technologies will be developed to further reduce AMCs. At present, cleanrooms with phosphates and phthalates of <10 ng/m^3 are available. Cleanliness will be further improved. One of the highlights is reduction of AMCs concentration in mini-environments, including Si wafer carrier.
- On the other hand, it is also important to define and maintain an appropriate level of air cleanliness, depending on the purpose of the cleanroom and the manufacturing process, by considering the type of products, substrate exposure time to the air, and AMC removal methods.

7.6.2 Advances in Analytical Methods

- Qualification: Due to improvement in instrumental analysis, the mechanisms of AMC adhesion and decomposition will be identified. It will also be to identify the mechanism of surface adsorption of AMCs and how AMCs cause contamination problems. It is also expected that the mechanism of organic compound carbonization on substrate surface will be defined.
- Quantification: It will become possible to determine AMCs at a level of pg/m^3 in the air and at a level of pg/cm^2 on Si wafer surface.

There are a few standardized ultra-microchemical analysis methods for contaminants of the order of nanogram or picogram in the air. (Only a few methods are standardized to determine trace contaminants in water and for electronic grade chemicals and trace acids in water at water storage source locations.) This is partly because inter-laboratory variation is significant for trace analysis and partly because very few laboratories are capable of analyzing analytes of extremely low concentration without contaminating or losing them. We can learn from the experience of pictogram level dioxin analysis when standardizing analysis of cleanroom air, since accuracy control and standardization are significantly more advanced in the dioxin analysis field.[268] It is essential in dioxin analysis to specify exactly the determination limits of the procedure, the analytical method, and the instrument, as well as the blank value, surrogate reference materials, control chart, check interval, and follow exactly what is specified, otherwise satisfactory inter laboratory reproducibility for the pico-level analysis cannot be achieved. The lower the concentration of dioxin gets, the more difficult it becomes to obtain mutually reliable data. Standardization and development of standards have just begun based on repeated crosschecks and accuracy control studies in the international arena. These experiences in dioxin analysis will set a good example to those who are engaged in cleanroom air analysis.

Control methods for analysis of electronic grade ultrapure water and chemicals are important and difficult. For analysis of the air, there are more factors to induce errors that affect reliability. Not only those who analyze cleanroom air and Si wafer surface but also those who receive and study measurement data will learn a lot from the experiences in dioxin analysis on how to maintain reliability and how to control accuracy of the measurement and data.

These analytical activities are quite different from instrumental analysis in which instrument performance governs the determination limit. Specifically, these analytical activities fall in the wet chemical analysis category which requires a lot of pre-treatments such as sampling, condensation, and preparation. It is essential to handle analytes without losing or contaminating them and to keep blank values as low as possible.

This is the detection limit of the third type, according to Kaiser[16] a model "to measure weight of captain (analyte) on a boat (blank value)." Figure 7.80 illustrates this model. Here is a case where it is not the blank value itself, rather its variation that affects the detection limit.

7.6.3 Application of Air Cleanliness Technology to Various Fields

The technologies to control AMCs in cleanroom air will be increasingly deployed in various fields. Below are some examples that illustrate how the technologies to achieve air cleanliness are being applied to other fields than the semiconductor industry. The methods to analyze AMCs in cleanroom air are also being applied to trace contaminant analysis in the environment conservation field.

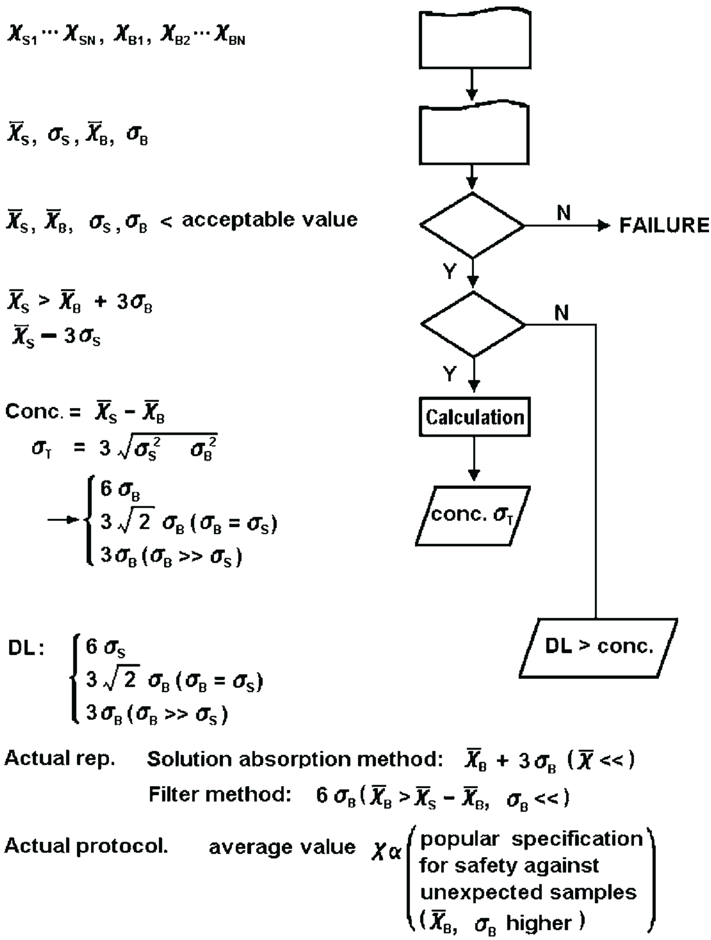


FIGURE 7.80 Schematic model for calculating the determination limit, the calculation limit, and the blank value for analysis of AMCs.

7.6.3.1 Sick House Syndrome

A variety of compounds can trigger sick house syndrome in indoor dwellings. The guidelines specify DBP of <220 μg/m^3 (December 2000) and DEHP of <1200 μg/m^3 (July 2001).[269] In the future, a method will be required to quantify compounds of the ng/m^3 order. Fujimoto et al. have reported the vapor pressure estimation method for AMCs at room temperature.[162]

7.6.3.2 Endocrine Disrupters

Phthalates, DBP in particular, are strong estrogenic chemicals. As endocrine disrupters, their trace analysis for environmental samples including air and water will become increasingly important. Figure 7.81 shows the Organic

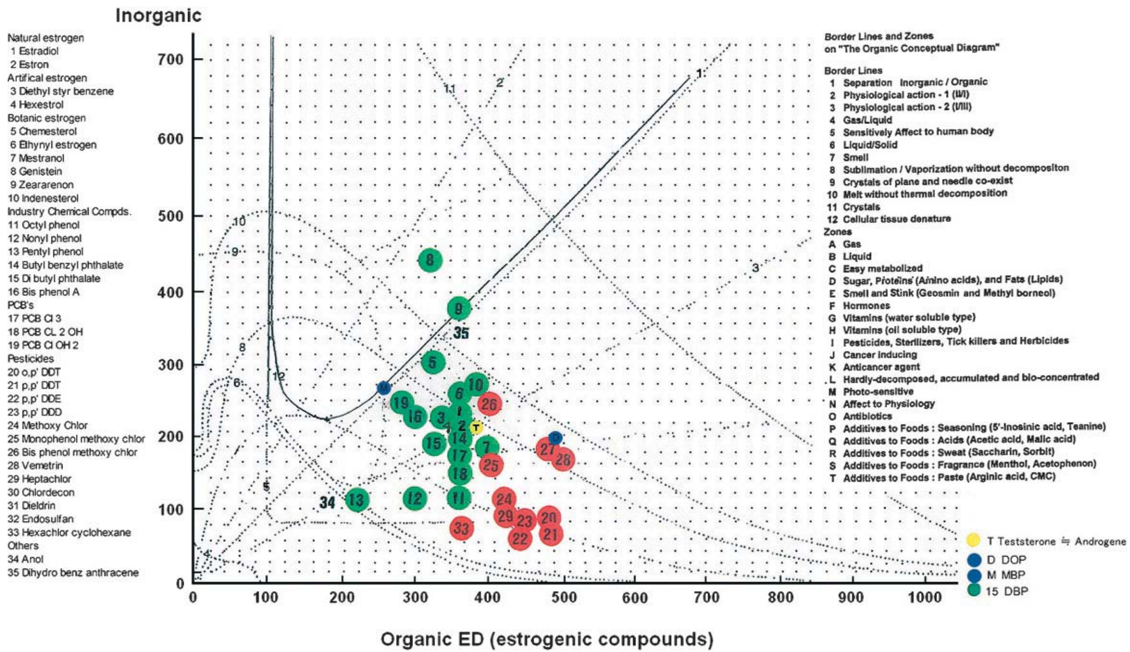


FIGURE 7.81 The organic conceptual diagram for endocrine disruptor materials.

Conceptual Diagram for estrogenic endocrine disrupters. Estrogen (No. 1) and DBP (No. 15) are found to be located very close to each other.[188]

7.6.3.3 Cleanliness of Experiment and Analysis Environments

Cleanliness of cleanroom in terms of ester phthalate contamination is very critical in reducing and maintaining blank values, not only in the experiments but also in the analysis of experimental results. It is found that the blank value may be affected by the environment for drying and storage of the vessel and apparatus, as well as by the atmosphere for sampling, reduction and condensation. So-called “once-through draft” measures have been reported that perform analysis in outdoor air atmosphere and that take in (and not recycle) fresh outdoor air.[188,202,270–273]

7.6.3.4 Other Applications

- Biology: In bio-cleanrooms, siloxanes and phosphates will become problematic.
- In museums, art museums and large-scale PC rooms, specific compounds such as acids, bases and organics will need to be better controlled.
- In cleanrooms, testing rooms, and laboratories of other industries than the semiconductor industry, it will be necessary to give consideration to cleaning, storage and transportation of containers for trace analysis and to control the background level of AMCs.
- It will be essential to establish theories, models and equations in moving from technology to science. The know-how for interpreting and handling a phenomenon within a limited range will be replaced by universal and generic models, theories or equations. Detailed discussion about the proper cleanliness level for a specific contaminant in various manufacturing processes will be consequential in making further advances in measurement and evaluation methods. This is expected to reveal the mechanisms of various phenomena, including the fruit-basket phenomenon, diffusion from the interior to the surface of a solid material, adsorption/desorption on the substrate surface, and carbonization of organic compounds.

7.6.4 Contribution of the Analyst/Chemist to Troubleshooting

Problems in leading-edge technologieCXs are sometimes solved by the inspiration of production experts. It is also common that these problems are often solved by a group effort. A cross-functional team meets to identify potential factors and to work out various actions that are combined to tackle all the potential factors. This type of approach has often been taken to actually solve a number of problems. This approach has often proven to be effective in promptly and efficiently taking appropriate actions, but sometimes this approach does not work.

When the same problem recurs, it is unclear which actions taken before were effective. When the nature of this problem is slightly different from that of the previous problem, the measures designed precisely for the previous problem do not work at all this time. There is a concern in these cases that the past experiences of problem-solving are not at all helpful and that new solutions need to be worked out from square one.

On the other hand, the approach to collect measurement data and to chemically analyze them requires time and labor, and it is not efficient. This approach, however, enables us to accumulate data, conduct and brainstorm technical discussions, and to establish common rules and practices. This approach helps us to deal with a wide range of variables.

Specialists in ultra-microanalysis, chemists, and experts in troubleshooting in semiconductor fabs described above have extensive experience to successfully take the latter approach. In many cases, we have realized that "chemist sense" has more to contribute to breakthrough ultra clean technology than an analyst. These people helped us a lot in preparing this chapter.

7.7 SUMMARY OF THE CHAPTER

The analytical methods for AMCs have been presented. The main analysis for AMCs are (a) on Silicon wafer surface, (b) in the cleanroom environment, and (c) outgassing amounts from raw materials for cleanroom construction materials, such as sealants, plastics, and so on. The analytical methods consist of quantification of inorganic compounds, organic compounds, and identification of abnormal results with local/surface analysis.

The investigation of AMCs has been discussed based on the analytical results, including the source of the contaminants and how AMCs cause contamination problems. This discussion also leads to the methods for preventing contamination and to mitigation solutions. The sticking probability map is useful for selecting target AMCs which have high sticking probability, found usually in cleanroom atmosphere and on Silicon wafer surface, and induce trouble in fab and product quality. Various other interesting findings from the analytical data have been observed and investigated.

ACKNOWLEDGMENTS

The present authors would like to emphasize that solving AMC-induced contamination problems requires people who (a) are experts in trace analysis, (b) are capable of interpreting data to explore chemistry, (c) understand well how semiconductor devices are made, and (d) have extensive experience in successful problem solving. Some overseas physicists are still convinced that having high-sensitivity analysis systems is sufficient.

The authors have had a number of repeated fruitful discussions about AMCs with Dr. Mark Camenzind and Dr. Marjorie K. Balazs of Air Liquide Balazs, Dr. Klaus J. Budde of Siemens AG, and Dr. Laszlo Fabry of Wacker Siltronic

AG. These discussions have enabled us to complete this chapter. The above researchers are also authorities on AMCs who have published clear and in-depth primers on AMCs.

The AMC research described here has been greatly facilitated by a number of stimulating meetings with Professor Tadahiro Ohmi and other Ultra Clean Technology members, the Standardization Committee (chaired by Professor Susumu Yoshizawa) of the Japan Air Cleaning Association organized by Professor Shuji Fujii, Enhanced Clean Technology working group of the Japan Association of Aerosol Science and Technology organized by Professor Toshiaki Nishioka, and the Clean Technology Study Group organized by Professor Shuji Fujii in Tokyo Institute of Technology. Ultra microanalysis methods have been developed based on our repeated discussions with Professor Taketoshi Nakahara at Osaka Prefecture University. The authors would like to extend their heartfelt gratitude to them.

NOTES

a. BPSG is an unstable glass. Pyrex® glass is the so-called "boro-phospho-silicate glass," but it contains Al which makes it stable with good properties as laboratory glassware. It is quite different from BPSG in its stability, melting point and other characteristics. The melting point of BPSG glass is controlled by boron and/or phosphorus concentration to give a low-temperature product to make glass fiber for HEPA filters.

b. Each table of the cleanliness requirement in the 1999 edition of ITRS is compared to the corresponding table of the 2001 edition of ITRS.

Roadmap	ITRS 1999	ITRS 2001
Overall roadmap	Over all	Over all
Starting material roadmap	Table 32	Table 49
Surface preparation technology roadmap	Table 33	Table 50
Defect prevention and technology Requirement (wafer environment control)	Table 80	Table 95

c. This standard was withdrawn in 2003 and transferred to SEMI.

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